Dissertation Proposal

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Abstract

This dissertation consists of three essays in environmental and public economics. Chapter 1 studies how natural disasters affect U.S. presidential elections, asking whether disaster exposure changes electoral support and how voters respond to environmental shocks. Chapter 2 examines the health effects of PFAS exposure, focusing on infant birth outcomes and the role of hydrological conditions and rainfall in shaping exposure risk. Chapter 3 studies the policy allocation problem created by persistent pollutants. Using PFAS as the empirical setting, it asks why current policy is concentrated on downstream treatment and cleanup rather than upstream source control, and calibrates the welfare gains from reallocating public resources toward prevention. Together, the chapters examine how environmental risks affect political behavior, human health, and the design of public policy.

Keywords: environmental economics, natural disasters, presidential elections, PFAS, infant health, persistent pollutants, policy allocation.

JEL: Q53, Q54, Q58, I12, H23, H41, D72.

1 Dissertation Introduction

Environmental risks affect economic and political life in different ways. Some are sudden, like hurricanes and floods. Others build up slowly, through chemicals that move through water, soil, and the body without anyone noticing for years. This dissertation looks at both kinds. It asks how voters react to environmental shocks, how persistent pollutants affect infant health, and how public spending should be allocated when environmental damages are hard to see, hard to trace, and hard to prevent.Chapter 1 looks at natural disasters and U.S. presidential elections. Disasters are visible, but their political fallout doesn't follow mechanically from physical damage. Voters may punish or reward incumbents based on the costs they experienced, on how well the government appeared to respond, or on whether federal aid actually reached them. I estimate how disaster exposure shifts presidential vote shares, and use the results to ask what voters are actually responding to.Chapter 2 shifts to PFAS contamination and birth outcomes. Living near a contamination source isn't the same as being exposed to PFAS. These chemicals travel through groundwater and surface water, so actual exposure depends a lot on local hydrology and rainfall. I study whether PFAS contamination worsens birth outcomes, and whether the effect is larger in places and periods where environmental conditions make exposure more likely.Chapter 3 takes up the policy question that follows. Most PFAS spending in the U.S. goes to downstream treatment—filtering drinking water, cleaning up contaminated sites—even though damages from new emissions will persist for decades. I build a stock-pollution model to compare the marginal value of prevention against the marginal value of remediation, then calibrate it with U.S. PFAS data. The goal is to see whether the current mix is roughly right, or whether shifting resources toward source control would yield large welfare gains.The three chapters share a common question: who ends up bearing the damage from an environmental risk, and who is held responsible for it? For disasters, the question is whether voters can tell the difference between bad luck and bad governance. For PFAS, it is whether the harms of contamination fall on the people closest to the source or on those further along the exposure pathway. For policy, it is whether we are spending public money in the right place. The dissertation contributes to environmental economics by treating these as connected questions, rather than as separate problems about damages, health, and regulation.

Credible Recognition: Disaster Declarations, Party Lines, and Electoral Accountability

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Abstract

Government responses to shocks allocate resources and produce public signals. This paper studies this distinction in federal disaster policy. Presidential disaster declarations are not only gateways to FEMA aid; they are public acts of federal recognition. Using a county-by-presidential-election panel from 1992 to 2020, I link election returns to storm events, FEMA declarations, aid obligations, and gubernatorial partisanship. Federal disaster action is primarily predicted by realized harm rather than by swing-county status or governor-president co-partisanship along the observed county-level margins. Yet voters do not interpret damage-based recognition uniformly. Among casualty disasters, the vote-share slope with respect to human harm is more positive when the event receives a presidential declaration, and this recognition premium is concentrated when recognition crosses party lines. The pattern is difficult to reconcile with a simple co-partisan favoritism account: the premium is larger when governors and presidents are from different parties, not when they are aligned. Mechanism and placebo tests support a signal-extraction interpretation. The premium is strongest for severe casualty disasters and prompt declarations, while future-treatment and no-damage placebos are close to zero. Aid timing further separates recognition from transfer: post-election obligations remain predictive of vote-share changes even though they could not have entered voters' pre-election information sets. Federal disaster declarations therefore operate as public signals whose electoral value depends on partisan credibility, while aid obligations partly measure latent severity and recovery complexity.

Keywords: disaster politics, presidential declarations, FEMA, electoral accountability, retrospective voting, cross-partisan recognition.
JEL Codes: D72, H84, Q54.

1 Introduction

Governments often respond to local shocks by doing two things at once: they allocate resources, and they produce public information. These two functions are closely linked in practice but conceptually distinct. A government action may transfer value to affected groups, but it may also reveal how the state interprets the underlying event: whether the event is severe, whether it warrants public attention, and whether public authority is willing to recognize local harm. This distinction matters for political accountability. If voters reward government action because it delivers resources, the relevant object is the allocation of transfers. If voters respond because government action conveys information, the relevant object is the credibility of the signal.

Federal disaster policy provides a natural setting in which to study this distinction. After a hurricane, flood, tornado, or severe storm, the president may issue a disaster declaration for affected counties. The declaration can open access to federal assistance, but it is also a public act of recognition: it announces that the federal government views local harm as severe enough to warrant national involvement. FEMA aid obligations occur later through administrative processes that depend on damage assessment, eligibility, project formulation, documentation, and recovery needs. Declarations and aid are therefore related but not equivalent. A declaration is discrete, visible, and attributable to the federal executive. Aid obligations are continuous administrative outcomes that combine transfers with information about latent severity and recovery complexity.

This paper studies the political economy of this recognition-transfer distinction. I ask two questions. First, are federal disaster declarations and aid allocated according to political incentives once realized damage is held fixed? Second, conditional on disaster harm and federal recognition, do voters interpret that recognition differently depending on the partisan environment in which it is delivered? The argument is that a presidential disaster declaration is electorally informative not only because it opens access to federal dollars, but also because it publicly recognizes local harm. The informational value of that recognition depends on whether voters can attribute it to partisan accommodation. A declaration issued when the governor and president are from the same party may reflect severe underlying damage, but it may also be interpreted as a lower effective threshold for an allied state. A declaration issued across party lines is harder to dismiss in this way. It should therefore be more diagnostic of underlying severity and federal responsiveness.

This argument combines two lines of work that are usually studied separately. A large literature in political economy emphasizes the allocation of public resources: politicians may target transfers, grants, or public goods toward swing voters, core supporters, or electorally pivotal places (Lindbeck and Weibull, 1987; Dixit and Londregan, 1996, 1998; Strömberg, 2004; Ansolabehere and Snyder, 2006; Kriner and Reeves, 2015). Disaster policy is a natural application because presidents and federal agencies exercise discretion over declarations, eligibility, and relief. Consistent with this view, Garrett and Sobel (2003) document political influences in FEMA payments, and Reeves (2011) shows that presidents are more likely to issue disaster declarations in electorally competitive states. A second literature studies disasters as tests of retrospective accountability: voters may punish incumbents for adverse shocks or reward visible government response (Key, 1966; Kramer,

1971; Fiorina, 1981; Ferejohn, 1986; Achen and Bartels, 2004; Healy and Malhotra, 2009; Gasper and Reeves, 2011; Bechtel and Hainmueller, 2011; Cole et al., 2012; Healy and Malhotra, 2013). These approaches show that disaster politics combines local harm with evaluations of government action. But the relevant government action is not a single object. A declaration is a public recognition decision; aid obligations are administrative transfers that also contain information about latent severity and recovery burden. Treating them as a single relief bundle obscures whether voters respond to material transfers, to public recognition, or to the information contained in each.

The paper separates these margins. On the supply side, I test whether declarations and aid are politically selected once realized damage is held fixed. On the demand side, I test whether voters interpret recognized harm differently depending on partisan context. This distinction allows damage-based allocation and partisan-conditioned voter response to coexist. Even if federal action is largely explained by realized harm along the observed county-level margins, voters may still discount a same-party declaration if they cannot rule out partisan accommodation. A cross-party declaration is harder to dismiss and is therefore more credible as a signal of severity and federal attention.

This credibility mechanism links retrospective voting to models of political information. Voters often infer competence, responsibility, or the state of the world from noisy public signals (Downs, 1957; Popkin, 1991; Lupia, 1994; Zaller, 1992; Persson and Tabellini, 2000; Ashworth, 2012). In this setting, the informativeness of the signal depends on whether the action cuts across partisan lines. I use a simple signal-extraction framework to organize the empirical predictions. The framework is not estimated structurally. Its role is to clarify why the recognition premium should be larger when recognition crosses party lines, and why the premium should be strongest when recognition is most informative: severe human harm, prompt declarations, and environments where voters may rely more heavily on public signals.

I test these predictions using a county-by-presidential-election panel from 1992 to 2020. The data link county presidential election returns to NOAA storm events, FEMA disaster declarations, FEMA aid obligations, gubernatorial partisanship, and lagged county economic and political controls. The outcome is the change in the incumbent presidential party's county vote share between consecutive presidential elections. The main damage measure for the recognition analysis is casualty harm, defined from deaths and injuries and scaled by lagged population. The empirical design proceeds in three steps. I first estimate allocation equations for declarations and aid. I then estimate whether the vote-share slope with respect to casualty damage differs when a disaster receives federal recognition. Finally, I test whether this recognition premium is larger when recognition crosses party lines.

The allocation results do not support a simple political-targeting interpretation at the county-election level. Conditional on realized casualty and property damage, county fixed effects, election-year fixed effects, and lagged controls, neither swing-county status nor governor-president co-partisanship systematically predicts declaration probability or FEMA aid obligations. Damage strongly predicts both outcomes. Partial-R^2 decompositions and equivalence tests show that most

of the explanatory power in the allocation equations comes from realized harm rather than from the political variables. The evidence is strongest against large swing-county targeting effects and less restrictive for co-partisan declaration effects, but the estimates do not indicate systematic co-partisan allocation. These results do not imply that disaster policy is never political. They show that, in this panel, the observed allocation margin is better described as damage-based than as politically targeted along the county-level margins considered here.

The vote-share results show that recognition matters for casualty disasters. Among county-years with casualty harm, the interaction between casualty damage and a presidential disaster declaration is 1.94. Thus, relative to casualty disasters without federal recognition, a one-log- point increase in casualty harm is associated with a 1.94 percentage-point higher incumbent-party vote-share change when the disaster receives a declaration. This coefficient is not the average effect of a declaration. It is the difference in the vote-share slope with respect to casualty damage between recognized and non-recognized casualty disasters.

The recognition premium is concentrated in cross-partisan settings. When the governor and president are from different parties, the declaration-related casualty-damage slope is about 3.9 percentage points. When they are from the same party, the corresponding estimate is about 0.6 and is not statistically distinguishable from zero. The pooled triple-interaction estimate also points to a larger premium in cross-party settings, although the formal difference is less precisely estimated under conservative inference. I therefore interpret the cross-partisan result as a pattern across complementary evidence rather than as resting on a single coefficient: the split-sample contrast, pooled interactions, placebo tests, mechanism tests, and a close-gubernatorial-election subsample all point in the same direction.

This pattern is difficult to reconcile with a simple favoritism story. If the electoral value of declarations came mainly from presidents delivering benefits to aligned state governments, the premium should be larger when governors and presidents are from the same party. The data point in the opposite direction. Recognition is more electorally valuable when it is harder to attribute to partisan accommodation.

The mechanism tests support the same interpretation. The cross-partisan recognition premium is concentrated among severe casualty disasters, where public recognition should be most informative about underlying harm. It is largest for prompt declarations, especially when they cross party lines, consistent with the idea that quick recognition is harder to attribute to prolonged bargaining or delayed administrative processing. It is also larger in lower-income counties, where public signals may carry more weight because private information about local severity is less precise. Lead-treatment, future-partisanship, and non-disaster placebo tests are close to zero.

Aid timing provides a final distinction between recognition and transfer. I split FEMA obligations into amounts obligated before and after Election Day. Post-election aid, which could not have entered voters' pre-election information sets, remains at least as predictive of vote-share changes as pre-election aid. This pattern cautions against interpreting aid coefficients as direct rewards for observed federal dollars. Aid appears to contain information about unobserved severity, adminis-

trative complexity, and recovery needs. Declarations, by contrast, are the public signal through which federal recognition becomes politically visible.

The contribution is therefore not that disaster policy is free of politics. Rather, it is that the politics documented here appears on the interpretation margin. The allocation of federal action is largely explained by realized harm along the observed county-level margins, but voters interpret that action through a partisan informational lens. The paper contributes to distributive politics by separating political selection in the supply of public action from voter interpretation of that action. It contributes to retrospective voting by showing that government response matters not only as a transfer but also as a public signal. And it contributes to models of political information by showing that the credibility of a public signal depends on whether the action cuts across partisan lines.

The empirical design is best interpreted as a set of mutually reinforcing tests rather than a single natural experiment. The paper does not claim that declarations are randomly assigned, nor does it estimate the unconditional causal effect of a declaration. The allocation analysis asks whether the political variables that would generate targeted selection predict declarations and aid once damage and fixed effects are held constant. The vote-share analysis studies how the electoral slope of casualty damage differs with recognition and partisan context. Weather instruments address measurement error in reported damage, not declaration assignment. The close-gubernatorial-election exercise is diagnostic rather than a stand-alone regression-discontinuity design. The interpretation rests on the combined evidence from allocation tests, recognition premiums, cross-partisan heterogeneity, mechanism tests, placebo tests, aid-timing diagnostics, and robustness exercises.

The remainder of the paper proceeds as follows. Section 2 develops the conceptual framework and predictions. Section 3 describes the data and measurement. Section 4 presents the empirical design. Section 5 studies the allocation of declarations and aid. Section 6 estimates the recognition premium and its cross-partisan heterogeneity. Section 7 tests when recognition is most informative. Section 8 examines aid timing. Section B reports robustness checks. Section 10 discusses interpretation and limitations, and Section 11 concludes.

2 Conceptual Framework and Predictions

This section presents a simple framework for organizing the empirical tests. The framework is deliberately reduced-form. It is not a structural model of FEMA decision-making and it is not a Bayesian estimation of voter beliefs. Its purpose is to clarify three distinctions that guide the empirical analysis: allocation versus interpretation, recognition versus transfer, and same-party versus cross-party credibility.

The key friction is informational. Voters observe local harm and federal recognition, but they do not observe the effective threshold that generated the declaration. If a same-party state may be perceived as facing a lower threshold, then a same-party declaration is less diagnostic of underlying severity. A cross-party declaration is harder to attribute to partisan accommodation and therefore

carries more information. The empirical implication is not an unconditional declaration effect. It is a slope-difference: casualty harm should have a more positive association with incumbent-party vote share when it is federally recognized, and this recognition premium should be larger when recognition crosses party lines.

2.1 Recognition, Transfer, and the Allocation Margin

Consider a county c in presidential election year t. A local disaster generates underlying harm S_{ct}. The public observes noisy measures of harm, including casualties and property damage. The federal government may then take two related but distinct actions. It may issue a presidential disaster declaration, T_{ct}, and it may later obligate federal aid, M_{ct}.

The declaration is the paper's measure of federal recognition. It is a discrete, public, and politically attributable act. It identifies the affected jurisdiction and announces that the federal government recognizes the event as serious enough to warrant national involvement. Aid obligations are different. They are continuous administrative outcomes that depend on damage assessment, eligibility, project formulation, documentation, infrastructure needs, and recovery timing. They may arrive before or after the election and may be visible to different constituencies in different ways.

The first empirical margin is an allocation test:

$$\{T_{ct}, M_{ct}\} = f(D^P_{ct}, D^C_{ct}, \text{Swing}_{c,t-1}, \text{SameParty}_{st}, X_{ct}, \mu_c, \tau_t).$$

This margin asks whether declarations and aid are politically selected after conditioning on realized damage. It does not establish random assignment. It disciplines the alternative interpretation that any vote-share premium reflects the political allocation of federal action rather than voter interpretation of that action.

2.2 Recognition as a Credibility Signal

The second margin is interpretive. Voters observe local harm and whether the president issues a declaration, but they do not observe the effective threshold behind the recognition decision. A parsimonious representation is

$$T_{ct} = 1 \quad \Longleftrightarrow \quad S_{ct} + \xi \cdot \mathbf{1}\{\text{SameParty}_{st} = 1\} + \nu_{ct} \geq c,$$

where c is the recognition threshold, ν_{ct} is an administrative or political shock, and $\xi \geq 0$ captures the voter's perceived possibility that aligned states face a lower effective threshold. The parameter ξ is not estimated and is not assumed to be the true degree of federal bias. It represents the possibility that voters may entertain when interpreting recognition.

This formulation separates actual allocation from perceived credibility. The federal actor need not be biased in fact for voters to discount same-party recognition. It is enough that voters cannot

fully rule out a lower threshold for aligned states. When $\xi > 0$, a same-party declaration can be rationalized either by severe underlying harm or by partisan accommodation. A cross-party declaration is harder to rationalize as favoritism. Conditional on observed harm, it is therefore more diagnostic of severity and federal attention.

This informational logic can be summarized by a local linear representation of voter updating:

$$\mathbb{E}[S_{ct} \mid D_{ct}, T_{ct}, \text{Cross}_{st}] = a + w_D D_{ct} + w_T^0 T_{ct} + w_T^1 (T_{ct} \times \text{Cross}_{st}),$$

with

$w_T^1 > 0$ when voters assign positive probability to partisan accommodation in same-party settings.

This expression is not estimated. It is a compact way to state the comparative static: cross-party recognition carries more information than same-party recognition.

2.3 Vote-Share Slopes

The empirical outcome is the change in the incumbent presidential party's county vote share, Y_{ct}. The framework does not require voters to reward disaster harm itself. Instead, voters may update their evaluation of the incumbent party's response when observed harm is publicly recognized. For casualty disasters, the relevant reduced-form object is therefore the vote-share slope with respect to casualty harm when that harm receives federal recognition.

Let D_{ct}^C denote casualty harm. The baseline recognition specification is

$$Y_{ct} = \alpha_C D_{ct}^C + \beta T_{ct} + \delta (D_{ct}^C \times T_{ct}) + X_{ct}'\theta + \mu_c + \tau_t + u_{ct}.$$

The coefficient δ is the recognition premium. It measures whether the electoral slope of casualty harm differs between recognized and non-recognized casualty disasters. It is not the average causal effect of a declaration.

The credibility prediction is tested by allowing this premium to vary with partisan context:

$$\begin{aligned} Y_{ct} = {} & \alpha_C D_{ct}^C + \beta T_{ct} + \lambda \text{Cross}_{st} + \lambda_D (D_{ct}^C \times \text{Cross}_{st}) \\ & + \lambda_T (T_{ct} \times \text{Cross}_{st}) + \delta (D_{ct}^C \times T_{ct}) \\ & + \phi (D_{ct}^C \times T_{ct} \times \text{Cross}_{st}) + X_{ct}'\theta + \mu_c + \tau_t + u_{ct}. \end{aligned}$$

The key coefficient is ϕ. A positive ϕ means that the recognition premium is larger when the governor and president are from different parties. This is the core implication of the credibility mechanism.

2.4 Additional Predictions and the Role of Aid

The same logic yields additional predictions. First, recognition should matter more when human harm is severe. When casualty harm is limited, local information may already reveal that the event was minor. When casualty harm is severe, federal recognition provides a more valuable public signal that the event crossed a national threshold.

Second, prompt recognition should carry a larger premium than delayed recognition. A fast declaration arrives while the emergency is salient and before a long process of bargaining, documentation, and project review can unfold. A prompt cross-party declaration is therefore especially informative: it is both quick and difficult to attribute to co-partisan accommodation.

Third, recognition should matter more where voters have less precise private information about local severity. If private information is noisy, public recognition receives greater weight in voters' updated assessment. As a suggestive proxy, the empirical analysis uses baseline income, which may capture information access and administrative capacity but may also capture vulnerability, fiscal capacity, and recovery needs.

Finally, the framework clarifies why aid is analyzed separately. If voters directly reward federal dollars, aid obligated before Election Day should have a stronger electoral association than aid obligated afterward. Post-election aid could not have entered voters' pre-election information sets. By contrast, if aid obligations partly capture latent severity and recovery complexity, then post-election obligations may remain predictive of vote-share changes. This motivates the aid-timing diagnostic:

$$Y_{ct} = \alpha_P D_{ct}^P + \gamma_1 M_{ct}^{pre} + \gamma_2 M_{ct}^{post} + \varepsilon_1 (D_{ct}^P \times M_{ct}^{pre}) + \varepsilon_2 (D_{ct}^P \times M_{ct}^{post}) + X_{ct}'\theta + \mu_c + \tau_t + u_{ct}.$$

The comparison between ε_1 and ε_2 is not a treatment effect of aid. It is a diagnostic for whether aid coefficients can plausibly be interpreted as voter rewards for observed transfers. If post-election aid remains predictive, aid is better understood as an administrative marker of severity and recovery burden.

The framework produces four empirical predictions. Federal action should be predicted by damage rather than by the observed political targeting variables. The vote-share slope with respect to casualty harm should be more positive when the event receives federal recognition. This recognition premium should be larger when recognition crosses party lines. Finally, post-election aid should remain predictive of vote-share changes if aid obligations primarily capture latent severity and recovery complexity rather than voter-observed transfers.

The next section describes the county-election panel used to take these predictions to the data.

3 Data and Measurement

The analysis uses a county-by-presidential-election panel covering U.S. presidential elections from 1992 through 2020. The panel links four sets of variables: presidential election returns, disas-

ter harm, federal recognition, and federal aid obligations. The empirical unit is a county in a presidential election year. The baseline sample contains 24,871 county-election-year observations.

The measurement choices follow the paper's conceptual distinction between harm, recognition, and transfer. Disaster harm is measured from local storm-event records. Federal recognition is measured by presidential disaster declarations. Federal transfer is measured by FEMA obligations and split by whether obligations occur before or after Election Day. The timing of population, GDP, and control variables is fixed at $t-4$, the previous presidential election year, to avoid scaling or controlling for outcomes that may themselves respond to disasters or recovery activity during the election cycle.

3.1 Electoral Outcome

The outcome variable Y_{ct} is the change in the incumbent presidential party's vote share in county c between consecutive presidential elections, measured in percentage points. For election year t, Y_{ct} compares the incumbent party's county vote share in year t with that party's vote share in the previous presidential election. Using vote-share changes rather than levels focuses the analysis on within-county electoral movement and absorbs persistent differences in baseline partisanship.

3.2 Disaster Harm

Disaster harm is constructed from the NOAA Storm Events Database, which records event-level information on weather-related hazards, including deaths, injuries, property damage, and crop damage. Event records are aggregated to the county-election-year level using events that occur during presidential election year t. The analysis distinguishes human harm from property harm because casualty losses and economic losses may carry different political information and may enter the declaration process differently.

Casualty harm is defined as

$$D_{ct}^{C} = \log\left(1 + 10{,}000 \cdot \frac{\text{Deaths}_{ct} + \text{Injuries}_{ct}}{\text{Population}_{c,t-4}}\right).$$

Property harm is defined as

$$D_{ct}^{P} = \log\left(1 + 1{,}000 \cdot \frac{\text{PropertyDamage}_{ct} + \text{CropDamage}_{ct}}{\text{Population}_{c,t-4}}\right).$$

The denominators use lagged population from the previous presidential election year. This timing avoids scaling disaster harm by population that may itself respond to the disaster, recovery conditions, or other within-cycle shocks. The logarithmic transformation limits the influence of extreme events while preserving intensive-margin variation in harm.

The main recognition analysis focuses on casualty disasters, $D_{ct}^{C} > 0$. This restriction follows the theory: the recognition premium is defined as the difference in the vote-share slope with respect to human harm when that harm receives federal recognition. Property harm is used both in the

allocation equations and in the aid-timing diagnostic, where aid obligations are more naturally linked to physical damage and recovery needs.

3.3 Federal Recognition

Federal recognition is measured using FEMA Disaster Declarations Summaries. FEMA's OpenFEMA documentation describes this dataset as listing official FEMA disaster declarations beginning in 1953 and including major disaster, emergency, and fire-management declaration types. The baseline recognition measure uses major disaster declarations only. Specifically, T_{ct} equals one if county c receives at least one FEMA DR declaration in presidential election year t, and zero otherwise.

This variable maps directly to the theory's recognition object. A major disaster declaration is discrete, public, and politically attributable to the federal executive. It is not treated as a proxy for aid dollars. Rather, it is the public act through which the federal government formally recognizes a local event as severe enough to warrant major federal involvement. Aid obligations are measured separately below.

3.4 Federal Transfers and Aid Timing

Federal transfers are measured using FEMA obligations from Public Assistance, Individual Assistance, and Hazard Mitigation programs. These variables record obligations rather than final disbursements. This distinction is important: obligations reflect approved federal commitments and may occur after the initial disaster, after the declaration, or after the election.

The baseline aid measure is

$$M_{ct} = \log\left(1 + 100 \cdot \frac{\text{Aid}_{ct}}{\text{GDP}_{c,t-4}}\right).$$

Normalizing by lagged county GDP scales federal obligations by the size of the local economy and avoids conflating aid levels with county size. The denominator is measured at $t-4$ to avoid conditioning on economic outcomes that may themselves respond to disaster exposure or recovery activity. The logarithmic transformation reduces the influence of extreme obligation amounts.

To distinguish aid that could enter the pre-election information environment from later administrative obligations, aid is split into pre-election and post-election components:

$$M_{ct}^{pre} = \log\left(1 + 100 \cdot \frac{\sum_i Aid_i \mathbf{1}\{ObligationDate_i \leq ElectionDay_t\}}{GDP_{c,t-4}}\right),$$

$$M_{ct}^{post} = \log\left(1 + 100 \cdot \frac{\sum_i Aid_i \mathbf{1}\{ObligationDate_i > ElectionDay_t\}}{GDP_{c,t-4}}\right).$$

This split is used as a diagnostic rather than as a treatment design. Aid obligated before Election Day could in principle be observed before voting. Aid obligated after Election Day could not. If aid coefficients capture direct voter rewards for federal dollars, pre-election obligations should have

the clearer electoral association. If aid obligations instead proxy for latent severity, administrative complexity, and recovery needs, post-election obligations may remain predictive of vote-share outcomes.

3.5 Partisan Environment

The main partisan-context variable is Cross_{st}, an indicator equal to one when the governor of state s and the president are from different parties in presidential election year t. Same-party observations are the complement. This variable measures the partisan environment in which federal recognition is delivered.

The interpretation of Cross_{st} is not that partisan alignment causes disasters or declarations. Rather, it identifies whether a given act of presidential recognition occurs across party lines. The theory predicts that cross-partisan recognition is more informative because voters have more difficulty attributing it to co-partisan accommodation.

For the close-gubernatorial-election robustness exercise, each county-election-year observation is matched to the most recent prior gubernatorial election in the state. Gubernatorial returns are taken from Algara and Amlani (2021). These data identify state-years in which gubernatorial control was decided by a narrow margin and allow the analysis to examine whether the cross-partisan recognition premium remains in settings where governor party is less tightly linked to persistent state characteristics.

3.6 Lagged Controls

The baseline control vector X_{ct} includes unemployment, log wages, turnout, and log per-capita income, all measured at $t-4$. These variables capture pre-existing economic conditions and political participation before the presidential election cycle in which the disaster occurs. Measuring controls before the cycle avoids conditioning on variables that may themselves be affected by disaster exposure, federal recovery activity, or election-cycle shocks.

4 Empirical Design and Estimands

The empirical design follows the logic of the framework. I first test whether federal disaster action is politically selected along the observed county-level margins. I then estimate whether federal recognition changes the vote-share slope with respect to casualty harm. Finally, I test whether this recognition premium is larger when recognition crosses party lines. The design therefore separates three objects: the allocation of federal action, the electoral interpretation of recognized harm, and the credibility of recognition across partisan environments.

Throughout the analysis, the unit of observation is a county-election year. The baseline specifications include county fixed effects μ_c, presidential-election-year fixed effects τ_t, and lagged county controls X_{ct}. County fixed effects absorb persistent differences in partisan alignment, disaster exposure, geography, and local institutions. Election-year fixed effects absorb national electoral

conditions common to all counties. Lagged controls account for pre-cycle economic and political conditions without conditioning on variables that may respond to disasters or federal recovery activity during the election cycle.

4.1 The Allocation Margin

The first step asks whether federal disaster action is predicted by political variables after conditioning on realized damage. This test is not meant to prove that declarations or aid are randomly assigned. It instead evaluates a necessary condition for a simple political-targeting interpretation of the voter-response results: if presidents or agencies systematically direct recognition or aid toward electorally valuable or co-partisan jurisdictions, then those political variables should predict federal action conditional on damage and fixed effects.

Define

$$\text{Swing}_{c,t-1} = \mathbf{1}\{|\text{margin}_{c,t-1}| < 10\,\text{pp}\}, \qquad \text{SameParty}_{st} = \mathbf{1}\{\text{governor}_{st} = \text{president}_t\}.$$

I estimate

$$M_{ct} = \pi_1 D^P_{ct} + \pi_2 D^C_{ct} + \pi_3\,\text{Swing}_{c,t-1} + \pi_4\,\text{SameParty}_{st} + X'_{ct}\theta + \mu_c + \tau_t + u_{ct}, \tag{1}$$

$$T_{ct} = \pi_1 D^P_{ct} + \pi_2 D^C_{ct} + \pi_3\,\text{Swing}_{c,t-1} + \pi_4\,\text{SameParty}_{st} + X'_{ct}\theta + \mu_c + \tau_t + v_{ct}. \tag{2}$$

The first outcome, M_{ct}, is logged FEMA obligations scaled by lagged county GDP. The second outcome, T_{ct}, is an indicator for a presidential DR declaration and is estimated by a linear probability model. The coefficients π_3 and π_4 measure whether the observed political variables predict aid or recognition after conditioning on realized casualty and property damage.

A political-targeting account predicts nonzero π_3 or π_4. A damage-based allocation account predicts that D^P_{ct} and D^C_{ct} predict federal action, while the political variables add little explanatory power. In addition to conventional tests, I report partial-R^2 decompositions and equivalence tests. These exercises distinguish imprecise null estimates from estimates that are small enough to rule out substantively large targeting effects along the observed county-level margins.

4.2 The Recognition Premium

The second step estimates the electoral value of federal recognition. The object of interest is not the average effect of receiving a declaration. Declarations are not randomly assigned, and a declared disaster differs from an undeclared disaster in many ways. The estimand is instead a slope-difference: whether the relationship between casualty harm and incumbent-party vote share is different when that harm is federally recognized.

The baseline recognition specification is

$$Y_{ct} = \alpha_C D^C_{ct} + \beta T_{ct} + \delta(D^C_{ct} \times T_{ct}) + X'_{ct}\theta + \mu_c + \tau_t + u_{ct}.$$

The outcome Y_{ct} is the change in the incumbent presidential party's county vote share between consecutive presidential elections, measured in percentage points. The sample for the main recognition analysis is the casualty-disaster subsample, $D_{ct}^{C} > 0$.

The coefficient δ is the recognition premium. It compares the casualty-damage slope among recognized disasters with the corresponding slope among non-recognized casualty disasters:

$$\frac{\partial Y_{ct}}{\partial D_{ct}^{C}} = \begin{cases} \alpha_C, & T_{ct} = 0, \\ \alpha_C + \delta, & T_{ct} = 1. \end{cases}$$

A positive δ means that casualty harm has a more favorable association with incumbent-party vote share when the federal government publicly recognizes the event. This is the reduced-form implication of the recognition mechanism.

I do not treat aid as a preferred mediator in this specification. Aid is a downstream administrative outcome that may itself proxy for unobserved severity and recovery complexity. Specifications that control for aid are therefore reported as robustness checks rather than as the main mediation analysis.

4.3 Credibility Across Party Lines

The third step tests whether the recognition premium depends on partisan credibility. The model predicts that recognition should be more informative when it crosses party lines, because a cross-party declaration is harder to attribute to co-partisan accommodation. I estimate

$$\begin{aligned} Y_{ct} = {} & \alpha_C D_{ct}^{C} + \beta T_{ct} + \lambda \text{Cross}_{st} + \lambda_D (D_{ct}^{C} \times \text{Cross}_{st}) + \lambda_T (T_{ct} \times \text{Cross}_{st}) \\ & + \delta (D_{ct}^{C} \times T_{ct}) + \phi (D_{ct}^{C} \times T_{ct} \times \text{Cross}_{st}) + X_{ct}' \theta + \mu_c + \tau_t + u_{ct}. \end{aligned} \tag{3}$$

The coefficient ϕ is the difference in recognition premiums between cross-partisan and same-partisan environments. The lower-order interactions allow baseline casualty slopes and baseline declaration associations to differ by partisan environment. Thus $\phi > 0$ means that the additional vote-share slope associated with recognized casualty harm is larger when the governor and president are from different parties.

I report both pooled triple-interaction estimates and split-sample estimates. The pooled specification provides the formal test of the difference in premiums. The split-sample estimates show where the recognition premium is concentrated. Because Cross_{st} varies at the state-election-year level while D_{ct}^{C} and T_{ct} vary at the county-year level, inference must allow residual dependence within state-election-year cells. Baseline standard errors are therefore clustered at the state-by-election-year level. I also report state-level clustered and wild-cluster-bootstrap inference for the triple interaction.

4.4 Aid Timing as a Diagnostic

The final step distinguishes recognition from transfer. If voters directly reward federal dollars, aid obligated before Election Day should have a clearer electoral association than aid obligated afterward. Post-election aid could not have entered voters' pre-election information sets. If aid obligations instead proxy for latent severity and recovery complexity, post-election aid may remain predictive of vote-share changes.

I estimate

$$Y_{ct} = \alpha_P D^P_{ct} + \gamma_1 M^{pre}_{ct} + \gamma_2 M^{post}_{ct} + \varepsilon_1 (D^P_{ct} \times M^{pre}_{ct}) + \varepsilon_2 (D^P_{ct} \times M^{post}_{ct}) + X'_{ct}\theta + \mu_c + \tau_t + u_{ct}.$$

The comparison of interest is

$$H_0 : \varepsilon_1 = \varepsilon_2.$$

This is not a treatment-effect test for aid. It is a diagnostic for whether aid coefficients can plausibly be interpreted as rewards for voter-observed transfers. A persistent post-election association indicates that aid also contains information about unobserved severity, administrative complexity, and recovery burden.

4.5 Measurement Error and Inference

A remaining concern is that reported disaster damage is measured with error. NOAA Storm Events damage measures may reflect local reporting capacity, media attention, and administrative effort in addition to true harm. Classical measurement error would attenuate damage coefficients, while systematic reporting differences could bias the estimated recognition premium.

To address this concern, I report weather-based IV specifications that replace reported damage variation with variation predicted by non-monetary storm-intensity measures. The instrument Z^W_{ct} is constructed from county-cycle measures of severe-weather intensity in the NOAA Storm Events Database: the maximum tornado intensity rating, the maximum wind-speed magnitude for wind events, and the maximum hail-size magnitude for hail events. Each component is standardized over the full panel and summed:

$$Z^W_{ct} = z(\text{maxTor}_{ct}) + z(\text{maxWind}_{ct}) + z(\text{maxHail}_{ct}).$$

County-cycles with no event of a given type receive a zero value for that component before standardization. The instrument is designed to use physical storm intensity rather than reported dollar losses or reported casualties. Details on the construction of Z^W_{ct} are reported in Appendix **??**.

For damage measure $h \in \{C, P\}$, the first stage is

$$D^h_{ct} = \rho^h Z^W_{ct} + X'_{ct}\gamma + \mu_c + \tau_t + \nu^h_{ct}.$$

The first-stage F-statistics are strong in the main specifications, ranging from $F = 144$ in the

casualty-disaster recognition sample to $F = 3{,}076$ in the full-sample property-damage specification. The IV estimates are not used to instrument declarations or aid. They are diagnostic tests for whether the main vote-share patterns are sensitive to measurement error and reporting heterogeneity in the damage measures.

The exclusion restriction is that, conditional on county fixed effects, election-year fixed effects, and lagged controls, severe-weather intensity affects vote-share changes through realized disaster harm rather than through differential reporting of damage or federal recognition itself. This assumption is more plausible for the weather-IV exercises than for an IV design that would attempt to instrument declarations or aid. The weather instrument is therefore used only as a robustness check for damage measurement, not as the paper's main source of identification.

A separate HURDAT2 hurricane-track exercise provides an additional source-independence diagnostic based on storm position and intensity rather than local damage reports; details are reported in the robustness section.

The main inference issue is the level at which residual shocks may be correlated. Baseline standard errors are clustered by state-election year, matching the level of the partisan environment in the cross-partisan analysis. The robustness section reports alternative clustering and spatial HAC procedures. When inference for the pooled triple interaction is sensitive to the clustering level, I interpret that coefficient together with the split-sample estimates, placebo tests, mechanism tests, and close-gubernatorial-election exercise rather than as a stand-alone decisive test.

The empirical analysis begins with the allocation margin.

5 Is Federal Disaster Action Politically Selected?

Before studying how voters interpret federal recognition, I first examine whether federal disaster action is politically selected along the observed county-level margins. This step is necessary because a recognition premium could otherwise reflect selection rather than interpretation. If presidents or agencies direct declarations toward electorally valuable or co-partisan jurisdictions, then later vote-share differences may partly capture the political allocation of federal action rather than voters' interpretation of recognized harm.

The allocation tests focus on two political margins. Swing-county status captures the electoral incentive to direct visible federal action toward competitive local constituencies. Governor-president co-partisanship captures the intergovernmental channel in the declaration process: major disaster declarations are requested by governors and approved by the president. These variables do not exhaust all possible forms of political influence. They instead capture the two margins most directly connected to electoral and partisan targeting accounts in this setting.

The question is therefore not whether disaster policy is ever political. The narrower question is whether the political variables most relevant to the paper's main alternative explanation predict federal recognition or aid after conditioning on realized damage, county fixed effects, election-year fixed effects, and lagged controls.

5.1 Baseline Allocation Tests

Tables 1 and 2 report estimates of equations (1) and (2). The two outcomes are normalized FEMA aid obligations and the probability of a presidential DR declaration. The pattern is the same for both outcomes: realized harm predicts federal action, while the observed political variables add little explanatory power.

In the aid equation, both property damage and casualty damage are positive predictors of obligated FEMA aid. In the declaration equation, the same damage variables strongly predict whether a county receives a DR declaration. By contrast, swing-county status and same-party gubernatorial alignment are small and statistically indistinguishable from zero in the full specifications.

The joint tests reinforce this pattern. The damage variables are jointly predictive of both outcomes, with $F = 8.3$ for aid and $F = 36.9$ for declarations. The political variables are not jointly predictive, with corresponding F-statistics of 0.24 and 0.30. These tests should not be read as proving that political considerations never enter disaster administration. They show that, in this county-election-year panel and conditional on realized damage, county fixed effects, election-year fixed effects, and lagged controls, the observed political variables explain little of the variation in federal recognition or aid.

The partial-R^2 decomposition leads to the same conclusion. The damage block accounts for roughly 91% of the explained within-model variation in aid and 90% of the explained within-model variation in declarations. The political block accounts for about 5% of the explained variation in the aid equation and 3% in the declaration equation. Although the overall within-R^2 values are modest, the explanatory power that is present comes primarily from realized disaster harm rather than from swing-county status or governor-president co-partisanship.

5.2 Bounding Political Selection

The absence of statistically significant political coefficients could reflect either small effects or insufficient precision. I therefore report two one-sided equivalence tests for the swing and same-party coefficients in the full allocation specifications. These tests ask whether the confidence intervals are tight enough to reject political effects larger than pre-specified economically meaningful bounds.

For normalized aid, the 95 percent equivalence bounds imply that the data rule out swing-county effects larger than 0.011 log-units and same-party effects larger than 0.024 log-units in absolute value. For declarations, measured on the probability scale, the corresponding bounds are 4.2 percentage points for swing counties and 8.4 percentage points for same-party alignment. The swing-county declaration bound and the aid bounds are sufficiently tight to reject several substantively meaningful forms of political targeting. The same-party declaration bound is less restrictive, so the evidence is weaker on that margin.

The equivalence results sharpen the interpretation of the allocation tests. The evidence does not merely fail to reject zero political effects. For swing-county status, and for the aid equation more generally, the estimates are precise enough to rule out effects of a size that would support a strong county-level targeting interpretation. For same-party declaration effects, the estimates

Table 1: Aid Allocation: Damage and Political Predictors

	(1) Damage only	(2) Politics only	(3) Politics +ctrls	(4) Damage+ Politics	(5) Full
π_1 (D^P)	0.0020***	—	—	0.0020***	0.0020***
	(0.0007)	—	—	(0.0008)	(0.0008)
π_2 (D^C)	0.0295***	—	—	0.0293***	0.0293***
	(0.0082)	—	—	(0.0082)	(0.0082)
π_3 (Swing)	—	0.0008	0.0009	0.0008	0.0010
	—	(0.0051)	(0.0051)	(0.0051)	(0.0050)
π_4 (SameParty)	—	0.0069	0.0067	0.0064	0.0062
	—	(0.0093)	(0.0091)	(0.0093)	(0.0092)
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Lag-4 controls	No	No	Yes	No	Yes
Within-R^2	0.0064	0.0005	0.0006	0.0068	0.0070
Observations	24871	24871	24871	24871	24871

Notes: Outcome is $M = \log(1 + 100 \times \text{aid}/\text{GDP}_{t-4})$. The sample consists of county-election-year observations for presidential elections from 1992 to 2020. D^P denotes property damage and D^C denotes casualty damage. Swing equals one if the previous presidential election margin in the county was less than 10 percentage points in absolute value. SameParty equals one if the governor and president are in the same party. Partial R^2 of damage variables: 0.0064 (91% of full). Partial R^2 of political variables: 0.0004 (5% of full). Joint Wald test for $\pi_1 = \pi_2 = 0$: $F = 8.3$, $p < 0.001$. Joint Wald test for $\pi_3 = \pi_4 = 0$: $F = 0.24$, $p = 0.786$. Standard errors clustered at state × election-year. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

do not indicate systematic co-partisan allocation, but the confidence interval leaves more room for moderate effects. Thus, the allocation evidence rules out several large targeting effects along the observed margins while leaving open other forms of political discretion not captured by these variables.

5.3 What the Allocation Tests Do and Do Not Show

The allocation evidence plays a specific role in the paper's design. It does not establish that declarations or aid are randomly assigned. Nor does it rule out all possible forms of political discretion. It shows that, conditional on realized damage and fixed effects, the observed political variables most closely tied to electoral and partisan targeting do not explain much of the variation in federal recognition or aid.

This matters for the demand-side analysis. If the cross-partisan recognition premium below were driven primarily by politicians steering declarations or aid toward favorable counties, one would expect the political variables to predict federal action in the allocation equations. Instead, the allocation results indicate that federal action is mainly predicted by realized harm along these margins. This makes it more plausible that the heterogeneity documented below reflects differences

Table 2: Declaration Allocation: Damage and Political Predictors

	(1) Damage only	(2) Politics only	(3) Politics +ctrls	(4) Damage+ Politics	(5) Full
π_1 (D^P)	0.0129***	—	—	0.0128***	0.0130***
	(0.0022)	—	—	(0.0022)	(0.0022)
π_2 (D^C)	0.0896***	—	—	0.0893***	0.0890***
	(0.0160)	—	—	(0.0161)	(0.0161)
π_3 (Swing)	—	-0.0109	-0.0105	-0.0092	-0.0084
	—	(0.0177)	(0.0177)	(0.0173)	(0.0172)
π_4 (SameParty)	—	0.0222	0.0222	0.0202	0.0197
	—	(0.0331)	(0.0330)	(0.0330)	(0.0329)
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Lag-4 controls	No	No	Yes	No	Yes
Within-R^2	0.0192	0.0007	0.0022	0.0198	0.0216
Observations	24871	24871	24871	24871	24871

Notes: Outcome is T_{ct}, an indicator for a FEMA DR declaration in a presidential election year, estimated by a linear probability model. The sample consists of county-election-year observations for presidential elections from 1992 to 2020. Partial R^2 of damage variables: 0.0194 (90% of full). Partial R^2 of political variables: 0.0005 (3% of full). Joint Wald test for $\pi_1 = \pi_2 = 0$: $F = 36.9$, $p < 0.001$. Joint Wald test for $\pi_3 = \pi_4 = 0$: $F = 0.30$, $p = 0.738$. Standard errors clustered at state × election-year. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

in voter interpretation of recognition rather than direct political selection into recognition.

Appendix Table 11 shows that this allocation pattern is not driven by a single administration period. Re-estimating the allocation equations separately for pre-2001 elections, the Bush administration elections, and the modern period yields the same broad pattern: damage variables are more consistently predictive of federal action than the political block. The only marginal exception is the declaration equation during the Bush administration block, where the political block is jointly significant at the 10 percent level.

The allocation tests also clarify the role of the weather instrument used later in the paper. The allocation equations are not IV specifications; they are selection diagnostics. They ask whether observed political variables predict federal action after conditioning on reported damage and fixed effects. The weather IV is used later to address measurement error in reported damage in the vote-share equations, not to generate quasi-random variation in declarations or aid.

The next section turns to the demand side and asks whether voters respond to federal recognition uniformly or whether the electoral meaning of recognition depends on the partisan environment in which it is delivered.

6 Federal Recognition and Vote-Share Slopes

Having shown that federal recognition and aid are more closely associated with realized disaster harm than with the political variables considered above, this section turns to voter response. The demand-side analysis asks whether voters respond uniformly to disaster harm, or whether the electoral meaning of a disaster depends on federal recognition and the partisan environment in which that recognition occurs.

The section proceeds in five steps. Section 6.1 first shows that average disaster damage has little systematic electoral association once county and election-year fixed effects are included. Section 6.2 then shows that the recognition channel becomes visible for casualty disasters: presidential declarations change the vote-share slope with respect to casualty harm. Section 6.3 tests the main heterogeneity prediction, namely that the casualty-recognition premium is larger in cross-partisan settings. Section **??** explains why this pattern is difficult to reconcile with a simple co-partisan favoritism account. Section 6.4 reports placebo tests that address persistent county-level confounds and future political variables.

6.1 Average Damage Effects Do Not Reveal the Recognition Channel

Table 3 reports the damage-only estimates. Across both casualty damage and property damage, the average relationship between disaster harm and the change in incumbent-party vote share is small and statistically imprecise after county fixed effects, election-year fixed effects, and lagged county controls are included. The weather-IV estimates are also imprecise and do not reveal a strong average electoral effect of disaster harm.

This absence of a strong average effect is consistent with the framework. Disaster harm need not have a single unconditional electoral meaning. The same local harm may be interpreted differently depending on whether it is federally recognized and on whether that recognition is credible. The rest of the section therefore moves from average damage effects to recognition-conditioned and party-conditioned vote-share slopes.

6.2 The Casualty-Recognition Premium

Table 4 estimates equation (4.2) on the casualty-disaster subsample, $D^C_{ct} > 0$. The coefficient of interest is the interaction $D^C_{ct} \times T_{ct}$. This coefficient is not the average effect of receiving a declaration. It is a difference in slopes: it asks whether the vote-share response to casualty harm differs when the event receives federal recognition.

The estimated recognition premium is positive. In the preferred OLS specification, $\hat{\delta} = 1.937$. Thus, relative to casualty disasters without federal recognition, a one-log-point increase in casualty harm is associated with a 1.94 percentage-point larger increase in incumbent-party vote-share change when the disaster receives a DR declaration. The Wald test rejects $\alpha_C + \delta = 0$ at the 5 percent level ($p = 0.017$), indicating that the casualty-damage slope among declared disasters is positive.

Table 3: Baseline: Disaster Damage and Incumbent-Party Vote Share

	Casualty D^C		Property D^P	
	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)
α (D^h)	0.192	0.545	0.002	0.033
	(0.203)	(0.725)	(0.021)	(0.043)
County FE	Yes	Yes	Yes	Yes
Election-year FE	Yes	Yes	Yes	Yes
Lag-4 controls	Yes	Yes	Yes	Yes
First-stage F	—	1766	—	3076
Observations	24,871	24,871	24,871	24,871

Notes: The outcome is the change in incumbent presidential party vote share between consecutive presidential elections, measured in percentage points. The sample consists of county-election-year observations for presidential elections from 1992 to 2020. $D^C = \log(1 + 10{,}000 \times \text{casualties}/\text{pop}_{t-4})$ and $D^P = \log(1 + 1{,}000 \times \text{damage}/\text{pop}_{t-4})$. Lag-4 controls include the unemployment rate, log wages, turnout, and log per-capita income. Z^W is the standardized weather-severity instrument. Standard errors, clustered at state × election-year, are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4: Casualty Damage and Federal Recognition

	OLS			IV
	(1)	(2)	(3)	(4)
α_C (D^C)	0.424	0.399	-0.008	-0.021
	(0.348)	(0.346)	(0.364)	(1.234)
β_C (T)	—	0.237	0.526	0.528
	—	(0.423)	(0.463)	(0.532)
δ_C ($D^C \times T$)	—	—	1.937**	1.943*
	—	—	(0.865)	(1.107)
γ (M)	—	—	-0.232	-0.231
	—	—	(0.570)	(0.569)
Wald: $\alpha_C + \delta_C = 0$ (p)	—	—	0.017	—
First-stage F	—	—	—	144
County FE / Election-year FE / Lag-4 controls	Yes	Yes	Yes	Yes
Observations	1,965	1,965	1,965	1,965

Notes: The sample is restricted to casualty disasters ($D^C > 0$). The outcome is the change in incumbent presidential party vote share, measured in percentage points. Column (4) instruments D^C with Z^W. All variables are mean-centered. Standard errors, clustered at state × election-year, are reported in parentheses. Specifications with aid obligations are reported as robustness checks because aid is a downstream administrative outcome of the disaster response process. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

This result is the first piece of evidence for the recognition channel. Casualty harm has little electoral association when it is not federally recognized, but the slope becomes more positive when the same type of harm receives public federal recognition. The pattern is not evidence of an unconditional declaration effect. It is evidence that the electoral interpretation of human harm differs when that harm is formally recognized by the federal government.

6.3 Recognition Across Party Lines

The credibility mechanism predicts that the recognition premium should be larger when recognition crosses party lines. A declaration issued when the governor and president are from different parties is harder to attribute to co-partisan accommodation and should therefore be more informative about underlying severity and federal attention.

I test this prediction in two complementary ways. Panel A of Table 5 estimates the recognition specification separately in cross-partisan and same-partisan environments. These split-sample estimates show where the recognition premium is concentrated. Panel B estimates the pooled triple-interaction specification in equation (3). This pooled specification provides the formal test of whether the recognition premium differs across partisan environments.

The two panels use related but not identical identifying variation. Panel A uses within-county variation within each partisan subsample. Counties with only one election cycle within a given subsample are fixed-effect singletons and do not contribute to the split-sample estimates. Panel B uses within-county variation across changes in partisan environment and therefore retains observations that are informative for the pooled triple-interaction design. The table reports both input observations and effective observations to make this distinction explicit.

The split-sample estimates show that the recognition premium is concentrated in cross-partisan settings. In the cross-partisan subsample, the OLS estimate is $\hat{\delta}^{cross} = 3.97$ ($p < 0.01$). In the same-partisan subsample, the corresponding estimate is 0.57 and is not statistically distinguishable from zero. The IV estimates show the same qualitative pattern: $\hat{\delta}^{cross,IV} = 4.43$, while $\hat{\delta}^{same,IV} = -0.08$.

The split-sample contrast should not be interpreted as a formal difference-in-coefficients test by itself. A coefficient that is statistically significant in one subsample and insignificant in another does not imply that the two coefficients are statistically different. The formal difference test is the pooled triple interaction in Panel B.

Panel B estimates this difference directly. In the pooled OLS specification, the baseline recognition premium is $\hat{\delta} = 2.04$, and the additional cross-partisan premium is $\hat{\phi} = 1.97$. Thus, the implied cross-partisan premium is $\hat{\delta} + \hat{\phi} = 4.02$, close to the split-sample estimate. The pooled IV estimates are similar. The estimated ϕ is positive in both OLS and IV, as predicted by the credibility mechanism, but it is less precisely estimated than the split-sample coefficients.

The pooled coefficient becomes stronger when aid obligations are included as an additional control: the OLS estimate of ϕ rises to 3.05 ($SE = 1.26$, $p = 0.016$). I treat this aid-control specification as a robustness check rather than the preferred mediation specification because aid is downstream of the declaration process and may proxy for latent severity and recovery complexity.

Table 5: Cross-Partisan Heterogeneity in the Recognition Premium

Panel A: Split-Sample Estimates ($D^C > 0$)				
	Cross-partisan		Same-partisan	
	OLS	IV	OLS	IV
δ ($D^C \times T$)	3.970***	4.432***	0.567	-0.078
	(1.300)	(1.594)	(0.810)	(1.205)
First-stage F	—	43	—	59
Input observations	1,620	1,620	1,017	1,017
Effective observations	882	882	598	598
Panel B: Pooled Triple-Interaction				
	OLS		IV	
δ ($D^C \times T$)	2.044**		2.064*	
	(0.828)		(1.056)	
ϕ ($D^C \times T \times$ Cross)	1.972		1.960	
	(1.234)		(1.272)	
Observations	1,946		1,946	

Notes: The sample is restricted to casualty-disaster county-years ($D^C > 0$). Panel A estimates the recognition specification separately by partisan environment. Input observations are observations available before fixed-effect singletons are removed; effective observations are observations used in estimation after removing county-fixed-effect singletons. The sum of effective observations in Panel A is smaller than the pooled sample in Panel B because some counties have only one election cycle within a given partisan subsample. These observations contribute to the pooled triple-interaction design through within-county changes in Cross_{st}, but not to the within-subsample fixed-effect estimates. Panel B estimates the pooled triple-interaction specification. In Panel B, δ is the baseline recognition premium when $\text{Cross}_{st} = 0$, and ϕ is the additional recognition premium when $\text{Cross}_{st} = 1$; the cross-partisan premium is therefore $\delta + \phi$. IV specifications instrument D^C with the weather-severity measure Z^W. Standard errors are clustered at state × election-year. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Thus, the evidence indicates that the recognition premium is concentrated in cross-partisan settings, while the formal pooled difference is positive and economically meaningful but less precise in the baseline specification. I therefore interpret the cross-partisan result as a pattern across complementary evidence rather than as resting on a single coefficient.

This pattern is difficult to reconcile with a simple co-partisan favoritism account. If the electoral value of declarations came mainly from presidents delivering benefits to aligned state governments, the premium should be larger when governors and presidents are from the same party. The data point in the opposite direction. Combined with the allocation results above, the pattern is more consistent with a credibility mechanism: voters place greater weight on public recognition when it is harder to attribute to partisan accommodation.

6.4 Placebo and Falsification Tests

The recognition interpretation implies that the effect should be tied to current, voter-observed recognition of current local harm. I therefore report three placebo tests. The first replaces the contemporaneous treatment $D^C_{ct} \times T_{ct}$ with its next-cycle analog, $D^C_{c,t+4} \times T_{c,t+4}$. Future disasters and future declarations cannot affect past votes. The second replaces the partisan moderator Cross_{st} with its lead, $\text{Cross}_{s,t+4}$. Future partisan alignment cannot retroactively change the credibility of current recognition. The third restricts the sample to county-years with no contemporaneous NOAA-reported damage but with a DR declaration. With no local harm to interpret, declarations alone should not generate the same recognition premium.

Table 6 reports the placebo coefficients alongside the real-data baseline.

Table 6: Placebo and Falsification Tests

	Coefficient	p-value
Panel A: Real treatment ($D^C > 0$ subsample)		
$D^C \times T$, cross-partisan ($n = 882$)	3.914^{***}	0.003
$D^C \times T$, same-partisan ($n = 598$)	0.583	0.467
Panel B: Lead-treatment placebo ($D^C_{lead} > 0$ subsample)		
$D^C_{lead} \times T_{lead}$, cross-partisan ($n = 1{,}620$)	0.312	0.766
$D^C_{lead} \times T_{lead}$, same-partisan ($n = 1{,}017$)	0.608	0.589
Panel C: Future-Cross placebo		
$D^C \times T \times \text{Cross}_{lead}$	-1.673	0.162
Panel D: Non-disaster placebo ($D^C = D^P = 0$ subsample)		
T only, no contemporaneous damage	0.028	0.936

Notes: The outcome is the change in incumbent presidential party vote share, measured in percentage points. All specifications include county fixed effects, election-year fixed effects, lag-4 controls, and standard errors clustered at state × election-year. Panel A reports the real cross-partisan and same-partisan recognition estimates. Panel B replaces contemporaneous treatment with the next-cycle lead. Panel C replaces Cross_{st} with $\text{Cross}_{s,t+4}$. Panel D restricts to county-years with no contemporaneous NOAA-reported damage but with a DR declaration. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The placebo estimates are close to zero. The real cross-partisan recognition premium is +3.91 percentage points and statistically significant ($p = 0.003$). The corresponding lead-treatment placebo is +0.31 and far from significant ($p = 0.77$). Same-partisan estimates are small in both the real and placebo specifications. Future partisan alignment does not retroactively moderate current recognition, and declarations without contemporaneous local damage have no detectable electoral association.

These tests reduce the concern that the recognition premium is driven by persistent county-level confounds, future political conditions, or administrative declarations unrelated to local harm. The results are instead consistent with voter response to contemporaneous casualty harm that receives

federal recognition, especially when that recognition crosses party lines.

The next section tests when recognition should be most informative.

7 When Is Recognition Most Informative?

The previous section shows that the recognition premium is concentrated in cross-partisan settings. This section asks whether the premium is also larger in the cases where the signal-extraction framework predicts recognition should be most informative.

The tests use three margins. First, recognition should matter more when human harm is severe, because a declaration then provides public confirmation that the event crossed a high-severity threshold. Second, recognition should matter more when it is prompt. A declaration is not automatic: it follows local and state assessment, a gubernatorial request, and federal review. A prompt declaration therefore indicates that the event cleared the federal recognition threshold quickly. Third, recognition should matter more where voters have less precise private information about local severity, because public recognition then receives greater weight in voter inference.

These tests are not separate identification strategies. They are over-identifying evidence for the mechanism. The empirical question is whether the recognition premium appears in the cells where the model predicts the declaration signal should carry the most information: severe casualty disasters, prompt cross-partisan declarations, and counties where voters may rely more heavily on public signals.

7.1 Severity of Human Harm

The first prediction concerns the severity of casualty harm. If declarations operate as public recognition of local harm, their informational value should be greater when human losses are large. For minor casualty events, local information may already convey that the event was limited. For severe casualty events, federal recognition provides a stronger public signal that the event crossed a national threshold.

To test this prediction, I split the casualty-disaster sample at the median of D^C. I then estimate the recognition specification separately for bottom-half and top-half casualty disasters and separately within cross-partisan and same-partisan environments.

The results line up with the severity prediction. In cross-partisan settings, the recognition premium is concentrated among high-severity casualty disasters:

$$\hat{\delta}^{cross,top} = 3.52 \quad (p = 0.002).$$

For low-severity casualty disasters, the cross-partisan estimate is negative and statistically indistinguishable from zero. In same-partisan settings, the recognition premium is close to zero in the top half of the severity distribution and imprecisely estimated in the bottom half.

This pattern is informative because it shows that the cross-partisan premium is not a generic

Table 7: Recognition Premium by Casualty Severity and Partisan Status

	Cross-partisan		Same-partisan	
	Bottom D^C	Top D^C	Bottom D^C	Top D^C
$\hat{\delta}_C$ $(D^C \times T)$	−2.381	3.524***	−10.569	0.208
	(5.717)	(1.087)	(7.736)	(1.592)
p-value	0.678	0.002	0.175	0.897
Observations	865	797	569	636

Notes: D^C is split at its median value within the casualty-disaster subsample ($D^C > 0$). The recognition equation is estimated separately within each cell. All specifications include county fixed effects, election-year fixed effects, lag-4 controls, and standard errors clustered at state × election-year. The outcome is the change in incumbent presidential party vote share, measured in percentage points. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

cross-party effect. It appears in the cell where the model predicts recognition should be most diagnostic: severe human harm that receives federal recognition across party lines.

7.2 Speed of Recognition

The second prediction concerns the timing of recognition. Declaration delay is informative because federal recognition is not automatic. It follows damage assessment, a request from the governor or tribal chief executive, and federal review. FEMA guidance indicates that requests should generally be submitted promptly once the need for assistance is apparent. A fast declaration therefore provides a stronger signal that the event crossed the recognition threshold with little delay.

The credibility mechanism predicts that this timing margin should matter especially in cross-partisan settings. A prompt declaration that crosses party lines is difficult to attribute to co-partisan accommodation or to extended intergovernmental bargaining. It combines two features of a strong signal: speed and partisan distance.

I split recognized disasters at the median declaration delay W, where W is the number of days between the incident date and the declaration date. The median delay among recognized disasters is 9 days. Define

$$\text{Fast}_{ct} = 1\{W_{ct} \leq 9\}, \qquad \text{Slow}_{ct} = 1\{W_{ct} > 9\}.$$

I estimate

$$Y_{ct} = \alpha D^C_{ct} + \delta_{\text{fast}}(D^C_{ct} \times \text{Fast}_{ct}) + \delta_{\text{slow}}(D^C_{ct} \times \text{Slow}_{ct}) + X'_{ct}\theta + \mu_c + \tau_t + u_{ct},$$

separately for cross-partisan and same-partisan observations.

The timing results are consistent with the prediction. The largest coefficient is the fast cross-partisan recognition premium:

$$\hat{\delta}^{cross}_{fast} = 4.83 \quad (p < 0.01).$$

Table 8: Recognition Premium by Speed of Declaration and Partisan Status

	Cross-partisan	Same-partisan
$\hat{\delta}_{\text{fast}}$ ($D^C \times$ Fast, $W \leq 9$ d)	4.831^{***}	0.111
	(1.151)	(0.887)
$\hat{\delta}_{\text{slow}}$ ($D^C \times$ Slow, $W > 9$ d)	2.355^{**}	1.276^{*}
	(1.027)	(0.697)
Observations	1,662	1,205

Notes: The sample is restricted to $D^C > 0$. Declaration delay W is the number of days between the incident date and declaration date. *Fast* equals one if $W \leq 9$ days and *Slow* equals one if $W > 9$ days, where 9 is the median delay among recognized disasters ($T = 1$). Each column is a separate regression with county fixed effects, election-year fixed effects, lag-4 controls, and standard errors clustered at state × election-year. The outcome is the change in incumbent presidential party vote share, measured in percentage points. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Slow cross-partisan declarations also carry a positive premium, but the magnitude is roughly half as large:

$$\hat{\delta}^{cross}_{slow} = 2.36 \quad (p < 0.05).$$

In same-partisan settings, fast declarations have essentially no premium, while slow declarations have a smaller and only marginally significant association.

The contrast supports the signal interpretation. Recognition is most valuable electorally when it is both prompt and cross-partisan. A fast cross-party declaration is precisely the type of action that is hardest to dismiss as routine partisan accommodation.

7.3 Local Information Environment

The third prediction concerns voters' information environment. In the model, public recognition receives greater weight when private information about local severity is less precise. The empirical proxy is baseline per-capita income. This proxy is imperfect: income may capture access to local information, media, and administrative capacity, but it may also capture vulnerability, fiscal capacity, or recovery needs. I therefore interpret this test as suggestive heterogeneity rather than as a causal test of information.

The prediction is that the recognition premium should be larger in lower-income counties if public signals matter more where private information is less precise.

The income-split evidence is consistent with this prediction. In low-income counties, the recognition premium is

$$\hat{\delta} = 2.02$$

and statistically significant at the 5 percent level. In high-income counties, the estimate is smaller, $\hat{\delta} = 0.96$, and statistically indistinguishable from zero. The IV estimates show the same ordering:

Table 9: Recognition Premium by Baseline Per-Capita Income

	Low income		High income	
	OLS	IV	OLS	IV
δ_C $(D^C \times T)$	2.020**	2.747**	0.958	0.793
	(0.824)	(1.254)	(0.739)	(0.792)
First-stage F	—	65	—	73
Observations	731	731	1,234	1,234

Notes: The sample is restricted to casualty disasters ($D^C > 0$). Counties are split at the median baseline per-capita income, $25,868. The outcome is the change in incumbent presidential party vote share, measured in percentage points. All specifications include county fixed effects, election-year fixed effects, lag-4 controls, and standard errors clustered at state × election-year. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

2.75 in low-income counties and 0.79 in high-income counties.

This result should not be read as evidence that income itself causes voters to respond differently to recognition. The narrower interpretation is that the recognition premium is larger in the group where public signals may carry more weight, which is consistent with the signal-extraction mechanism.

7.4 Synthesis

The mechanism tests point in the same direction. The recognition premium is largest when federal recognition should be most informative: when human harm is severe, when recognition is prompt, and when voters may rely more heavily on public signals. These patterns use different margins of variation and are not mechanical consequences of the baseline cross-partisan result.

The results also help distinguish the credibility mechanism from alternative explanations. A simple co-partisan favoritism account predicts a larger premium in same-party settings, which is not what the data show. A pure media-salience account can explain why severe disasters matter, but it does not naturally predict why fast cross-partisan declarations dominate fast same-partisan declarations. Persistent county-level confounds are less plausible because the placebo tests in Section 6.4 show no comparable effects for future disasters, future partisan alignment, or declarations without contemporaneous local damage.

The evidence is therefore consistent with voters extracting information from federal recognition rather than merely rewarding disaster aid or reacting to damage itself. Recognition is most electorally meaningful when it validates severe harm, arrives quickly, and crosses party lines.

The next section turns to federal aid and shows why aid obligations should not be interpreted in the same way as public recognition.

8 Aid Timing and the Recognition–Transfer Distinction

The preceding sections show that federal recognition is electorally meaningful in casualty disasters, especially when recognition crosses party lines. This section turns to federal aid obligations. The goal is not to estimate the causal effect of aid dollars. Aid obligations are downstream administrative outcomes: they depend on damage assessment, program eligibility, project formulation, documentation, and recovery needs. In FEMA Public Assistance, for example, FEMA obligates funds based on the eligible total of an approved project, after which the recipient disburses funding to the applicant. Aid obligations therefore need not coincide with what voters observe before Election Day.

The timing of aid provides a diagnostic for the paper's central distinction between recognition and transfer. A presidential declaration is a discrete public act of recognition. Aid obligations are continuous administrative outcomes that may be recorded before or after voters cast their ballots. If voters reward federal dollars because they observe and value transfers before the election, then pre-election obligations should have the clearer electoral association. If aid obligations instead proxy for latent severity, administrative complexity, and recovery burden, then post-election obligations may remain predictive of vote-share outcomes even though they could not have affected voters' pre-election information sets.

8.1 Pre- and Post-Election Obligations

Table 10 reports the timing diagnostic. Aid obligations are split into amounts obligated before Election Day, M_{ct}^{pre}, and amounts obligated after Election Day, M_{ct}^{post}. The key comparison is between the interaction of property damage with pre-election aid, $D_{ct}^{P} \times M_{ct}^{pre}$, and the interaction of property damage with post-election aid, $D_{ct}^{P} \times M_{ct}^{post}$. Property damage is the relevant damage margin for this diagnostic because aid obligations are more closely tied to physical losses, project eligibility, infrastructure repair, and recovery costs than to the public recognition of human harm.

The estimates do not support a simple reward-for-dollars interpretation. The interaction between property damage and pre-election aid is positive but imprecisely estimated:

$$\hat{\varepsilon}_1 = 0.197.$$

The interaction between property damage and post-election aid is negative and statistically distinguishable from zero:

$$\hat{\varepsilon}_2 = -0.277 \quad (p < 0.05).$$

The IV specification produces the same pattern, with a post-election interaction of -0.296. The post-election coefficient is also similar under Conley spatial HAC inference at a 50 km bandwidth.

The sign of the post-election interaction should not be interpreted as voters punishing aid they could not observe. Instead, it indicates that, conditional on reported contemporaneous property damage, later obligations are negatively selected on unobserved recovery burden. Counties with

Table 10: Aid Timing Diagnostic: Pre- and Post-Election Obligations

	OLS	IV
	(1)	(2)
α (D^P)	0.014	0.061
	(0.038)	(0.083)
γ_1 (M^{pre})	1.333	1.139
	(2.019)	(2.039)
γ_2 (M^{post})	0.322	0.317
	(0.851)	(0.854)
ε_1 ($D^P \times M^{pre}$)	0.197	0.201
	(0.351)	(0.348)
ε_2 ($D^P \times M^{post}$)	-0.277**	-0.296**
	(0.131)	(0.135)
Wald: $\varepsilon_1 = \varepsilon_2$ (p-value)	0.249	—
Conley HAC 50 km on ε_2	-0.277**	—
SE [p-value]	(0.116) [p=0.017]	—
First-stage F	—	1562
County FE / Election-year FE / Lag-4 controls	Yes	Yes
Observations	14,913	14,913

Notes: The sample is restricted to county-years with positive property damage, $D^P > 0$. The outcome is the change in incumbent presidential party vote share, measured in percentage points. M^{pre} denotes aid obligated by Election Day, and M^{post} denotes aid obligated after Election Day. Aid is measured as $\log(1 + 100 \times \text{aid}/\text{GDP}_{t-4})$. The IV specification instruments D^P with the weather-severity instrument Z^W. Standard errors are clustered at state × election-year unless otherwise noted. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

larger post-election obligations appear to be places where the disaster's full costs were revealed gradually through damage assessment, project approval, eligibility determination, infrastructure repair, and administrative processing.

The Wald test does not reject equality between the pre- and post-election aid interactions ($p = 0.249$). The evidence therefore does not show that post-election aid has a statistically larger association than pre-election aid. The diagnostic point is narrower: aid that could not have entered voters' pre-election information sets remains at least as predictive of vote-share changes as aid that could have been observed before the election. This pattern is difficult to reconcile with a pure voter-reward mechanism and is more consistent with aid serving as an ex post measure of disaster severity and recovery complexity.

8.2 Interpretation

The aid-timing evidence reinforces the distinction between recognition and transfer. Declarations and obligations are both parts of federal disaster response, but they are politically different objects. A presidential declaration is discrete, public, and immediately observable as federal recognition. Aid obligations are continuous, administratively processed, and partly realized after the election.

Their levels combine relief with information about latent severity, administrative delay, and recovery complexity.

This distinction is important for interpreting the paper's main results. The recognition premium documented above is not well understood as a simple reward for federal dollars. If voters were directly rewarding aid, pre-election obligations should have the clearer association with vote-share changes. Instead, post-election obligations remain predictive despite being outside voters' pre-election information sets. Recognition is therefore the voter-facing signal, while aid obligations are better interpreted as administratively generated measures of recovery burden.

The next section examines the robustness of the recognition premium and the cross-partisan interpretation.

9 Robustness and Alternative Explanations

This section summarizes additional checks on the recognition premium and the cross-partisan interpretation. The purpose is not to introduce a new identification strategy, but to assess whether the main pattern is sensitive to inference choices, state-level political confounding, or measurement error in reported disaster damage.

9.1 Inference and Specification Checks

The baseline casualty-recognition premium is stable across alternative inference procedures and specifications. Appendix Table 12 reports estimates of $D_{ct}^{C} \times T_{ct}$ under state-by-election-year clustering, state clustering, county clustering, Conley spatial HAC corrections, state-by-year fixed effects, exclusion of the 2020 election, and the weather-IV specification.

Across these checks, the coefficient remains positive and economically meaningful. The baseline estimate is 1.94. It remains 1.94 under state and county clustering, 1.94 under Conley corrections at 50 km and 100 km, 2.28 when 2020 is excluded, and 1.94 in the weather-IV specification. Precision weakens under the most conservative spatial correction and under state-only clustering, but the sign and magnitude are preserved. With state-by-year fixed effects, the point estimate falls to 0.96 but remains statistically distinguishable from zero. These results indicate that the recognition premium is not driven by a single clustering choice, the 2020 election, or the baseline fixed-effect structure.

9.2 Close Gubernatorial Elections

A separate concern is that Cross_{st} may be correlated with state-level political environments that also shape voter responses to disasters. Cross-partisan configurations may be more common in competitive states with distinctive media markets, political attention, administrative capacity, or voter polarization. County fixed effects absorb persistent county differences and election-year fixed effects absorb national shocks, but time-varying state-level factors could still contribute to the estimated cross-partisan asymmetry.

To probe this concern, I restrict the sample to state-years in which the most recent gubernatorial election was decided by less than five percentage points, using gubernatorial returns from Algara and Amlani (2021). The logic is modest: within close gubernatorial-election environments, governor party is less tightly tied to persistent state characteristics. This exercise is not a stand-alone regression-discontinuity design; it does not estimate a local-linear RD, nor does it rely on a density or covariate-balance test at the threshold. It asks whether the cross-partisan asymmetry survives in a sample where governor party is less obviously selected by long-run state differences.

In this restricted sample, the pooled triple-interaction coefficient is $\hat{\phi} = 6.50$, with a standard error of 3.12. The estimate is larger than the full-sample estimate despite the smaller sample size ($N = 625$). Appendix Table 13 reports the corresponding specification. The split-sample estimates in this subsample are extremely imprecise and are therefore not interpreted separately.

This result should be read as supporting evidence rather than as a separate causal design. Its value is diagnostic: the cross-partisan asymmetry does not disappear when attention is restricted to state-years in which gubernatorial party control was determined by a close election. This pattern is difficult to reconcile with an explanation based solely on persistent differences between cross-partisan and same-partisan states.

9.3 Weather-Based Measurement Diagnostics

The weather-IV estimates throughout the paper address measurement error and reporting heterogeneity in disaster damage, not declaration assignment. NOAA Storm Events damage measures may vary with local reporting capacity, media attention, and administrative effort. Replacing reported damage variation with weather-predicted damage provides a check on whether the main patterns are driven by reporting differences. In the main casualty-disaster specifications, the weather instrument has a strong first stage; for example, the first-stage F-statistic is 144 in the recognition specification.

As an additional source-independence check, I construct hurricane exposure from HURDAT2, the National Hurricane Center's revised hurricane database based on best-track storm position and intensity records. HURDAT2 is operationally distinct from local damage reports, but it is not a strong instrument for total disaster damage in this setting because much of the sample's disaster damage comes from non-hurricane events. This is exactly what the first-stage diagnostics show: the HURDAT2 first stages are weak. I therefore do not use HURDAT2 as an alternative IV design. Instead, I use it to define a hurricane-exposed subsample and re-estimate the recognition specification within that subsample.

Appendix Table 14 reports the HURDAT2 diagnostics. In the hurricane-exposed subsample, the cross-partisan recognition premium is 5.52, compared with 0.93 in same-partisan settings, and the pooled triple-interaction coefficient is 3.67. The point estimates are larger than in the full sample. Because the HURDAT2 measure is based on storm tracks and intensity rather than local damage reports, this replication provides additional evidence that the cross-partisan recognition premium is not mechanically generated by NOAA Storm Events damage reporting.

10 Discussion

The results change the interpretation of disaster accountability in two ways. First, federal disaster response should not be treated as a single political object. A presidential declaration is a public act of recognition: it is discrete, visible, and attributable to the federal executive. Aid obligations are administratively generated transfers that also encode damage assessment, eligibility, project approval, and recovery burden. Declarations and aid are institutionally connected, but they do not carry the same electoral information.

Second, the politics of disaster response does not appear only on the supply side. The allocation tests show that, along the observed county-level margins, declarations and aid are better predicted by realized harm than by swing-county status or governor-president co-partisanship. This does not imply that disaster policy is never political. It means that the main politics documented here lies in the interpretation of recognition. Voters respond more strongly when federal recognition is harder to attribute to partisan accommodation.

10.1 Interpretation

The central empirical pattern is that the recognition premium is concentrated in cross-partisan settings. This pattern is difficult to reconcile with a simple co-partisan favoritism account. If the electoral value of declarations came mainly from presidents delivering benefits to aligned state governments, the premium should be larger when governors and presidents are from the same party. The data point in the opposite direction. Recognition is most electorally meaningful when it crosses party lines.

The mechanism tests are consistent with this interpretation. The recognition premium is larger when human harm is severe, when recognition is prompt, and in counties where public signals may carry more weight. These are precisely the settings in which the signal-extraction framework predicts federal recognition should be most informative. The placebo tests further reduce the concern that the result is driven by future disasters, future partisan alignment, or administrative declarations unrelated to local harm.

Aid timing reinforces the recognition-transfer distinction. Post-election aid obligations remain predictive of vote-share changes even though voters could not have observed those obligations before casting their ballots. This association cannot be interpreted as voters rewarding post-election dollars. It indicates that aid amounts partly capture unobserved severity, administrative complexity, and recovery burden. Declarations are therefore the voter-facing signal; aid obligations are better understood as administratively generated measures of recovery needs.

10.2 Relation to Existing Work

The findings help reconcile three strands of research. Work on distributive politics shows that governments may allocate public resources for electoral or partisan reasons. Studies of disaster politics document that declarations and FEMA spending can be shaped by political incentives in

some settings. The allocation results here do not overturn that literature. They show that, in this county-election-year panel and along the observed political margins, realized harm is the dominant predictor of federal disaster action.

The results also refine the retrospective-voting literature. Existing work asks whether voters punish disaster losses or reward government response. The evidence here suggests that the relevant response is not only spending. Public recognition of human harm has a distinct electoral association, while aid obligations are difficult to interpret as direct voter rewards because they also proxy for latent recovery burden.

Finally, the paper connects disaster accountability to models of political information. Voters observe government actions without observing the full state of the world that generated them. In this setting, the credibility of a public signal depends on partisan context. A same-party declaration may be discounted because voters cannot rule out co-partisan accommodation. A cross-party declaration is harder to attribute to partisan loyalty and is therefore more diagnostic.

10.3 Limits of the Evidence

The empirical design should be interpreted as a set of mutually reinforcing tests rather than a single natural experiment. The paper does not claim that declarations are randomly assigned, nor does it estimate the unconditional causal effect of a declaration. The allocation tests discipline the targeting interpretation, but they do not establish full conditional exogeneity of declaration assignment.

The formal pooled triple-interaction test is positive but not always precisely estimated under the most conservative inference. The evidence for the cross-partisan mechanism therefore comes from the broader pattern: split-sample estimates, pooled interactions, mechanism tests, placebo tests, the close-gubernatorial-election subsample, and measurement-error diagnostics.

The close-gubernatorial-election exercise is also diagnostic rather than a complete regression-discontinuity design. It restricts attention to state-years in which governor party is less tightly linked to persistent state characteristics, but it does not estimate a full local-linear RD around the gubernatorial margin. The resulting evidence is useful because the cross-partisan asymmetry does not disappear in close-election environments, but it should not be read as a stand-alone source of identification.

Finally, the mechanism tests should be read as mechanism-consistent evidence rather than structural estimation of voter beliefs. Baseline income is an imperfect proxy for information environments, and partisan context is richer than a binary cross-party indicator. Future work could use more direct measures of media exposure, administrative capacity, or voter information to study how public recognition is processed in local political environments.

11 Conclusion

This paper studies federal disaster response by separating two objects that are often bundled together: public recognition and federal transfer. Presidential disaster declarations are discrete, visible acts of federal recognition. FEMA aid obligations are continuous administrative outcomes that partly reflect damage assessment, eligibility, project approval, and recovery burden. Treating these objects as a single relief bundle obscures the distinction between the allocation of federal action and the electoral interpretation of that action.

The first result is that federal disaster action is better described as damage-based than politically targeted along the observed county-level margins. Conditional on realized damage, fixed effects, and lagged controls, swing-county status and governor-president co-partisanship do not explain much of the variation in declarations or aid. Equivalence tests rule out several large political-targeting effects, especially for swing counties and aid obligations. This evidence does not prove that disaster policy is apolitical, but it weakens a simple allocation-based explanation for the vote-share patterns.

The second result is that voters do not interpret damage-based recognition uniformly. The vote-share slope with respect to casualty harm is more positive when the harm receives federal recognition, and the recognition premium is concentrated when the governor and president are from different parties. The premium is also larger for severe casualty disasters, prompt declarations, and lower-income counties. These patterns are consistent with a signal-extraction mechanism: cross-partisan recognition is more credible because it is harder to dismiss as partisan accommodation.

Aid obligations play a different role. Post-election obligations remain predictive of vote-share changes even though they could not have entered voters' pre-election information sets. Aid amounts therefore appear to capture latent severity and recovery complexity, not only voter-observed transfers. Recognition is the voter-facing signal; aid is the administratively generated recovery measure.

The broader implication is that disaster accountability is not only about who receives relief or whether voters reward spending. It also depends on how voters interpret public acts of government recognition. Federal disaster declarations are not merely gateways to aid. They are public signals, and their electoral meaning depends on whether voters view them as credible recognition rather than partisan accommodation.

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A Stability Across Administrations

A further concern is that the damage-based allocation pattern may be specific to a particular administration or institutional period. This concern is relevant because earlier studies document political influence in disaster declarations and FEMA payments in earlier samples (**??**), and because post-Katrina institutional reforms may have changed the administration of federal disaster policy. To examine this possibility, the allocation equations are estimated separately for three administration blocks: pre-2001 elections (1992, 1996, 2000), the Bush administration elections (2004, 2008), and the modern period (2012, 2016, 2020). Table 11 reports the joint Wald statistics for the damage and political blocks in each period.

The period-specific estimates are consistent with the baseline result. The damage block is more informative than the political block in each administration period. For aid, the political F-statistics range from 0.32 to 1.82. For declarations, they range from 0.79 to 2.58 and do not reach the 5 percent level. The one marginal case is the declaration equation during the Bush administration block, where the political block is jointly significant at the 10 percent level. This result is not strong enough to overturn the broader pattern, but it is consistent with the view that political allocation may have been more relevant in some institutional periods.

Table 11: Allocation by Administration Block: Damage and Politics Joint Wald F

	Pre-2001 (1992–2000)		Bush 43 (2004–2008)		Modern (2012–2020)	
	F_{dam}	F_{pol}	F_{dam}	F_{pol}	F_{dam}	F_{pol}
M (aid amount)	2.43^{*}	0.32	2.23	1.82	7.62^{***}	0.38
T (declaration)	12.1^{***}	1.22	6.60^{***}	2.58^{*}	7.65^{***}	0.79
Observations	9,327		6,217		9,327	

Notes: Each cell reports the joint Wald F-statistic for either the damage variables (D^P, D^C) or the political variables (Swing, SameParty) in equation (1) for M or equation (2) for T. The Bush administration block covers the 2004 and 2008 presidential elections. The modern period covers the 2012, 2016, and 2020 presidential elections. County fixed effects, election-year fixed effects, and lag-4 controls are included in every specification. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A pooled specification with administration-by-politics interactions also fails to reject the hypothesis that the political variables matter in any administration block: $F = 0.91$ for aid and $F = 0.32$ for declarations. These results suggest that the damage-based allocation pattern is not driven by a single administration. Within the county-election-year panel studied here, declarations and aid are consistently better explained by realized disaster harm than by swing-county status or gubernatorial co-partisanship.

B Robustness

This section reports additional robustness checks for the main casualty-recognition premium and for the cross-partisan interpretation. Section B.1 examines the sensitivity of the baseline recognition premium to alternative inference procedures, spatial correlation corrections, fixed effects, exclusion of 2020, and the weather-IV specification. Section B.2 uses close gubernatorial elections as a supporting quasi-experimental robustness check for the cross-partisan premium. Section B.3 clarifies the interpretation of the weather instrument and reports a HURDAT2-based hurricane-exposure diagnostic.

B.1 Clustering, Spatial Dependence, and Fixed Effects

Table 12 reports robustness checks for the main $D^C \times T$ coefficient under alternative inference choices and specifications. The coefficient is stable in sign and remains economically meaningful across all specifications.

The recognition premium is robust to clustering at the state-election-year level, state level, and county level; to Conley spatial HAC corrections at shorter bandwidths; to state-by-year fixed effects; to dropping 2020; and to the weather-IV specification. Precision weakens under the most conservative spatial correction, with Conley 200 km standard errors yielding $p = 0.120$, and under

Table 12: Robustness of the Casualty-Recognition Premium $\hat{\delta}_C$

Specification	$\hat{\delta}_C$	SE	p
(1) Main: state × year cluster	1.937**	0.865	0.026
(2) State cluster	1.937*	1.043	0.070
(3) County cluster	1.937***	0.596	0.001
(4) Conley 50 km	1.937**	0.958	0.043
(5) Conley 100 km	1.937*	1.056	0.067
(6) Conley 200 km	1.937	1.247	0.120
(7) State × year FE	0.964**	0.423	0.024
(8) Drop 2020	2.284**	0.937	0.015
(9) IV ($D^C \sim Z^W$)	1.943*	1.107	0.080

Notes: The sample is restricted to casualty disasters ($D^C > 0$). The outcome is the change in incumbent presidential party vote share, measured in percentage points. Specifications (1)–(7) and (9) use the recognition specification with aid obligations included as a linear control. Specification (8) excludes the 2020 presidential election. Column (9) instruments D^C with the weather-severity instrument Z^W; the first-stage F-statistic is 144. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

state-only clustering, where $p = 0.070$. The state-by-year fixed-effects specification reduces the point estimate to 0.964, but the coefficient remains statistically distinguishable from zero at the 5 percent level. Thus, the recognition premium is not equally precise under every inference procedure, but its sign and economic magnitude are preserved across demanding alternatives.

B.2 Close Gubernatorial Elections

A separate concern is specific to the cross-partisan interpretation in Section 6.3. The indicator Cross_{st} may be correlated with state-level political environments that also shape voter responses to disasters. Cross-partisan configurations may be more common in competitive states with distinctive media markets, levels of political attention, administrative capacity, or patterns of voter polarization. County fixed effects absorb time-invariant county differences, and election-year fixed effects absorb national shocks, but unobserved time-varying state-level factors could still contribute to the estimated cross-partisan asymmetry.

To probe this concern, the analysis uses close gubernatorial elections as a supporting quasi-experimental robustness check. Each presidential- election county-year is matched to the most recent prior gubernatorial election in the same state, using the returns reported by Algara and Amlani (2021). The sample is then restricted to state-years in which that gubernatorial election was decided by less than 5 percentage points. The identifying idea is modest: within close gubernatorial elections, the party of the governor is less tightly tied to persistent state characteristics than in the full sample. If the cross-partisan premium were entirely driven by fixed or slowly moving differences between deep-red, deep-blue, and competitive states, it should weaken in this close-

election subsample.

Within this restricted sample, the pooled triple-interaction specification from equation (3) is re-estimated. The exercise should not be interpreted as a full regression-discontinuity design. It does not estimate local linear specifications around a gubernatorial vote-margin cutoff, nor does it rely on a formal density or covariate-balance test at the threshold. Its purpose is to ask whether the cross-partisan asymmetry survives in a sample where the partisan identity of the governor is less obviously selected by persistent state-level differences.

Table 13: Close-Gubernatorial-Election Robustness: Cross-Partisan Recognition Premium

	Casualty subsample ($D^C > 0$, close5)	Pooled triple interaction
$\hat{\delta}_C$ ($D^C \times T$, cross-partisan)	—	−1.64
		(1.60)
$\hat{\phi}$ ($D^C \times T \times$ Cross)	—	6.50**
		(3.12)
$\hat{\delta}_C$, cross subsample	−5.02	—
	(21.4)	—
$\hat{\delta}_C$, same subsample	−0.53	—
	(3.07)	—
N	625	625
Cross-partisan n	340	—
Same-partisan n	285	—

Notes: The sample is restricted to county-years with $D^C > 0$ in which the state's most recent prior gubernatorial election was decided by less than 5 percentage points. Gubernatorial returns are from Algara and Amlani (2021). All specifications include county fixed effects, election-year fixed effects, lag-4 controls, and standard errors clustered at the state-election-year level. The split-sample estimates are imprecise because of small effective sample sizes; the pooled triple interaction is the main statistic in this close-election robustness exercise. $^{**}p < 0.05$.

The close-election exercise supports the cross-partisan interpretation. Within this restricted sample, the pooled triple-interaction coefficient is $\hat{\phi} = 6.50$, with a standard error of 3.12, yielding $p = 0.043$. The estimate is larger than the corresponding full-sample estimate ($\hat{\phi} \approx 1.97$), despite the smaller sample size ($N = 625$). The split-sample estimates are extremely imprecise, as expected given the limited number of casualty-disaster observations within close-election cells, and should not be interpreted separately. The pooled interaction is the relevant statistic.

This result should be read as supporting rather than definitive evidence. It does not replace the full-sample specification and does not by itself establish a complete RD design. Its value is diagnostic: the cross-partisan asymmetry does not disappear when attention is restricted to state-years in which gubernatorial party control is determined by a close election. If anything, the estimated asymmetry is larger in that subsample. This pattern is difficult to reconcile with an explanation based solely on persistent state-level differences between cross-partisan and same-

partisan environments.

Two interpretations are consistent with the larger close-election estimate. One is that state-level confounding attenuates the full-sample estimate, so restricting to close gubernatorial elections reveals a sharper cross-partisan premium. The other is that close-election states are precisely the environments in which voters are most attentive to partisan signals, making declarations by an opposite-party federal executive especially informative. Both interpretations are consistent with the paper's central mechanism: voters respond not only to the fact of federal recognition, but also to the partisan context in which that recognition is delivered.

B.3 Weather IV and HURDAT2 Hurricane Exposure

The IV estimates reported throughout the paper are interpreted as robustness for weather-induced harm, not as causal estimates of declaration assignment. The reason is conceptual. The supply-side allocation tests in Section 5 do not indicate systematic allocation of declarations or aid along the observed political margins once realized damage and fixed effects are controlled. The remaining concern is therefore measurement error and reporting heterogeneity in disaster damage itself. NOAA Storm Events reports may vary with county capacity, local news coverage, and reporting intensity. Instrumenting reported damage with objective weather severity helps address this source of measurement error.

The weather-IV estimates are less precise than OLS estimates, especially in the smaller casualty-disaster subsamples, but they are consistent in sign and magnitude with the baseline results. The first-stage F-statistics are well above conventional weak-instrument thresholds in the main samples, including $F = 144$ in the $D^C > 0$ subsample. The IV results therefore provide a severity-of-harm robustness check while leaving the interpretation of declarations and aid grounded in the supply-side allocation evidence and the demand-side heterogeneity tests.

As an additional diagnostic, the analysis constructs a hurricane-exposure measure from HURDAT2 hurricane-track data. For each county-election year, the measure is the maximum sustained wind speed of any Atlantic hurricane track point passing within 200 km of the county centroid during presidential election year t. HURDAT2 is operationally distinct from NOAA Storm Events because it records storm position and intensity rather than local damage reports. As a single instrument for total county damage, however, the HURDAT2 measure has a weak first stage. This is expected: most disaster damage in the sample comes from non-hurricane events such as tornadoes, severe storms, floods, and winter weather. Accordingly, HURDAT2 is not used as the primary instrument. Its role is diagnostic: it identifies a hurricane-exposed subsample in which the recognition specification can be re-estimated using exposure defined from a source independent of local damage reporting.

The hurricane-exposed subsample reproduces the cross-partisan asymmetry. The recognition premium is $\hat{\delta}_C^{cross} = 5.52$ in cross-partisan settings and $\hat{\delta}_C^{same} = 0.93$ in same-partisan settings, with a pooled triple-interaction coefficient of $\hat{\phi} = 3.67$. The point estimates are larger than in the full sample, although the HURDAT2 exercise should not be interpreted as an alternative IV design.

Table 14: HURDAT2 Diagnostic: First Stage and Recognition Premium in Hurricane-Exposed Counties

Panel A: First-stage relevance, $D^h \sim Z^H_{ct} + X + \text{FE}$				
	Full sample		Casualty subsample	
	D^P	D^C	D^P	D^C
Z^H coefficient	0.134	0.001	0.170	0.015
	(0.064)	(0.003)	(0.109)	(0.010)
Partial F	4.3	0.3	2.4	2.1
N	24,871	24,871	2,867	2,867
Panel B: Recognition premium in HURDAT2-exposed subsample ($Z^H > 0$, $D^C > 0$)				
	Split-sample		Pooled triple	
	Cross-partisan	Same-partisan	interaction	
$\hat{\delta}_C$ ($D^C \times T$)	5.523***	0.929	0.428	
	(1.686)	(0.707)	(0.785)	
$\hat{\phi}$ ($D^C \times T \times$ Cross)	—	—	3.667*	
	—	—	(1.889)	
N	937	807	1,744	

Notes: Z^H_{ct} is the log of one plus the maximum sustained wind speed, in knots, of any Atlantic hurricane track point passing within 200 km of the county centroid during presidential election year t, constructed from HURDAT2 (Landsea and Franklin, 2013). Panel A reports first-stage regressions of damage on Z^H in the full sample and the casualty subsample. Panel B re-estimates the OLS recognition equation and the pooled triple-interaction model on the subsample of county-years with positive HURDAT2 hurricane exposure. All regressions include county fixed effects, election-year fixed effects, lag-4 controls, and standard errors clustered at state × election-year. The HURDAT2 first stage is too weak to support an operationally independent IV for total damage; Panel B is therefore reported as an OLS subsample replication, not as an alternative IV. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Because the hurricane-exposure measure is constructed from satellite, radar, and reconnaissance-aircraft observations rather than local damage reports, this replication provides additional evidence that the cross-partisan recognition premium is not a mechanical artifact of NOAA Storm Events damage reporting.

Regulatory Borders or Watershed Boundaries? State PFAS Standards, Hydrologic Spillovers, and Infant Health

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Abstract

Environmental regulation is assigned to political jurisdictions, but pollution often moves through physical systems. This paper studies the consequences of this mismatch using state drinking-water standards for per- and polyfluoroalkyl substances adopted before the 2024 federal rule. I combine state monitoring records with EPA UCMR5 data to construct a county-by-chemical-by-year panel of 29 PFAS compounds in public water systems and link it to county-year natality outcomes. State maximum contaminant levels reduce regulated PFAS after a compliance lag. A staggered-adoption event study shows no adoption-year response, followed by a 2.6–3.8 point decline in the regulated PFAS6 hazard index per unit of state stringency during the first three post-adoption years. The early decline is concentrated among regulated chemicals, consistent with treatment installation rather than changes in monitoring or reporting. The health effects depend on whether the state border is also an exposure boundary. In adjacent border counties draining to different terminal basins, MCL adoption reduces low birth weight by 0.48 percentage points, or 6.0 percent relative to the sample mean, and raises mean birth weight by about 17 grams. In same-basin pairs where the treated county lies downstream of an unregulated control county, effects are near zero or wrong-signed. Spatial-gradient estimates and rainfall interactions point to hydrologic transport as the mechanism. The findings show that the effectiveness of subnational environmental policy depends on whether the regulated jurisdiction contains the physical exposure pathway. For persistent water pollutants, unilateral state standards can reduce local exposure, but federal or basin-level regulation is needed when contamination crosses state borders.

Keywords: PFAS; drinking water regulation; infant health; border-county design; hydrologic spillovers; transboundary pollution.

1 Introduction

Environmental regulation is written over political jurisdictions, but many pollutants move through physical systems. This mismatch creates a central limitation for subnational environmental policy. A state can regulate firms, utilities, and public water systems inside its borders, but it may not control the upstream sources that determine exposure for downstream residents. For drinking-water contaminants, the relevant exposure boundary is often a watershed rather than a state line. This paper studies how that mismatch affects the returns to state environmental regulation.

PFAS provide a useful setting because they are persistent, increasingly regulated, and transported through groundwater and surface-water systems (Agency for Toxic Substances and Disease Registry, 2021a; National Academies of Sciences, Engineering, and Medicine, 2022). Drinking water is an important exposure pathway in contaminated communities, and epidemiological evidence links prenatal PFAS exposure to fetal-growth outcomes (Stein et al., 2009; Gundacker et al., 2022; National Academies of Sciences, Engineering, and Medicine, 2022). Before the 2024 federal rule, enforceable drinking-water standards were adopted state by state. This decentralized policy environment creates variation in regulatory timing and stringency, while hydrologic transport creates a natural test of whether state-level regulation is effective when exposure crosses state borders.

Two empirical challenges make the policy effect difficult to identify. The first is policy endogeneity. Early-adopting states were not randomly selected; many adopted PFAS standards after visible contamination events, litigation, local activism, or heightened regulatory attention. A comparison between adopting and non-adopting states would therefore confound regulation with baseline exposure risk and political demand for cleanup. The second challenge is hydrologic spillover. PFAS moves through connected groundwater and surface-water systems. A downstream county in a regulated state may continue to receive contamination from an upstream county in an unregulated state. In that case, the policy changes at the state border, but exposure need not. The central empirical problem is therefore not only whether treated and untreated states are comparable, but whether the policy border is also an exposure border.

I address these problems using state PFAS drinking-water maximum contaminant levels adopted before the federal rule. Between 2018 and 2023, nine states adopted enforceable standards that differed in timing, chemical coverage, and numerical stringency. I combine state monitoring records with EPA's Fifth Unregulated Contaminant Monitoring Rule data to construct a county-by-chemical-by-year panel of 29 PFAS compounds in public water systems from 2018 to 2024, and link the exposure panel to county-year birth outcomes. UCMR5 requires monitoring for 29 PFAS and lithium in drinking water during the 2023–2025 monitoring period, providing a national post-2022

monitoring frame (U.S. Environmental Protection Agency, 2021, 2024c).

The empirical design has two parts. First, I estimate a chemical-stacked event study for PFAS concentrations. The design compares regulated and unregulated PFAS within county-by-chemical panels, using variation in policy timing, chemical coverage, and numerical stringency. County-by-chemical fixed effects absorb persistent differences in contamination and monitoring intensity. Chemical-by-year fixed effects absorb national changes in testing, reporting limits, and regulatory attention. Because state rules target only a subset of the monitored PFAS compounds, unregulated chemicals provide a placebo group measured in the same water systems.

Second, I estimate infant-health effects using adjacent counties on opposite sides of treated-control state borders. Border-pair-by-year fixed effects compare counties within the same local border region in the same year, absorbing local shocks common to both sides of the border. The key heterogeneity test classifies each border pair by HydroBASINS drainage connectivity. HydroBASINS provides hierarchically nested sub-basin polygons and topology for upstream–downstream connectivity analysis (Lehner and Grill, 2013; HydroSHEDS, 2024). In different-basin pairs, the state border approximately coincides with an exposure boundary: contamination from the control county cannot mechanically drain into the treated county through the same terminal basin. In same-basin pairs where the treated county lies downstream of an unregulated county, the legal treatment changes at the border but upstream contamination can still cross the state line.

State MCLs reduce regulated PFAS after a compliance lag. The regulated PFAS6 hazard index falls by 2.6 index points in the first post-adoption year, 3.8 points in the second year, and 3.1 points in the third year per unit of state stringency. The adoption-year coefficient is close to zero. This timing is consistent with the compliance process: public water systems must identify exceedances, secure financing, obtain permits, and install treatment before measured concentrations decline. The early decline is concentrated among regulated chemicals; unregulated PFAS do not show the same early post-adoption response.

Infant-health improvements are concentrated where hydrologic and regulatory boundaries align. In the pooled border-pair sample, MCL adoption reduces the low-birth-weight rate by 0.21 percentage points and raises mean birth weight by 8.6 grams, but both estimates are imprecise. In different-basin border pairs, the low-birth-weight rate falls by 0.48 percentage points, or 6.0 percent relative to the sample mean, and mean birth weight rises by about 17 grams. In same-basin pairs where the treated county lies downstream of an unregulated control county, the corresponding estimates are close to zero. The same legal treatment therefore generates detectable health gains only where the regulatory boundary also approximates the hydrologic exposure boundary.

Spatial-gradient and rainfall results provide supporting evidence for this interpretation. A geographic regression-discontinuity diagnostic shows little post-policy level jump at the state line but more favorable treated-side distance gradients for low birth weight and mean birth weight. Rainfall interactions also move in the direction predicted by hydrologic transport: wetter years attenuate fetal-growth benefits in hydrologically isolated pairs and have the opposite sign in connected downstream pairs. I interpret these results as mechanism evidence rather than as separate research designs.

The paper contributes to environmental economics by showing that the effectiveness of subnational regulation depends on the overlap between legal jurisdiction and the physical transport process. Border designs are common in applied work because political boundaries create sharp policy variation (Dube et al., 2010; Holmes, 1998). This paper shows that, for mobile pollutants, the same boundary can be a weak exposure discontinuity. The hydrologic split turns that concern into a testable prediction and provides evidence on when decentralized environmental regulation does and does not internalize damages.

The policy implication is direct. State PFAS MCLs can reduce regulated PFAS concentrations and improve infant health when the regulated jurisdiction controls the relevant exposure pathway. They cannot fully protect downstream residents when contamination originates upstream in an unregulated state. The 2024 federal PFAS drinking-water rule establishes enforceable national standards for several PFAS, including PFOA, PFOS, PFNA, PFHxS, HFPO-DA, and a hazard-index limit for mixtures (U.S. Environmental Protection Agency, 2024d). In this setting, the relevant policy boundary is the watershed, not only the state line.

2 Institutional setting

PFAS drinking-water regulation before 2024 was decentralized. States chose whether to regulate, when to regulate, which compounds to cover, and how stringent the numerical limits should be. At the same time, PFAS exposure was not confined to the jurisdiction that wrote the rule. PFAS can move through surface water, groundwater, soils, and sediment, often crossing state borders before reaching downstream public water systems (Agency for Toxic Substances and Disease Registry, 2021b; National Academies of Sciences, Engineering, and Medicine, 2022). The setting therefore combines policy variation across states with a physical exposure process that need not respect state boundaries.

Three institutional features are central for identification. First, state MCLs targeted only a subset of PFAS compounds, creating a regulated-versus-unregulated comparison within the same monitoring data. Second, compliance required water-system invest-

ment, so concentration responses should appear with a lag rather than immediately at the legal effective date. Third, state rules applied only within state jurisdiction, leaving downstream counties exposed to upstream contamination when drainage basins crossed the regulatory border.

2.1 PFAS exposure and drinking-water regulation

PFAS are synthetic fluorinated chemicals used in firefighting foam, industrial processes, and consumer products. Their persistence allows them to accumulate in soils, groundwater, and surface water. In contaminated communities, drinking water is an important exposure pathway, although total exposure also reflects diet, indoor dust, occupational sources, private wells, bottled-water use, and household avoidance behavior (National Academies of Sciences, Engineering, and Medicine, 2022; Agency for Toxic Substances and Disease Registry, 2021a). Epidemiological evidence links PFAS exposure to several adverse health outcomes, including immune-response attenuation and reduced fetal growth (Grandjean et al., 2012; Stein et al., 2009; Gundacker et al., 2022). For this paper, the central institutional feature is narrower: public water-system treatment can reduce PFAS at the point of distribution, but it does not remove the historical stock of contamination in upstream soils, aquifers, rivers, or source areas.

The monitoring panel contains the 29 PFAS compounds covered by EPA's Fifth Unregulated Contaminant Monitoring Rule. UCMR5 requires monitoring for 29 PFAS and lithium in public water systems during the 2023–2025 monitoring period, including all community water systems and non-transient non-community water systems serving at least 3,300 people and a nationally representative sample of smaller systems (U.S. Environmental Protection Agency, 2021, 2024c). State drinking-water standards adopted before the federal rule covered only a subset of these compounds. The most common targets were PFOA, PFOS, PFNA, PFHxS, PFHpA, and PFDA, either as single-compound MCLs or as group-sum rules. The gap between targeted and untargeted compounds provides within-system placebo variation: the same water systems report both regulated PFAS and chemically related but unregulated PFAS.

Federal enforceable regulation came later. EPA issued non-binding lifetime health advisories for PFOA and PFOS in 2016 and revised advisory levels in 2022, but a binding National Primary Drinking Water Regulation was not finalized until April 2024 (U.S. Environmental Protection Agency, 2016, 2022, 2024d). The 2024 federal rule set MCLs of 4 ng/L for PFOA and PFOS, 10 ng/L for PFHxS, PFNA, and HFPO-DA, and a hazard-index constraint for mixtures containing PFHxS, PFNA, HFPO-DA, and PFBS (U.S. Environmental Protection Agency, 2024d). Public water systems must complete initial monitoring by 2027 and comply with MCLs by 2029 under the rule's implementation schedule (U.S. Environmental Protection Agency, 2024b). The

pre-federal period from 2018 to 2023 therefore provides the policy window used in this paper: some states imposed enforceable PFAS drinking-water standards while neighboring states did not.

Nine states adopted enforceable drinking-water MCLs before the federal rule. New Jersey adopted a PFNA standard in 2018 and added PFOA and PFOS in 2020. Vermont adopted a sum-of-five PFAS rule in 2019. Massachusetts, Michigan, New Hampshire, and New York adopted standards in 2020. Rhode Island and Wisconsin adopted standards in 2022, and Pennsylvania followed in 2023. These states are concentrated in the Northeast and Midwest, where historical industrial activity, military installations, firefighting-foam use, and dense public water-system networks made PFAS contamination salient before federal standards became binding.

Table 1: State PFAS drinking-water standards adopted before the federal rule.

State	Effective date	PFAS covered	MCL (ng/L)
New York	2020-08-26	PFOA; PFOS	10; 10
Massachusetts	2020-10-02	PFAS6: PFOA, PFOS, PFNA, PFHxS, PFHpA, PFDA	20 (sum)
Michigan	2020-08-03	PFNA; PFOA; PFOS; PFHxS; PFBS; HFPO-DA; PFHxA	6; 8; 16; 51; 420; 370; 400,000
New Jersey	2018-09-04; 2020-06-01; 2020-06-01	PFNA; PFOA; PFOS	13; 14; 13
Vermont	2019-05-15	PFAS5: PFOA, PFOS, PFNA, PFHxS, PFHpA	20 (each or combined)
New Hampshire	2020-06-16	PFOA; PFOS; PFNA; PFHxS	12; 15; 11; 18
Rhode Island	2022-06-24	PFAS6 interim group standard	20 (sum)
Wisconsin	2022-08-01	PFOA; PFOS	70 (each or combined)
Pennsylvania	2023-01-14	PFOA; PFOS	14; 18

Notes: The table reports enforceable state drinking-water maximum contaminant levels adopted before EPA finalized the federal PFAS drinking-water rule in April 2024. Effective dates are used to assign treatment timing. MCL denotes maximum contaminant level. Some states regulate individual compounds; others regulate a sum of multiple PFAS compounds.

2.2 Compliance timing and co-removal

A drinking-water MCL affects exposure through compliance by public water systems, not through immediate removal of contamination sources. Systems that exceed a standard must monitor, identify affected sources, finance treatment, obtain permits, install equipment, and operate the treatment system before measured concentrations fall. The relevant empirical implication is dynamic: a treatment-based response should appear with a lag rather than mechanically in the adoption year.

The main PFAS treatment technologies include granular activated carbon, ion exchange, and high-pressure membranes. These technologies are effective at removing many PFAS from treated water, but their performance varies across compounds, influent chemistry, and operating conditions (Appleman et al., 2014; Dickenson and Higgins, 2016; U.S. Environmental Protection Agency, 2024a). These engineering constraints

make the legal effective date an imperfect proxy for the date at which treated water quality changes. An immediate drop in event year zero would raise concerns about reporting, sampling, or monitoring changes. A null adoption-year response followed by declines in later years is more consistent with physical treatment.

Treatment technology also complicates the regulated-versus-unregulated comparison. Systems installing granular activated carbon or ion exchange to comply with a PFOA, PFOS, PFNA, PFHxS, PFHpA, or PFDA standard may also remove chemically related PFAS not named in the state rule. Unregulated compounds therefore need not remain flat indefinitely. The sharper prediction is about timing: targeted compounds should fall first, at the horizon when compliance becomes binding; non-targeted compounds may decline later through co-removal.

2.3 Other state PFAS policies

MCL adoption occurred in a broader state PFAS policy environment. During the same period, states adopted restrictions on PFAS-containing firefighting foam, restrictions on PFAS in food packaging or consumer products, and cleanup or remediation legislation for contaminated sites (Safer States, 2024; U.S. Environmental Protection Agency, 2024e). These policies operate on different margins. Firefighting-foam restrictions reduce future AFFF use. Product restrictions reduce consumer-product sources. Cleanup legislation funds monitoring and remediation of contaminated sites. Drinking-water MCLs instead require public water systems to meet concentration limits at the point of delivery.

This overlap matters for identification because MCL adoption may coincide with broader PFAS policy activity. I therefore code three categories of non-MCL PFAS policies: firefighting-foam restrictions, product restrictions, and cleanup or remediation statutes. The empirical specifications include a state-year count of these non-MCL policy categories. This variable is not a structural measure of environmental effort; it is a control for contemporaneous PFAS policies that may move with MCL adoption. The remaining identification comes from within-county and within-chemical changes in the first stage, regulated-versus-unregulated comparisons measured in the same water systems, and adjacent border-county comparisons in the health analysis.

2.4 Hydrologic spillovers and regulatory borders

State regulation is jurisdictional, but PFAS transport is hydrologic. A state MCL applies to public water systems inside the state. It does not regulate an upstream source located across the border. A downstream county in a regulated state can therefore remain exposed to PFAS entering through a shared river, aquifer, or drainage basin

from an unregulated state. Local treatment may reduce the concentration delivered by the regulated water system, but the underlying source-water burden can still be shaped by upstream contamination outside the state's control.

This feature changes the interpretation of a border-county design. Standard border designs use adjacent counties on opposite sides of a policy boundary to absorb local shocks and isolate a discrete policy change. That logic is strongest when the policy boundary is also the relevant exposure boundary. PFAS weakens that condition. In a same-basin downstream pair, the treated county and untreated county are geographically adjacent, but their exposure relationship is directional: the untreated upstream county can affect the treated downstream county, while the treated county's MCL cannot regulate the upstream source.

Hydrologic geography determines when the border comparison is informative about the local policy margin. If two adjacent counties drain to different terminal basins, contamination from the control county cannot mechanically flow into the treated county through the same drainage system. The comparison then more closely resembles the canonical border design: the counties share local economic and demographic shocks, while only one side is subject to the MCL. If the counties share a basin and the treated county lies downstream, the border comparison combines the local effect of treatment with unregulated upstream exposure.

The empirical design uses this distinction directly. HydroBASINS provides hierarchically nested sub-basin polygons and topological information that supports upstream–downstream connectivity analysis (Lehner and Grill, 2013; HydroSHEDS, 2024). I use HydroBASINS Level-12 drainage data to classify each border pair by terminal basin and flow direction. Different-basin pairs identify the effect of state regulation when legal and hydrologic boundaries align. Same-basin treated-downstream pairs test the attenuation predicted when upstream contamination crosses the state line. Rainfall provides a further mechanism check because precipitation can change PFAS transport differently across these settings: it can mobilize residual PFAS into local source water in isolated basins, while higher streamflow can dilute upstream concentrations in connected downstream basins.

The policy implication follows immediately. State MCLs can reduce in-jurisdiction exposure when public water systems comply, but they do not internalize upstream contamination generated outside the state. When exposure pathways cross state lines, federal regulation or basin-level interstate coordination is required to align the regulatory boundary with the physical movement of pollution.

3 Empirical framework

The empirical object is the reduced-form effect of a binding state PFAS drinking-water standard on infant health. The parameter combines the effect of regulation on water-system PFAS concentrations with the effect of prenatal PFAS exposure on birth outcomes. It is the policy parameter relevant for evaluating state MCL adoption. I do not interpret the estimates as structural toxicological dose-response parameters, because drinking-water concentration is only one component of total maternal PFAS exposure.

Let E_{ct} denote birth-relevant PFAS exposure in county c and year t. Exposure has a local component and, for hydrologically connected counties, an upstream component:

$$E_{ct} = \mu_{ct} + \kappa_{cc'}\eta_{c't}, \tag{1}$$

where μ_{ct} is exposure generated by sources and water systems inside county c, $\eta_{c't}$ is contamination entering from an upstream county c', and $\kappa_{cc'}$ measures hydrologic connectivity. If the counties drain to different terminal basins, $\kappa_{cc'} = 0$. If county c lies downstream of county c' in the same basin, $\kappa_{cc'} > 0$.

A state MCL affects exposure by reducing the local component:

$$\mu_{ct} = \mu_{c0} - \beta_E M_{s(c)t} + \xi^E_{ct}, \tag{2}$$

where $M_{s(c)t}$ is the regulatory treatment in state $s(c)$ and $\beta_E > 0$ is the first-stage effect of the policy on local PFAS exposure. The treatment does not directly change $\eta_{c't}$ if the upstream source lies outside the regulated state. The same legal treatment can therefore generate different exposure effects depending on whether the treated county is hydrologically isolated from, or downstream of, the control side of the border.

Infant health is modeled as a function of exposure:

$$Y_{ct} = \beta_H E_{ct} + \xi^Y_{ct}, \tag{3}$$

where Y_{ct} is low birth weight, mean birth weight, or gestational age. For low birth weight, $\beta_H > 0$ if exposure worsens health. For birth weight and gestation, $\beta_H < 0$. The reduced form of the state MCL is the product of the first-stage exposure response and the health response to exposure. The sign and magnitude of this reduced form are expected to be clearest when $\kappa_{cc'} = 0$, because the policy changes local exposure without offsetting upstream contamination from the control county.

This framework delivers three empirical implications. First, regulated PFAS concentrations should fall after state MCL adoption, with a lag corresponding to treatment installation. Second, infant-health effects should be stronger in border pairs that

drain to different terminal basins than in same-basin pairs where the treated county is downstream of the control county. Third, rainfall should moderate the policy effect differently across hydrologic classes: in isolated basins, rainfall can mobilize residual PFAS into local source water; in connected downstream basins, higher flow can dilute upstream concentrations in shared streams. The empirical analysis tests these implications directly.

3.1 Regulatory stringency

State rules differ in chemical coverage and numerical strictness, so I construct a continuous stringency measure relative to the 2024 federal standard. For chemical $j \in \{\text{PFOA}, \text{PFOS}, \text{PFHxS}, \text{PFNA}\}$, define

$$S_{sj} = \frac{\text{MCL}_j^{\text{fed}}}{\text{MCL}_{sj}}, \tag{4}$$

where the federal benchmarks are 4 ng/L for PFOA and PFOS and 10 ng/L for PFHxS and PFNA. A larger value of S_{sj} denotes a more stringent state rule relative to the federal benchmark. A 70 ng/L standard for PFOA has $S_{sj} = 0.057$, while a 6 ng/L standard for PFNA has $S_{sj} = 1.667$.

The state-level index averages across federally benchmarked chemicals regulated by state s:

$$S_s = \frac{1}{|J_s|} \sum_{j \in J_s} S_{sj}, \tag{5}$$

where J_s is the set of benchmarked PFAS regulated by state s. This index preserves the intensity variation in the rules while putting heterogeneous MCLs on a common scale. It also separates treatment status from bindingness: Wisconsin is treated after 2022, but its 70 ng/L standard implies substantially lower stringency than Michigan's 2020 rule.

The empirical specifications use three treatment measures. The health regressions use the binary post-MCL indicator $D_{st} = \mathbf{1}\{t \geq \tau_s\}$ as the baseline policy variable. The exposure regressions use the stringency-scaled treatment $D_{st}S_s$ and, in chemical-stacked specifications, $D_{sjt}S_{sj}$. The chemical-level treatment equals zero for unregulated PFAS, which produces a built-in placebo group in the first-stage analysis.

3.2 First-stage exposure specification

The first empirical test asks whether state MCLs reduced measured PFAS concentrations in public water systems. The specification uses a county-by-chemical-by-year

panel:

$$y_{cjt} = \alpha_{cj} + \lambda_{jt} + \sum_{e \neq -1} \beta_e \mathbf{1}\{t - \tau_{s(c)j} = e\} S_{s(c)j} + \delta P^{\text{other}}_{s(c)t} + \varepsilon_{cjt}. \tag{6}$$

The outcome y_{cjt} is the measured concentration of PFAS chemical j in county c and year t, or the corresponding contribution to a hazard-index measure. County-by-chemical fixed effects absorb time-invariant spatial differences in PFAS exposure and testing intensity. Chemical-by-year fixed effects absorb national changes in laboratory methods, reporting limits, regulatory attention, and substitution across PFAS compounds. The omitted event year is $e = -1$.

The coefficients β_e measure the change in regulated PFAS concentration per unit of state chemical stringency relative to the year before adoption. Identification comes from differential timing and chemical coverage of state rules, after removing county-by-chemical levels and chemical-specific year shocks. New Jersey's 2018 rule is excluded from dynamic specifications because it lacks a clean pre-period, and Pennsylvania's 2023 rule is excluded because it has insufficient post-period observations. Static specifications use all adopting states.

The main identifying threat in (6) is that adopting states may have changed monitoring or reporting practices at the same time they adopted MCLs. The chemical-stacked design addresses this threat by comparing regulated and unregulated PFAS under the same monitoring environment. A reporting-limit change or general improvement in monitoring quality should affect both groups. A compliance response to MCL adoption should be concentrated among regulated chemicals at the relevant horizon.

3.3 Border-pair health specification

The health analysis uses adjacent counties on opposite sides of treated-control state borders. The estimating equation is

$$Y_{ct} = \alpha_c + \mu_{pt} + \beta D_{s(c)t} + \theta' X_{ct} + \varepsilon_{ct}, \tag{7}$$

where Y_{ct} is the low-birth-weight rate, mean birth weight, or gestational age in county c and year t. County fixed effects absorb time-invariant differences in baseline health, industrial history, water-system structure, and demographic composition. Border-pair-by-year fixed effects μ_{pt} compare counties within the same local border region in the same year. The control vector includes maternal age, smoking during pregnancy, ambient $PM_{2.5}$, and the count of non-MCL PFAS policies. Standard errors are clustered by state.

The coefficient β is identified from within-pair-year differences in MCL exposure across the state line. The identifying assumption is that, absent the MCL, adjacent

counties in the same border pair would have followed parallel infant-health trends after conditioning on county fixed effects, pair-year shocks, and observed controls. This assumption is weaker than a national staggered-adoption comparison because it does not require treated and untreated states to be comparable away from the border. It remains vulnerable if state-specific shocks to infant health change discontinuously at the border in the same year as PFAS regulation.

Hydrologic spillovers generate a separate threat. In a same-basin downstream pair, the treated county may continue to receive PFAS from the unregulated control side. In that case, the treatment indicator changes but birth-relevant exposure may not fall. This does not invalidate the reduced form; it changes the estimand. The pooled border estimate averages clean local treatment effects in hydrologically isolated pairs with attenuated or contaminated effects in connected downstream pairs. The framework therefore predicts that the pooled coefficient should be smaller than the coefficient in different-basin pairs.

3.4 Hydrologic heterogeneity

I classify border pairs using HydroBASINS Level-12 drainage polygons. Each county centroid is assigned to a Level-12 sub-basin, and each sub-basin is linked to a terminal drainage basin. Border pairs are classified as different-basin if the two counties drain to different terminal basins. Same-basin pairs are classified by flow direction using distance to the terminal sink: the county with the smaller distance is downstream.

The main heterogeneity specification is

$$Y_{ct} = \alpha_c + \mu_{pt} + \beta_1 D_{s(c)t} + \beta_2 D_{s(c)t} D_p^{\text{diff}} + \theta' X_{ct} + \varepsilon_{ct}, \tag{8}$$

where D_p^{diff} equals one for different-basin border pairs. The coefficient β_1 is the effect in same-basin pairs, and $\beta_1 + \beta_2$ is the effect in different-basin pairs. For low birth weight, the model predicts $\beta_2 < 0$; for birth weight and gestation, it predicts $\beta_2 > 0$.

This heterogeneity test is not a search over subgroups. It is the empirical counterpart of the exposure equation. The same state law should have larger health effects where the treated county's relevant water exposure is governed by in-state treatment and smaller effects where untreated upstream contamination crosses the border. A generic state-level health shock would not predict this sign pattern by drainage connectivity.

3.5 Rainfall moderation

Rainfall provides a physical-channel test because it changes PFAS transport differently across hydrologic classes. I estimate

$$Y_{ct} = \alpha_c + \mu_{pt} + \beta_1 D_{s(c)t} + \beta_2 D_{s(c)t} R_{ct} + \delta R_{ct} + \theta' X_{ct} + \varepsilon_{ct}, \tag{9}$$

where R_{ct} is annual precipitation standardized within county. The coefficient β_2 measures how the policy effect changes in wetter-than-normal years.

In different-basin pairs, rainfall can mobilize residual PFAS from contaminated soils, landfills, industrial sites, and fire-training areas into local source water. This mechanism predicts that wetter years attenuate the policy benefit: $\beta_2 > 0$ for low birth weight and $\beta_2 < 0$ for birth weight and gestation. In same-basin downstream pairs, higher flow can dilute upstream PFAS concentrations in shared streams. This mechanism predicts the opposite sign for fetal-growth outcomes. The identifying content of the rainfall test comes from this sign reversal. State policy selection can explain differences between adopting and non-adopting states, but it does not naturally explain why precipitation should moderate the same legal treatment in opposite directions across drainage classes.

I also estimate a pooled triple-interaction specification that interacts treatment, rainfall, and same-basin status. This specification tests directly whether rainfall moderation differs between hydrologically isolated and connected border pairs. The estimates are interpreted as mechanism evidence rather than as a separate causal estimand.

3.6 Geographic regression discontinuity

The border-pair design is supplemented with a geographic regression-discontinuity specification around treated-control state borders. Let r_c denote the signed distance from county c's centroid to the nearest treated-control border, positive on the treated side. Within bandwidth h, I estimate

$$Y_{ct} = \alpha + \beta(D_c \times Post_t) + \gamma_1 r_c + \gamma_2(D_c \times r_c) + \theta' X_{ct} + \lambda_t + \varepsilon_{ct}, \tag{10}$$

where $D_c = \mathbf{1}\{r_c > 0\}$ and $Post_t$ indicates years after the treated state's adoption date. The coefficient β is the post-policy discontinuity at the border. The coefficient γ_2 is the differential post-policy distance gradient on the treated side relative to the control side.

The RDD is not the primary design because county centroids are coarse and water-system service areas do not necessarily coincide with county borders. Its role is diagnostic. If the policy effect is attenuated at the border by hydrologic spillovers but

becomes clearer deeper inside treated states, the level discontinuity at $r = 0$ may be small while the treated-side gradient differs from the control-side gradient. This is the pattern the estimates test. I report $h = 200$ km as the headline bandwidth and show robustness at 100 and 300 km in the appendix.

3.7 Identification threats

The first threat is policy endogeneity. States adopted PFAS MCLs partly because contamination was visible, so comparisons between adopting and non-adopting states are not credible on their own. The design addresses this by using within-county changes in the first stage, adjacent counties in the health analysis, and border-pair-by-year fixed effects that absorb local shocks common to both sides of the border. Remaining bias would require a state-specific change in infant health that occurs at the border, coincides with MCL adoption, and is not captured by non-MCL PFAS policy controls.

The second threat is differential monitoring. Treated states may monitor more intensively or use lower reporting limits after adoption. The chemical-stacked first stage addresses this by comparing regulated and unregulated PFAS under the same state-year monitoring environment. A general monitoring change should appear across chemicals. A treatment response should appear first for regulated chemicals and only later, if at all, for unregulated chemicals through technology co-removal.

The third threat is exposure spillover. Standard border designs assume that the treatment changes discontinuously at the border and that the object affected by the treatment is locally contained. PFAS violates the second condition. The hydrologic split converts this violation into a testable implication: effects should be largest when legal and drainage boundaries align and attenuated when the treated county lies downstream of an unregulated control county. The rainfall interactions provide an additional check on the physical mechanism.

The fourth threat is small-sample inference. The border-pair health estimates are identified from a limited number of county-year observations and treated states. I therefore emphasize estimates that are consistent across outcomes, hydrologic classification, the RDD gradient, and the regulated-versus-unregulated first-stage placebo. The reduced form is the credible estimand; instrumental-variables estimates that use state stringency as an instrument for county-level hazard indices are reported only as auxiliary evidence because the county-year first stage is weak.

4 Data

The analysis combines public water-system PFAS monitoring records, state PFAS policy data, county-level natality outcomes, weather, and hydrologic geography. The unit

of observation differs across the two empirical stages. The first-stage exposure analysis uses a county-by-chemical-by-year panel. The health analysis uses county-year outcomes and, for the main design, adjacent county pairs on opposite sides of treated-control state borders. This section describes the construction of these panels and the sample restrictions used in the estimating equations.

4.1 PFAS monitoring data

The PFAS exposure panel combines EPA's Fifth Unregulated Contaminant Monitoring Rule data with state monitoring records. UCMR5 provides the national post-2022 monitoring frame. It requires all community water systems serving more than 3,300 people, and a sample of smaller systems, to test for 29 PFAS chemicals using EPA-approved analytical methods. UCMR5 has uniform chemical definitions and reporting conventions, but it begins too late to identify the pre-period for most state MCL adopters. I therefore supplement UCMR5 with state monitoring records collected between 2018 and 2022.

The state monitoring data cover 27 states and are heterogeneous by design. States differed in the chemicals they tested, the laboratories they used, reporting limits, sampling frequency, and whether testing targeted suspected contamination sites. I harmonize these records to the 29 UCMR5 chemicals, convert all concentration measures to ng/L, and preserve the row-specific reporting limit. The raw state files contain 211,519 monitoring rows. Of these, 198,900 rows, or 94 percent, map to UCMR5-compatible PFAS chemicals and survive the harmonization step. The remaining rows correspond to legacy isomers, internal standards, or analytes outside the UCMR5 scope.

The combined PFAS panel contains 1,594,437 sample rows from 2018 to 2024, covering 16,157 public water-system identifiers. UCMR5 contributes 1,396,437 rows and the harmonized state files contribute 198,900 rows. The panel covers 66 jurisdictions before the regional restrictions described below. Each row records a public water system, sample date, chemical, measured concentration or non-detect indicator, reporting limit, analytical source, and state.

Two features of the monitoring data matter for identification. First, state monitoring before UCMR5 is not a random sample of water systems. It was often concentrated near military bases, industrial facilities, fire-training areas, and communities with known or suspected PFAS exposure. The empirical design therefore does not use cross-sectional differences in PFAS levels as causal variation. It uses within-county and within-chemical changes around state MCL adoption, with fixed effects absorbing persistent differences in testing intensity and baseline exposure. Second, reporting limits vary across laboratories, methods, chemicals, and years. This makes left-censoring a first-order measurement issue rather than a minor data-cleaning choice.

4.2 Left-censored measurements

PFAS monitoring records contain many non-detects. A reported value below the reporting limit indicates that the true concentration lies somewhere between zero and the row-specific reporting limit. Treating all non-detects as zero understates concentrations. Treating them as the reporting limit overstates concentrations. Treating them as one-half the reporting limit imposes an arbitrary distributional assumption. These substitutions are especially problematic here because censoring rates and reporting limits vary systematically across chemicals, states, laboratories, and years.

The baseline exposure panel uses censored maximum likelihood under a log-normal model. Within each chemical-by-state-by-year-by-source stratum, let X_i denote the latent concentration and L_i the reporting limit. For detected observations, the likelihood contribution is the log-normal density evaluated at the observed concentration. For non-detects, the likelihood contribution is the probability that $X_i < L_i$. The estimator maximizes

$$\mathcal{L}(\mu, \sigma) = \sum_{i \in D} \left[\log \phi \left(\frac{\log X_i - \mu}{\sigma} \right) - \log \sigma \right] + \sum_{i \in ND} \log \Phi \left(\frac{\log L_i - \mu}{\sigma} \right), \tag{11}$$

where D and ND denote detected and non-detected observations. The stratum mean is then estimated as

$$\widehat{E[X]} = \exp \left(\hat{\mu} + \frac{\hat{\sigma}^2}{2} \right). \tag{12}$$

This procedure uses the observation-specific reporting limit rather than imposing a single censoring threshold within a chemical-year. That distinction is empirically relevant because state reporting limits often vary within the same chemical and year. The estimator also uses information from the censoring pattern itself, which matters in highly censored strata. I report robustness to three alternative non-detect treatments in the appendix: Kaplan–Meier estimation, regression on order statistics, and a threshold rule that sets concentrations below 4 ng/L to zero. The first-stage event-study pattern is stable across these alternatives.

The descriptive summary statistics in Table 2 use Kaplan–Meier imputation because it is non-parametric. The regression estimates use censored maximum likelihood as the baseline because it accommodates multiple reporting limits within a stratum while preserving a complete county-by-chemical-year panel. This distinction is mechanical and should be stated explicitly: Kaplan–Meier is used for descriptive comparability, while censored maximum likelihood is used for the headline event-study and health regressions.

4.3 Aggregation to county-year exposure

The monitoring records are reported at the public water-system level, while the health outcomes are observed at the county-year level. I map public water systems to counties using a two-step population-weighted crosswalk. First, EPA SDWIS Water System Detail records identify the ZIP codes served by each public water system. Second, the HUD ZIP-to-county crosswalk assigns ZIP codes to counties using residential address weights. For each chemical-year, public water-system observations are averaged to the county level using these residential weights.

The aggregation is intended to approximate resident exposure in public water systems. It does not observe individual household water sources, bottled-water use, point-of-use filters, or private wells. The estimates should therefore be interpreted as reduced-form effects of regulation on county-level exposure through public water systems, not as individual-level dose-response estimates. Measurement error from service-area approximation is likely to attenuate exposure-health relationships, which is one reason the paper emphasizes reduced-form policy effects rather than county-year 2SLS estimates.

The aggregation produces 2,434 county-year exposure cells across 947 counties. Hazard-index measures are constructed after aggregation. For the regulated PFAS group,

$$\mathrm{HI}_{ct}^{\mathrm{reg6}} = \sum_{j \in \mathrm{REG6}} \frac{C_{cjt}}{R_j}, \tag{13}$$

where $\mathrm{REG6} = \{\mathrm{PFOA}, \mathrm{PFOS}, \mathrm{PFNA}, \mathrm{PFHxS}, \mathrm{PFHpA}, \mathrm{PFDA}\}$ and R_j is the relevant reference concentration. The unregulated index is constructed analogously for the remaining 23 UCMR5 PFAS chemicals. The regulated index is the main first-stage exposure measure. The unregulated index is used as a placebo.

4.4 State PFAS policy data

I hand-code state PFAS policies from state statutes, administrative rules, and agency rulemaking documents. The policy database records the effective date, covered chemicals, and numerical standard for each enforceable state drinking-water MCL adopted before the 2024 federal rule. It also records three non-MCL PFAS policy categories: restrictions on PFAS-containing firefighting foam, restrictions on PFAS in food packaging or consumer products, and cleanup or remediation legislation.

The drinking-water MCL data determine treatment timing and stringency. The binary treatment equals one in years after a state's MCL becomes effective. The stringency-scaled treatment uses the state-by-chemical ratio of the 2024 federal benchmark to the state MCL. This construction makes a 10 ng/L PFOA rule more stringent than a 70 ng/L PFOA rule, even though both states are treated in the binary sense.

The chemical-level treatment equals zero for chemicals not regulated by the state rule.

The non-MCL policies are summarized by a state-year count ranging from zero to three. This variable is included in the health regressions to separate drinking-water standards from other PFAS policy activity. The count is not interpreted as a structural measure of regulatory effort. It is a control for contemporaneous state PFAS policies that could otherwise move with MCL adoption.

4.5 Birth outcomes and county controls

Birth outcomes come from CDC Natality records aggregated to the county-year level. The three outcomes are the low-birth-weight rate, mean birth weight, and obstetric-estimate gestational age. Low birth weight is measured as the share of live births below 2,500 grams and is reported in percentage points. The merged birth and PFAS panel contains 840 county-year observations from 2018 to 2024 after applying the geographic and data-availability restrictions.

The baseline health controls are maternal age, the share of mothers smoking during pregnancy, ambient $PM_{2.5}$, and the non-MCL PFAS policy count. Maternal age and smoking come from county-level health data. $PM_{2.5}$ controls for contemporaneous air-pollution exposure that is known to affect birth outcomes. Appendix specifications add uninsured rates, primary-care physician rates, child poverty, unemployment, and income inequality. The headline estimates are not driven by the richer control vector.

The county-year health panel is not large enough to support a credible national staggered-adoption health event study. County birth outcomes are noisy, and state adoption is selected on contamination risk. The main health analysis therefore restricts attention to adjacent counties along treated-control state borders and uses border-pair-by-year fixed effects.

4.6 Rainfall and hydrologic geography

Rainfall is measured using PRISM monthly precipitation grids. I aggregate monthly precipitation to annual county totals by areal weighting over county boundaries. The rainfall moderator is standardized within county:

$$R_{ct} = \frac{\text{Rain}_{ct} - \overline{\text{Rain}_c}}{\sigma_c}. \tag{14}$$

The standardization removes cross-county climate differences and uses deviations from a county's own normal rainfall to test the physical transport mechanism.

Hydrologic connectivity is measured using HydroBASINS Level-12 drainage polygons. Each county centroid is assigned to a Level-12 sub-basin. Each sub-basin is

linked to a terminal drainage basin and has a recorded distance to the terminal sink. For each cross-state border pair, I compare the terminal-basin assignments of the two counties. Pairs draining to different terminal basins are classified as different-basin pairs. For pairs in the same terminal basin, the county with the smaller distance to the terminal sink is classified as downstream.

This classification creates three border-pair types. Different-basin pairs are the main clean-comparison group because the control county is not upstream of the treated county. Same-basin treated-downstream pairs are the spillover group because the treated county may receive contamination from an unregulated upstream neighbor. Same-basin treated-upstream pairs are less central because the direction of potential spillover runs from treated to control, and they are not the focus of the main health tables.

4.7 Analytical samples

The first-stage exposure sample is the county-by-chemical-by-year panel constructed from the cleaned PFAS monitoring records. Dynamic event-study specifications exclude New Jersey's 2018 adoption because it lacks a clean pre-period and Pennsylvania's 2023 adoption because it has insufficient post-period support. The remaining dynamic cohorts are Vermont in 2019; Massachusetts, Michigan, New Hampshire, and New York in 2020; and Rhode Island and Wisconsin in 2022. Static specifications use all nine adopting states.

The main health sample restricts the merged county-year panel to adjacent cross-state border pairs in which one side is treated and the other side is untreated during the relevant period. The resulting border sample contains 95 cross-state border pairs and 80 unique counties. After merging with birth outcomes, controls, and fixed-effect support, the estimating sample contains 146 county-year observations. The principal hydrologic subsamples contain 66 county-year observations in different-basin pairs and 66 county-year observations in same-basin treated-downstream pairs.

Table 2: Summary statistics for the merged county-year panel.

	Mean	SD	Min	Median	Max	N
Panel A: PFAS exposure						
Regulated PFAS6 hazard index	1.78	1.46	0.00	1.41	12.61	840
Unregulated PFAS23 hazard index	4.32	3.13	0.00	3.41	28.74	840
Full PFAS29 hazard index	6.10	4.04	0.00	4.94	32.10	840
Panel B: Birth outcomes						
Low-birth-weight rate (%)	8.04	1.99	2.45	8.05	16.10	840
Mean birth weight (g)	3287	91	2762	3293	3614	840
Gestational age (weeks)	38.49	0.39	36.46	38.51	39.93	840
Panel C: Policy and environment						
MCL indicator	0.42	0.49	0	0	1	840
Stringency-scaled MCL treatment	0.21	0.32	0	0	1.05	840
PFAS chemical coverage	1.45	2.10	0	0	7	840
Other PFAS-policy count	0.74	0.93	0	0	3	840
Annual rainfall (mm)	1158	234	614	1147	1894	840

Notes: The table reports county-year observations in the merged PFAS–birth panel before imposing the border-pair fixed-effect sample restriction. Hazard indices are constructed from population-weighted county-year PFAS concentrations; descriptive exposure statistics use Kaplan–Meier non-detect handling. Regression specifications use censored maximum-likelihood exposure measures unless otherwise stated. Low birth weight is reported as the share of live births below 2,500 grams. Means are unweighted across county-year cells.

Table 2 reports summary statistics for the merged county-year panel before imposing the border-pair fixed-effect sample restriction. The mean regulated PFAS6 hazard index is 1.78 under Kaplan–Meier imputation, and the mean unregulated PFAS23 index is 4.32. The mean share of births below 2,500 grams is 8.04 percent, mean birth weight is 3,287 grams, and mean gestational age is 38.49 weeks. The mean MCL treatment indicator is 0.42, reflecting staggered adoption during the sample period. The smaller border-pair estimating sample used in the health regressions is determined by cross-state adjacency, hydrologic classification, outcome availability, and support for border-pair-by-year fixed effects.

5 State MCLs and PFAS concentrations

PFAS concentrations fall only after treatment systems have had time to respond. Figure 1 reports Sun–Abraham event-study estimates from equation (6). Coefficients are scaled by chemical-by-state stringency, so each estimate is the change in concentration, or hazard-index exposure, per unit of regulatory stringency. The omitted event year is $e = -1$. Panels (a) and (b) use chemical-stacked concentration sums in ng/L; panels (c) and (d) use county-year hazard indices. Regulated PFAS6 and unregulated PFAS23 are shown on the same event-time scale.

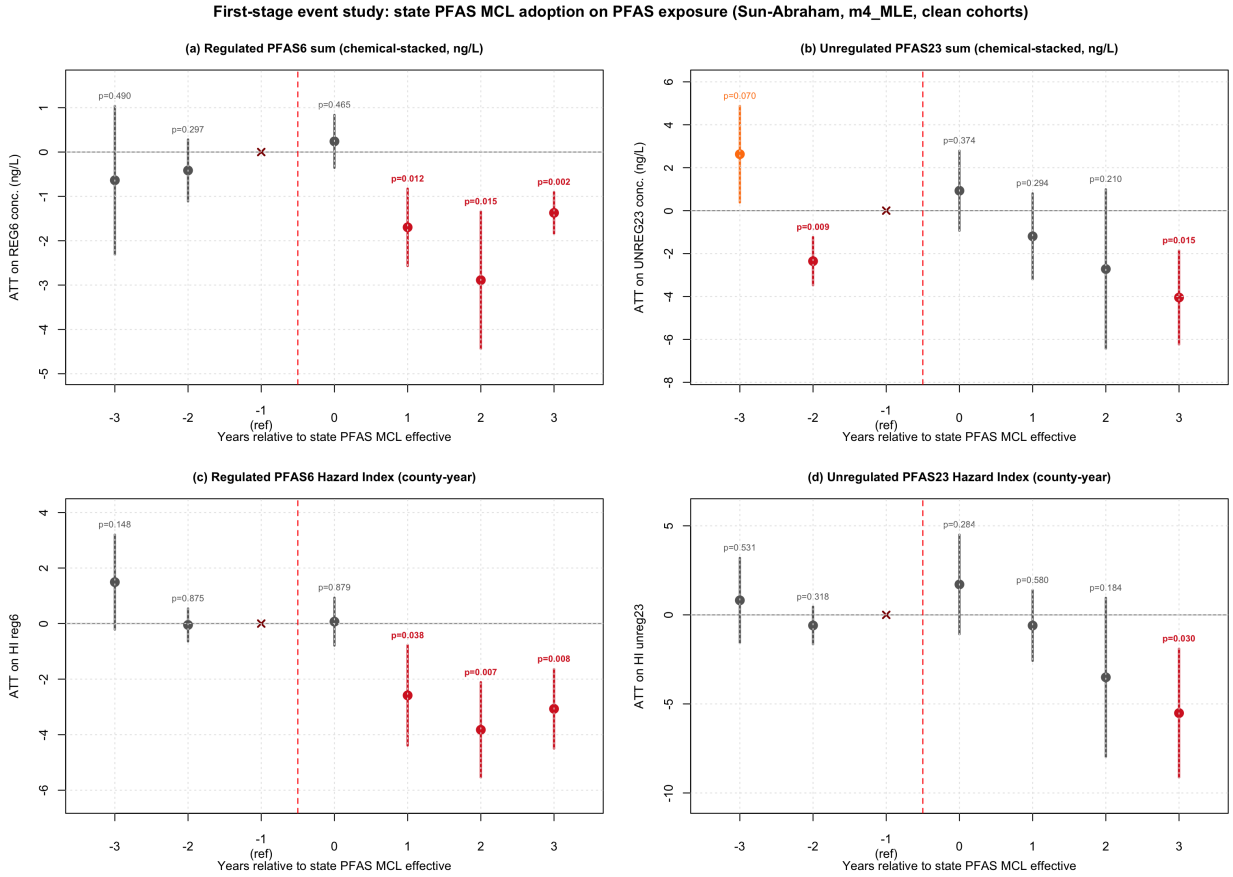


Figure 1: State MCL adoption and PFAS concentrations. The figure reports Sun–Abraham event-study coefficients from equation (6). Panels (a) and (b) report chemical-stacked concentration sums in ng/L for regulated PFAS6 and unregulated PFAS23. Panels (c) and (d) report county-year hazard indices for the same groups. Coefficients are scaled by chemical-by-state stringency, so each estimate is interpreted per unit of S_{sj}. The omitted event year is $e = -1$. Standard errors are clustered by state.

Regulated PFAS do not decline before adoption. In the hazard-index specification, coefficients are 1.49 in event year -3 ($p = 0.148$) and -0.05 in event year -2 ($p = 0.875$). Adoption itself produces essentially no change: the event-year-zero coefficient is 0.07 ($p = 0.879$). Declines begin in the first post-adoption year and persist through year 3. The regulated PFAS6 hazard index falls by 2.59 index points in event year 1 ($p = 0.038$), 3.82 points in event year 2 ($p = 0.007$), and 3.07 points in event year 3 ($p = 0.008$), all per unit of stringency. For a state with average treated-state stringency of about 0.41, the event-year-2 coefficient implies a decline of 1.57 hazard-index points, close to the mean regulated PFAS6 hazard index of 1.78 in Table 2.

Concentration outcomes tell the same story in physical units. Before adoption, regulated PFAS concentrations move by -0.62 ng/L in event year -3 ($p = 0.490$) and -0.39 ng/L in event year -2 ($p = 0.297$). In the adoption year, the coefficient is 0.24 ng/L ($p = 0.465$). After adoption, concentrations fall by 1.70 ng/L in event year 1 ($p = 0.012$), 2.89 ng/L in event year 2 ($p = 0.015$), and 1.37 ng/L in event year 3 ($p = 0.002$). Across both exposure measures, the pattern is the same: no pre-trend, no adoption-year drop, and a decline beginning one year after the rule becomes effective.

That delay is informative. Compliance with a drinking-water MCL is not instantaneous. Public water systems must identify exceedances, finance treatment, complete engineering design, obtain permits, install equipment, and begin operation before granular activated carbon or ion-exchange systems reduce measured concentrations. An accounting change, a reporting-limit change, or a shift in sample composition would be more likely to appear in the adoption year. The estimated dynamics instead line up with treatment installation.

Stringency matters as well as adoption. The event-year-2 hazard-index coefficient of -3.82 implies a 2.49-point decline for Michigan, where $S_s = 0.653$, but only a 0.22-point decline for Wisconsin, where $S_s = 0.057$. A binary treatment indicator would treat both rules as equivalent. Scaling by numerical stringency preserves the difference between stringent standards and weaker rules such as a 70 ng/L PFOA/PFOS limit.

Unregulated PFAS provide a placebo test measured in the same water systems. State MCLs directly regulate six chemicals in the monitoring panel: PFOA, PFOS, PFNA, PFHxS, PFHpA, and PFDA. The other 23 UCMR5 chemicals are observed under the same monitoring environment but are not directly constrained by the state rules. A general improvement in monitoring quality, a systematic change in reporting limits, or a broad state-year environmental shock should therefore move regulated and unregulated chemicals at the same time.

No such common early response appears. For the unregulated PFAS23 hazard index, coefficients are 1.71 in event year 0 ($p = 0.284$), -0.60 in event year 1 ($p = 0.580$), and -3.51 in event year 2 ($p = 0.184$). Concentration sums show the same ordering:

0.94 ng/L in event year 0 ($p = 0.374$), -1.14 ng/L in event year 1 ($p = 0.294$), and -2.70 ng/L in event year 2 ($p = 0.210$). At the horizon where regulated PFAS fall sharply, unregulated PFAS do not move with comparable precision.

By event year 3, unregulated chemicals also decline. The unregulated concentration sum falls by 4.05 ng/L ($p = 0.015$), and the unregulated hazard index falls by 5.52 points ($p = 0.030$). This later response is consistent with treatment co-removal. Water systems installing granular activated carbon or ion exchange for regulated PFAS can also remove chemically related compounds outside the legal target set. Because adsorption and breakthrough profiles differ across compounds, the placebo need not remain flat indefinitely. The sharper identifying restriction is timing: the regulated response should precede any broader decline among unregulated compounds. The event-study estimates satisfy that restriction.

The event studies therefore show that state MCLs moved the drinking-water exposure margin. Regulated PFAS fell by economically large amounts, with a lag consistent with treatment installation and with an early response concentrated in the chemicals named by the rules. The health analysis below asks whether those exposure changes translate into infant-health gains, and whether the gains survive when hydrologic spillovers cross the regulatory boundary.

6 Infant health effects: border-county estimates

Having established that state MCLs reduced regulated PFAS concentrations, I next ask whether those exposure changes appear in birth outcomes. The health estimates use adjacent counties on opposite sides of treated-control state borders. Border-pair-by-year fixed effects compare counties in the same local border region and year, absorbing shocks common to both sides of the border. County fixed effects absorb time-invariant differences in baseline health, industrial history, water-system structure, and demographic composition.

A standard border design would pool all adjacent treated-control county pairs. In this setting, pooling is not innocuous. PFAS can cross state lines through connected drainage systems, so the legal treatment boundary need not be an exposure boundary. I therefore begin with the pooled border-pair estimate and then split the sample by hydrologic connectivity.

6.1 Pooled border-pair estimates

Table 3 reports the pooled estimates from equation (7). The sample contains 146 county-year observations after imposing county fixed effects, border-pair-by-year fixed effects, outcome availability, and control availability. Point estimates have the expected

signs for low birth weight and mean birth weight, but are imprecise. MCL adoption lowers the low-birth-weight rate by 0.212 percentage points ($p = 0.170$) and raises mean birth weight by 8.55 grams ($p = 0.380$). Gestation changes by -0.016 weeks ($p = 0.617$).

Table 3: Pooled border-pair DD on infant health outcomes.

Outcome	$\hat{\beta}$	SE	p	N
Low-birth-weight rate (pp)	-0.212	0.143	0.170	146
Mean birth weight (g)	$+8.55$	9.31	0.380	146
Gestation (weeks)	-0.016	0.031	0.617	146

Notes: The table reports OLS estimates of equation (7) on the pooled border-pair sample. The treatment is the post-MCL indicator. All specifications include county fixed effects, border-pair-by-year fixed effects, maternal age, smoking during pregnancy, $PM_{2.5}$, and the state-year count of non-MCL PFAS policies. Standard errors are clustered by state. Low birth weight is reported in percentage points.

Small pooled effects are consistent with the exposure model. Some border pairs compare counties that drain to different terminal basins; in those pairs, contamination from the control county cannot mechanically enter the treated county through the same drainage system. Other pairs place the treated county downstream of an unregulated control county; in those pairs, upstream contamination can continue to cross the state line after MCL adoption. The pooled coefficient therefore averages clean local treatment effects with attenuated downstream effects.

6.2 Hydrologic heterogeneity

Hydrologic connectivity identifies where the state border is also an approximate exposure boundary. Figure 2 maps the treated-control border pairs and classifies them by drainage relation. Table 4 then estimates equation (7) separately for all pairs, different-basin pairs, and same-basin pairs in which the treated county lies downstream.

Health gains are concentrated in different-basin border pairs. In that subsample, MCL adoption lowers the low-birth-weight rate by 0.484 percentage points ($p = 0.025$). Relative to the sample mean of 8.04 percent, the estimate implies a 6.0 percent decline. Mean birth weight rises by 16.82 grams, though the estimate is less precise ($p = 0.336$). Gestation increases by 0.017 weeks ($p = 0.687$). I treat low birth weight as the primary health outcome because it is the sharpest estimate and because the sign is consistent with the birth-weight point estimate.

No comparable improvement appears when the treated county lies downstream in the same basin. In that subsample, MCL adoption changes low birth weight by +0.049

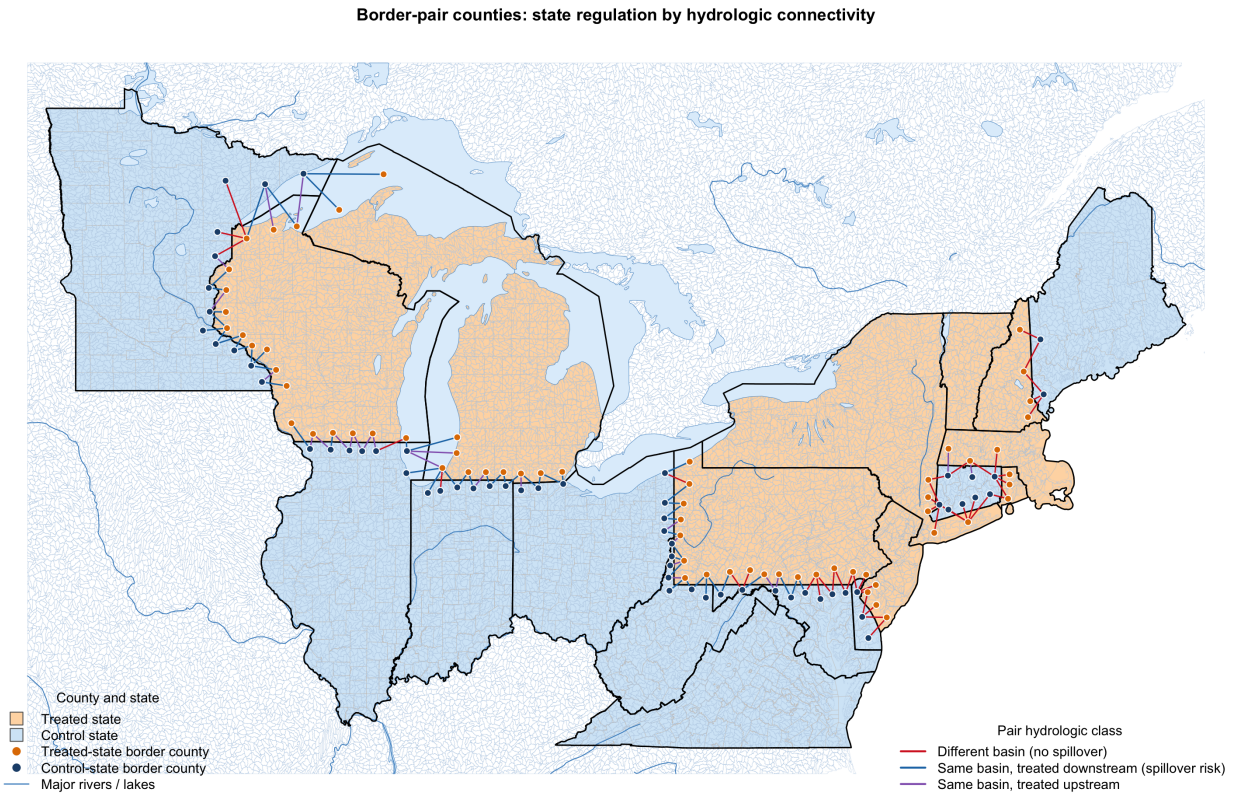


Figure 2: Cross-state border-pair counties and hydrologic connectivity. Treated states are shaded light orange and control states light blue. Border-pair counties are marked by filled circles. Connecting lines classify pairs by HydroBASINS Level-12 drainage relation: different terminal basins, same basin with the treated county downstream, or same basin with the treated county upstream.

Table 4: Border-pair DD by hydrologic class.

	All pairs		Different basin		Same basin, treated downstream	
Outcome	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p
Low-birth-weight rate (pp)	−0.212	0.170	−0.484**	0.025	+0.049	0.915
	(0.143)		(0.181)		(0.441)	
Mean birth weight (g)	+8.55	0.380	+16.82	0.336	−2.35	0.795
	(9.31)		(16.54)		(8.65)	
Gestation (weeks)	−0.016	0.617	+0.017	0.687	−0.062	0.431
	(0.031)		(0.042)		(0.073)	
N (county-years)	146		66		66	

Notes: The table reports OLS estimates of equation (7). The treatment is the post-MCL indicator. All specifications include county fixed effects, border-pair-by-year fixed effects, maternal age, smoking during pregnancy, $PM_{2.5}$, and the state-year count of non-MCL PFAS policies. Standard errors clustered by state are in parentheses. Low birth weight is reported in percentage points. $^{**}p < 0.05$.

percentage points ($p = 0.915$), mean birth weight by -2.35 grams ($p = 0.795$), and gestation by -0.062 weeks ($p = 0.431$). Precision is limited, but the contrast with different-basin pairs is clear: the same legal treatment produces health gains only where the control county is not hydrologically upstream of the treated county.

A pooled interaction specification gives the same message within a single equation. Estimating equation (8) yields a different-basin interaction of +32.4 grams for birth weight ($p = 0.043$), +0.125 weeks for gestation ($p = 0.072$), and -0.404 percentage points for low birth weight ($p = 0.114$). The same-basin baseline coefficients are -13.3 grams for birth weight ($p = 0.023$), -0.100 weeks for gestation ($p < 0.001$), and +0.061 percentage points for low birth weight. Although parameterized differently from Table 4, the interaction estimates again show a more favorable reduced form when the state border separates drainage systems.

Hydrologic class is not randomly assigned, and the estimates should not be read as if it were. Its role is to determine what the border comparison measures. In different-basin pairs, the policy changes the local treatment environment without mechanically importing upstream control-side contamination. In same-basin downstream pairs, the state MCL reduces only the in-state component of exposure; upstream PFAS remains outside the treated state's jurisdiction. Selection may also matter if downstream counties were precisely the places where contamination was severe enough to make regulation politically salient. Both forces predict weaker reduced-form effects downstream.

6.3 Geographic regression-discontinuity evidence

Spatial gradients around treated-control borders provide a second check on the same mechanism. The geographic RDD-DD specification uses signed distance to the nearest treated-control border as the running variable. The coefficient on $D_c \times Post_t$ measures the post-policy discontinuity at the border. The coefficient on $D_c \times r_c$ measures the difference in post-policy distance gradients on the treated side relative to the control side. Table 5 reports estimates using the headline 200 km bandwidth.

Border discontinuities are small. In the pooled sample, the level-jump estimates are +0.25 percentage points for low birth weight, -15.79 grams for birth weight, and +0.033 weeks for gestation; none is statistically different from zero. A muted discontinuity is not surprising. Counties immediately adjacent to the border may share hydrologic systems, and county centroids are coarse measures of water-system exposure.

Distance gradients are more informative. In the pooled sample, the treated-side slope difference is -0.030 percentage points per kilometer for low birth weight ($p = 0.021$) and +1.62 grams per kilometer for birth weight ($p = 0.025$). Gestation shows no comparable gradient. A 50 km movement farther into the treated state, relative to the

Table 5: Geographic RDD-DD estimates on level outcomes.

	All pairs		Different basin		Same basin, downstream	
Coefficient	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p
Level jump: $D_c \times Post_t$						
Low birth weight (pp)	+0.25	0.484	+0.46	0.270	−0.11	0.749
Mean birth weight (g)	−15.79	0.222	−24.39	0.139	+3.84	0.864
Gestation (weeks)	+0.033	0.276	−0.010	0.818	+0.113*	0.087
Slope difference: $D_c \times r_c$						
Low birth weight (pp/km)	−0.030**	0.021	−0.039	0.148	+0.013	0.732
Mean birth weight (g/km)	+1.62**	0.025	+2.37	0.143	−0.20	0.895
Gestation (weeks/km)	+0.0021	0.451	+0.0054	0.190	−0.0036	0.313
N (county-years)	365		162		132	

Notes: The table reports local-linear geographic RDD-DD estimates from equation (10) using a 200 km bandwidth. D_c indicates the treated side of the border and r_c is signed distance to the treated-control border. The level-jump coefficient is the post-policy discontinuity at the border. The slope-difference coefficient is the treated-side distance gradient minus the control-side distance gradient. Specifications include year fixed effects, maternal age, smoking during pregnancy, $PM_{2.5}$, and the state-year count of non-MCL PFAS policies. Standard errors are clustered by state. $^{*}p < 0.10$, $^{**}p < 0.05$.

same movement on the control side, corresponds to a 1.5 percentage-point lower low-birth-weight rate and an 81 gram higher mean birth weight. These are spatial gradients, not household-moving effects, but their signs match the border-pair estimates.

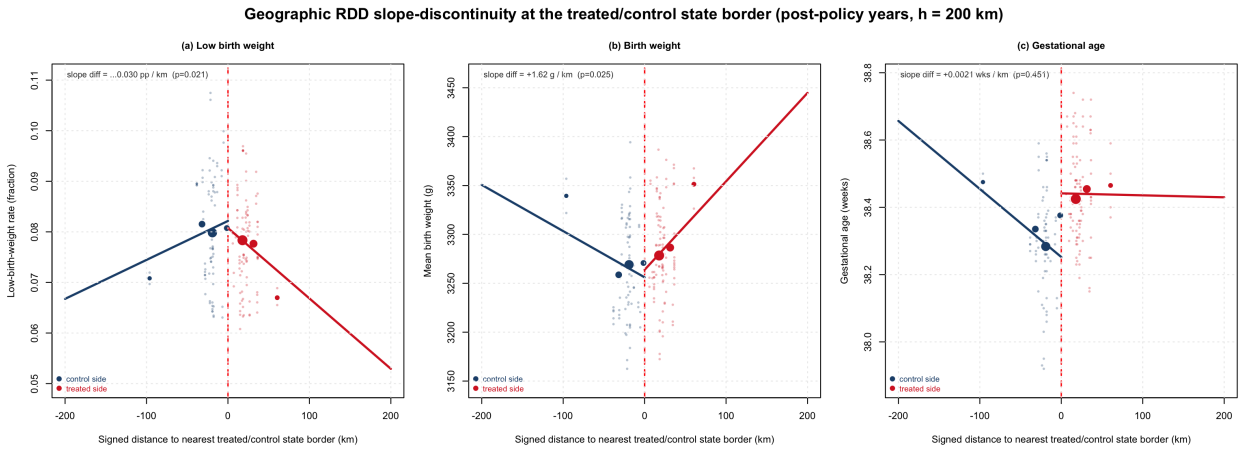


Figure 3: Geographic RDD slope differences at treated-control state borders. The figure plots post-policy county-year outcome means in signed-distance bins around treated-control state borders. Negative distances are on the control side; positive distances are on the treated side. Lines are local-linear fits on each side of the border. Marker size is proportional to the square root of cell counts.

Hydrologic heterogeneity appears in the RDD as well. For low birth weight, the

slope difference is −0.039 percentage points per kilometer in different-basin pairs and +0.013 in downstream pairs. For birth weight, the corresponding estimates are +2.37 grams per kilometer and −0.20 grams per kilometer. The RDD is not the primary design, because county centroids and county-level outcomes are coarse relative to water-system service areas. Its value is diagnostic: the post-policy spatial gradients move in the same direction as the hydrologic split.

6.4 Magnitudes

Different-basin estimates are large relative to baseline birth outcomes. A 0.484 percentage-point reduction in low birth weight amounts to a 6.0 percent decline relative to the sample mean of 8.04 percent. The associated 16.82 gram increase in mean birth weight is about 0.5 percent of the mean birth weight of 3,287 grams and 0.18 standard deviations of the cross-county distribution of mean birth weight. Because the birth-weight estimate is imprecise, inference rests primarily on low birth weight, with birth weight and gestation used as supporting outcomes.

RDD magnitudes are best read as gradients rather than average treatment effects. Moving 50 km farther into the treated state is associated with a 1.5 percentage-point lower low-birth-weight rate and an 81 gram higher mean birth weight relative to the control-side gradient. The stronger gradient than border jump is consistent with attenuation at the state line and cleaner exposure farther inside the treated-state interior.

Overall, the border-pair and RDD estimates point to the same regulatory interpretation. State PFAS MCLs generate infant-health gains when the treated jurisdiction controls the relevant exposure pathway. Benefits are not detectable when the treated county remains downstream of an unregulated control county. For this contaminant, the effective policy boundary is the overlap between the state line and the drainage basin.

7 Rainfall and the hydrologic mechanism

Annual rainfall provides a physical test of the hydrologic mechanism. If state MCLs affect infant health through drinking-water exposure, then weather shocks that change PFAS transport should also change the reduced form. More importantly, the sign of that moderation should differ across the hydrologic classes used in Section 6.

The prediction follows from the exposure environment. In different-basin border pairs, the treated county is not mechanically exposed to upstream contamination from the control county. Wet years can instead raise local source-water loading by mobilizing residual PFAS from contaminated soils, landfills, industrial sites, and AFFF training areas into runoff. Under this channel, rainfall should weaken the fetal-growth gains

from MCL adoption. In same-basin downstream pairs, by contrast, part of the treated county's exposure comes from upstream contamination on the unregulated side of the border. Higher streamflow can dilute upstream PFAS concentrations in the shared drainage system. Rainfall should therefore reduce the downstream spillover penalty. The mechanism predicts a sign reversal: wetter years should attenuate the policy effect in different-basin pairs and make it more favorable in same-basin downstream pairs.

I measure rainfall as annual precipitation standardized within county:

$$R_{ct} = \frac{\text{Rain}_{ct} - \overline{\text{Rain}_c}}{\sigma_c}.$$

Within-county standardization removes long-run climate differences across counties and identifies effects from deviations from a county's own rainfall distribution. Coefficients are therefore interpreted as changes in the MCL effect associated with a one-standard-deviation wetter year.

7.1 Mobilization in different-basin pairs

I first estimate the treatment-by-rainfall interaction in equation (9) on different-basin border pairs. In this subsample, runoff mobilization predicts negative coefficients for birth weight and gestation, and a positive coefficient for low birth weight. Because the hydrologic subsample is small, Table 6 focuses on the continuous fetal-growth outcomes.

Table 6: Rainfall moderation in different-basin pairs.

Outcome	$\hat{\beta}_2$ on MCL $\times R$	SE	p
Birth weight (g)	−31.64	18.30	0.118
Log birth weight (×100)	−1.97	1.03	0.122
Gestational age (weeks)	−0.118	0.078	0.165
Log gestation (×100)	−0.31	0.21	0.169

Notes: The table reports estimates of the treatment-by-rainfall interaction in equation (9), restricted to different-basin border pairs. Rainfall is standardized within county. All specifications include county fixed effects, border-pair-by-year fixed effects, maternal age, smoking during pregnancy, $PM_{2.5}$, and the state-year count of non-MCL PFAS policies. Standard errors are clustered by state.

All four coefficients have the predicted sign. A one-standard-deviation wetter year reduces the MCL effect on mean birth weight by 31.64 grams, on log birth weight by 1.97 percent, on gestation by 0.118 weeks, and on log gestation by 0.31 percent. None reaches conventional significance under the pair-by-year specification; the birth-weight interaction has $p = 0.118$. Precision is limited by the 66 county-year subsample and by

the demanding fixed-effect structure. Still, the direction of the estimates is consistent across every fetal-growth outcome.

The magnitude is economically meaningful. The different-basin estimate in Table 4 implies an average birth-weight gain of about 17 grams. The rainfall interaction in Table 6 implies that a one-standard-deviation wetter year more than offsets that gain. This is not evidence that rainfall permanently eliminates the policy effect. Rather, wet years raise PFAS loading into source water relative to normal operating conditions. Granular activated carbon and ion exchange remove PFAS after it enters the treatment system; they do not prevent new pulses of legacy contamination from being washed into source water.

7.2 Dilution in same-basin downstream pairs

Shared-basin downstream pairs imply the opposite interaction. If the treated county receives part of its exposure from an upstream control county, higher flow can dilute upstream concentrations and reduce the spillover penalty. I test this prediction with a pooled triple interaction between the post-MCL indicator, same-basin status, and standardized rainfall.

Table 7: Rainfall moderation by hydrologic connectivity.

Outcome	$\hat{\beta}_3$ on MCL $\times D^{\text{same}} \times R$	SE	p
Low birth weight (pp)	−0.027	0.071	0.715
Log low birth weight (×100)	−0.04	0.87	0.962
Birth weight (g)	+14.21*	7.18	0.076
Log birth weight (×100)	+0.44*	0.22	0.079
Gestation (weeks)	+0.038	0.029	0.221
Log gestation (×100)	+0.10	0.076	0.218
N (county-years)	146		

Notes: The table reports the triple interaction between the post-MCL indicator, same-basin status, and standardized rainfall. The omitted hydrologic category is different-basin pairs. All specifications include county fixed effects, border-pair-by-year fixed effects, maternal age, smoking during pregnancy, $PM_{2.5}$, and the state-year count of non-MCL PFAS policies. Standard errors are clustered by state. $^*p < 0.10$.

The fetal-growth interactions are positive. Relative to different-basin pairs, a one-standard-deviation wetter year in same-basin pairs raises the MCL effect on mean birth weight by 14.21 grams ($p = 0.076$), raises log birth weight by 0.44 percent ($p = 0.079$), and raises gestation by 0.038 weeks ($p = 0.221$). The low-birth-weight interaction is small and imprecise. For the continuous outcomes, however, the estimates move in the direction predicted by dilution of upstream contamination.

The dilution estimates are smaller than the mobilization estimates. For birth weight, the same-basin rainfall interaction is about 14 grams, whereas the different-basin mobilization interaction is about 32 grams in absolute value. Equality is not implied by the mechanism. Legacy PFAS stocks in soils and contaminated sites may accumulate over many years and be released in pulses during wet periods. Dilution, by contrast, changes upstream concentrations contemporaneously with flow. The relevant test is therefore the sign reversal, not equality of magnitudes.

7.3 Interpretation

Rainfall strengthens the hydrologic interpretation of the border results. Selection into state regulation can explain why early-adopting states differ from later adopters or non-adopters. It does not naturally predict that rainfall should weaken the MCL effect in hydrologically isolated pairs while improving it in connected downstream pairs. That sign pattern comes from the physical movement of PFAS through runoff and streamflow.

These estimates should not be read as standalone proof of the mechanism. Inference is limited by the small border-pair sample, and several rainfall coefficients are only moderately precise under the pair-by-year specification. The point estimates nevertheless line up across outcomes and across hydrologic classes. They also clarify why the pooled border estimate is small: pooling combines places where wet years mobilize local legacy contamination with places where wet years dilute upstream contamination.

A robustness check reaches the same conclusion. Adding annual rainfall as a level control to the different-basin border-pair regression leaves the low-birth-weight result intact and slightly larger in magnitude. The coefficient becomes -0.498 percentage points ($p = 0.004$), compared with -0.484 percentage points ($p = 0.025$) in Table 4. Cross-county rainfall differences or contemporaneous weather shocks are therefore unlikely to explain the hydrologic split.

8 Discussion

These estimates should be read as reduced-form effects of drinking-water regulation rather than as biological dose-response estimates. State MCLs change measured PFAS concentrations in public water systems, but maternal exposure also reflects food, indoor dust, occupational sources, private wells, bottled-water use, and the persistence of PFAS in serum. The coefficient therefore captures the policy-relevant effect of regulating the public-water exposure margin available to state governments.

On that margin, the first stage is large. By the second post-adoption year, the regulated PFAS6 hazard index falls by 3.82 index points per unit of stringency. For a state

with average treated-state stringency, this corresponds to a decline of about 1.57 index points, close to the mean regulated PFAS6 hazard index in the merged county-year panel. Concentration-based estimates tell the same story in physical units. Equally important, the timing matches the compliance process: no adoption-year response, followed by declines in years one through three.

Infant-health effects appear only where the border comparison corresponds to the exposure model. In different-basin pairs, MCL adoption reduces the low-birth-weight rate by 0.484 percentage points, a 6.0 percent decline relative to the sample mean. Mean birth weight also rises by about 17 grams, though that estimate is less precise. I therefore treat low birth weight as the main health outcome and use birth weight and gestation as supporting outcomes.

Hydrologic geography changes what the border estimate means. When adjacent counties drain to different terminal basins, the state border is close to an exposure boundary: contamination from the control county cannot mechanically flow into the treated county through the same drainage system. When the treated county lies downstream in a shared basin, the legal treatment changes at the border but exposure need not. PFAS can continue to enter from the unregulated upstream side. Null or wrong-signed estimates in that subsample therefore do not imply that treatment technologies fail. They imply that state-level regulation is incomplete when the exposure pathway crosses the state line.

Rainfall provides a useful check on this interpretation. Wetter years weaken the fetal-growth benefit of MCL adoption in different-basin pairs, consistent with runoff mobilizing residual PFAS from contaminated soils and source areas into local water supplies. In same-basin pairs, the rainfall interaction has the opposite sign for birth weight, consistent with dilution of upstream concentrations. Not all rainfall estimates are precise, but the sign reversal is difficult to reconcile with a simple story about state policy selection. It follows from the physical movement of water.

Several limitations remain. The health analysis relies on a small border-pair sample, especially after splitting by hydrologic class. The different-basin low-birth-weight estimate is statistically significant and robust to adding rainfall controls, but birth-weight and gestation estimates are less precise. Hydrologic classification is also measured at the county-centroid level, while actual exposure depends on public water-system service areas, surface-water intakes, groundwater sources, and engineered transfers. Such measurement error should blur differences across hydrologic classes, but it remains an important limitation.

Nor can the estimates separately identify mechanical spillover from selection on severe contamination. Same-basin downstream counties may show weaker effects because upstream contamination physically enters the treated county. They may also be

places where contamination was more severe or more persistent before regulation. The empirical claim is narrower: the reduced form is strongest where the exposure model predicts the state policy should be locally contained, and it weakens where the treated county remains hydrologically connected to an unregulated upstream neighbor.

For policy, the main lesson is that jurisdictional scale matters. State MCLs can reduce regulated PFAS in public water systems and improve infant health when the state controls the relevant exposure pathway. They are less effective when contamination is transported across state borders. In shared drainage basins, the relevant regulatory unit is not the state alone. It is the watershed, or a federal standard that applies to upstream and downstream systems at the same time.

9 Conclusion

State PFAS drinking-water standards reduced regulated PFAS concentrations before federal regulation became binding. The response appears after a one-year compliance lag, peaks in the second post-adoption year, and is concentrated in the chemicals targeted by state rules. This pattern is consistent with treatment-system installation rather than contemporaneous changes in reporting or monitoring.

Hydrologic geography determines whether those exposure reductions translate into detectable health gains. In border pairs that drain to different terminal basins, MCL adoption reduces the low-birth-weight rate by 0.484 percentage points and raises mean birth weight by about 17 grams. In same-basin pairs where the treated county lies downstream of an unregulated control county, the estimates are null or wrong-signed. Geographic RDD estimates and rainfall interactions point to the same interpretation: the policy effect is strongest when the legal boundary also approximates an exposure boundary.

Unilateral state regulation has an inherent limit in this setting. A state can require its own public water systems to treat PFAS, but it cannot fully control upstream contamination originating outside its jurisdiction. For PFAS, the relevant policy boundary is therefore not only the state line. It is the overlap between the state line and the watershed.

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A Robustness and data construction

This appendix reports supporting specifications and data-construction checks. Section A.1 documents the sample restrictions used in dynamic specifications. Section A.2 discusses alternative treatments of left-censored PFAS measurements. Section A.3 summarizes estimator choice. Section A.4 explains why the paper reports reduced-form

health effects rather than county-year two-stage least squares estimates. Sections A.5–A.7 report health robustness checks. Sections A.8–A.9 provide additional first-stage, placebo, mechanism, and data-quality figures.

A.1 Sample restrictions

Dynamic specifications use the clean-cohort sample. This restriction drops New Jersey's 2018 PFNA standard because the available monitoring data provide no clean pre-period, and it drops Pennsylvania's 2023 adoption because the sample contains too little post-period support. Static specifications use all nine adopting states. The qualitative first-stage pattern is unchanged when the full adopting-state sample is used in static regressions.

A.2 Non-detect handling

The PFAS monitoring data contain extensive left censoring, and reporting limits vary across chemicals, states, laboratories, methods, and years. The headline regressions use censored maximum likelihood under a log-normal model because it accommodates observation-specific reporting limits and preserves a complete county-by-chemical-year panel. Descriptive exposure statistics in Table 2 use Kaplan–Meier imputation because it is non-parametric.

The first-stage event-study pattern is stable across alternative non-detect treatments. Kaplan–Meier imputation, regression on order statistics, and censored maximum likelihood all produce a post-adoption decline in regulated PFAS beginning after the adoption year. A simple threshold rule that sets values below 4 ng/L to zero produces smaller estimates because the cutoff lies above many post-treatment regulated PFAS observations. Figure A6 reports yearly regulated-PFAS trends by imputation method.

A.3 Estimator choice

The main first-stage specification uses the Sun–Abraham staggered-adoption estimator. Callaway–Sant'Anna group-time average treatment effects give similar qualitative dynamics but are less stable in the small state-cohort panel. The Borusyak–Jaravel–Spiess imputation estimator produces large cohort-driven pre-period outliers in this application. For that reason, the Sun–Abraham estimator is used as the headline dynamic specification.

A.4 Two-stage least squares

Instrumental-variables estimates are not used as headline estimates. Specifications that treat the regulated PFAS6 hazard index as endogenous and instrument it with state stringency interacted with the post-MCL indicator produce Kleibergen–Paap first-stage F statistics below conventional weak-instrument thresholds across non-detect treatments and control sets. This weak first stage is consistent with measurement error in county-year exposure, imperfect mapping from public water systems to counties, and sparse pre-UCMR5 monitoring. The paper therefore emphasizes reduced-form border-pair estimates.

A.5 Health event study

Figure A1 reports the staggered-adoption health event study that motivates the border-pair design. The specification parallels the first-stage event study, but the dependent variables are county-year birth outcomes. Event-time coefficients are imprecise and do not show a stable post-treatment pattern. This result is consistent with the framework: a pooled health event study averages across hydrologically isolated counties and downstream connected counties, and it is more exposed to state-level selection into adoption.

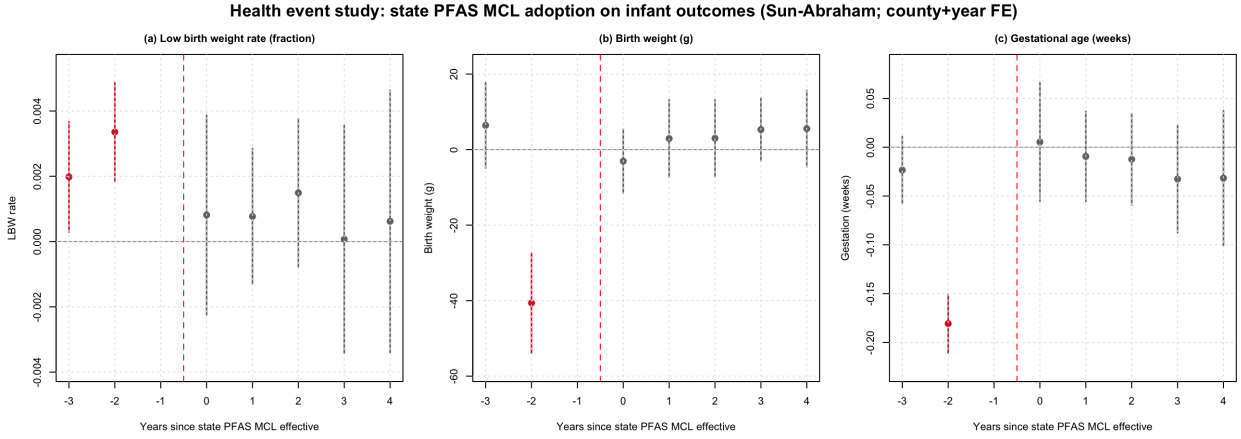


Figure A1: Health event study. The figure reports Sun–Abraham event-study estimates for low birth weight, mean birth weight, and gestational age. The sample, fixed-effect structure, and clean-cohort restriction parallel the first-stage event study. The pooled health event study is imprecise, which motivates the border-pair design with hydrologic heterogeneity.

A.6 Log outcomes

Table A1 reports the border-pair specification using log outcomes. The log low-birth-weight estimate in different-basin pairs is -6.43 percent ($p = 0.038$), close to the level

estimate of −0.484 percentage points relative to the sample mean low-birth-weight rate of 8.04 percent. Log birth weight and log gestation have the same qualitative pattern as the level outcomes but are imprecise.

Table A1: Border-pair DD: log outcomes.

	All pairs		Different basin		Same basin, downstream	
Outcome (×100)	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p
Log low birth weight	−2.76	0.185	−6.43**	0.038	+0.32	0.952
	(1.94)		(2.65)		(5.14)	
Log birth weight	+0.258	0.384	+0.514	0.339	−0.080	0.770
	(0.283)		(0.508)		(0.263)	
Log gestation	−0.041	0.624	+0.046	0.682	−0.161	0.430
	(0.082)		(0.108)		(0.190)	
N	146		66		66	

Notes: The table reports the same specification as Table 4, replacing level outcomes with 100 times the natural log of the outcome. All specifications include county fixed effects, border-pair-by-year fixed effects, maternal age, smoking during pregnancy, $PM_{2.5}$, and the state-year count of non-MCL PFAS policies. Standard errors clustered by state are in parentheses. $^{**}p < 0.05$.

First-differenced outcome specifications are not reported as tables because the border-pair sample is small and annual changes in county-level birth outcomes are noisy. These specifications remove much of the cumulative adjustment induced by treatment installation and produce imprecise estimates. The full output is included in the replication archive.

A.7 Rainfall moderation under a lighter fixed-effect structure

Table A2 reports the rainfall-by-treatment interaction in different-basin pairs using county and year fixed effects rather than border-pair-by-year fixed effects. This specification increases the effective sample and uses broader identifying variation, so it is not the headline rainfall specification. The estimates nevertheless have the same sign as the pair-by-year specification in Table 6: wetter years attenuate the fetal-growth benefit of MCL adoption in different-basin counties.

Table A2: Rainfall moderation under county and year fixed effects.

Outcome	$\hat{\beta}$ on MCL $\times$ R	SE	p
Birth weight (g)	-10.36^{***}	3.51	0.010
Log birth weight ($\times 100$)	-4.16^{***}	0.58	< 0.001
Gestation (weeks)	-0.0416^{***}	0.0058	< 0.001
Low birth weight (pp)	$+0.029$	0.025	0.262

Notes: The table reports treatment-by-rainfall interactions for the different-basin subsample using county and year fixed effects. Rainfall is standardized within county. Controls are maternal age, smoking during pregnancy, $PM_{2.5}$, and the state-year count of non-MCL PFAS policies. Standard errors are clustered by state. $N = 162$. $^{***}p < 0.01$.

A.8 Additional first-stage, placebo, and mechanism figures

Figure A2 reports the hazard-index version of the regulated-versus-unregulated first-stage event study. The left panel uses the regulated PFAS6 hazard index; the right panel uses the unregulated PFAS23 hazard index. This is the county-year counterpart to the chemical-stacked concentration event study. The regulated hazard index is flat before adoption and falls after the MCL becomes effective, while the unregulated index does not show the same early post-adoption response.

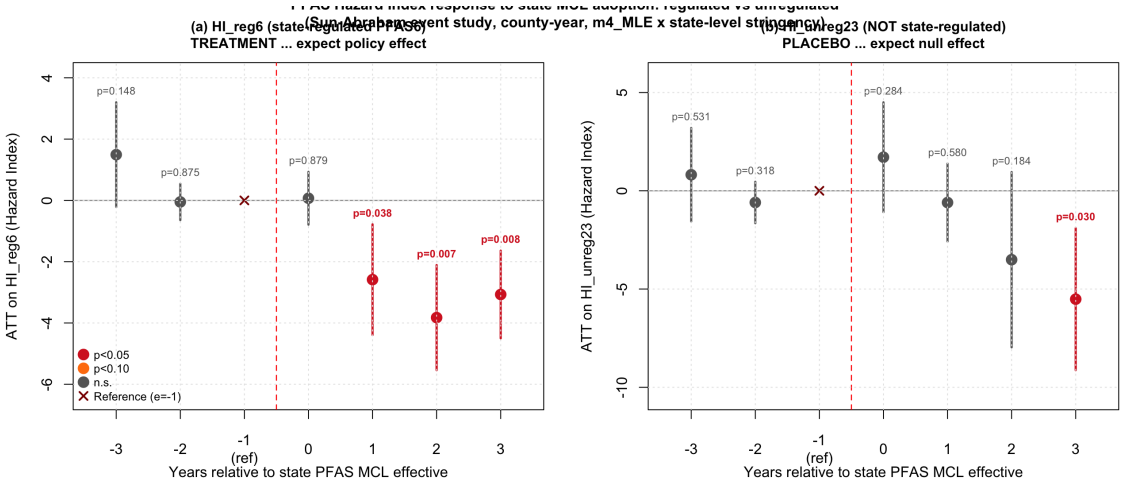


Figure A2: Hazard-index event study for regulated and unregulated PFAS. The figure reports Sun–Abraham event-study coefficients for county-year hazard-index outcomes. The left panel uses the regulated PFAS6 hazard index, and the right panel uses the unregulated PFAS23 hazard index. The omitted event year is $e = -1$. Coefficients are scaled by state-level stringency. Standard errors are clustered by state.

Figure A3 summarizes the rainfall and hydrologic heterogeneity estimates visually. Panel (1) reports the baseline border-pair estimates by hydrologic class. Panel (2) adds

annual rainfall as a level control. Panel (3) reports treatment-by-rainfall interactions in different-basin pairs. Panel (4) reports the pooled triple interaction between treatment, same-basin status, and rainfall. The figure is a visual summary of the mechanism evidence reported in the main text and in Table A2.

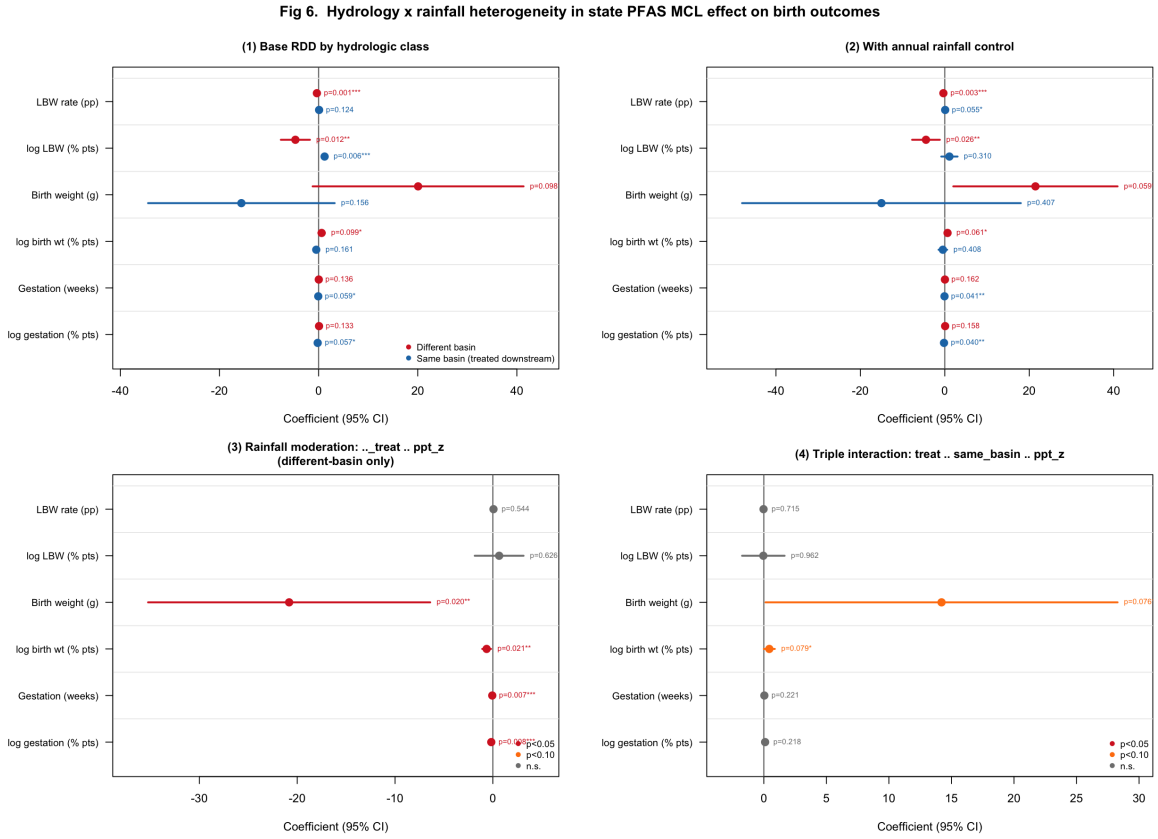


Figure A3: Hydrology and rainfall heterogeneity. The figure summarizes the estimated heterogeneity in the state PFAS MCL effect on birth outcomes. Panel (1) reports baseline border-pair estimates by hydrologic class. Panel (2) adds annual rainfall as a level control. Panel (3) reports the treatment-by-rainfall interaction in different-basin pairs. Panel (4) reports the pooled triple interaction between treatment, same-basin status, and rainfall. Points denote coefficients and horizontal bars denote 95 percent confidence intervals.

Figure A4 reports the chemical-stacked concentration version of the regulated-versus-unregulated placebo. The left panel stacks regulated PFAS6 chemicals; the right panel stacks the 23 unregulated PFAS chemicals. The regulated chemicals decline in the first two post-adoption years, while the unregulated chemicals do not show the same early compliance response.

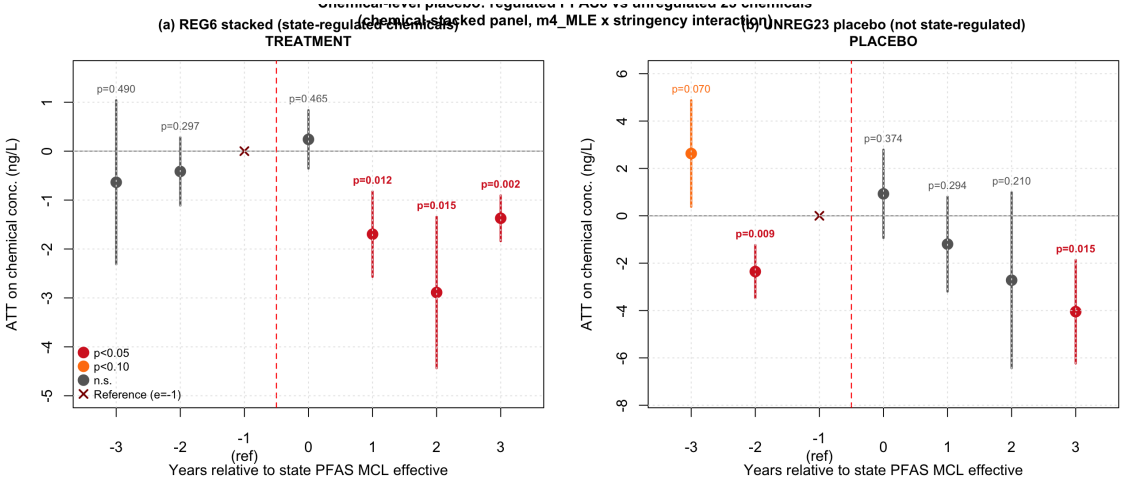


Figure A4: Chemical-stacked first-stage placebo. The figure reports Sun–Abraham event-study coefficients for chemical-stacked concentration outcomes. The left panel stacks regulated PFAS6 chemicals, and the right panel stacks the 23 unregulated PFAS chemicals. The omitted event year is $e = -1$. Coefficients are scaled by chemical-by-state stringency. Standard errors are clustered by state.

A.9 Reporting-limit heterogeneity and non-detect robustness

Figure A5 reports the distribution of reporting limits by PFAS chemical and data source. UCMR5 reporting limits are standardized by chemical, while state monitoring records exhibit substantial variation across laboratories and methods. This heterogeneity motivates the use of censored-data methods that preserve observation-specific reporting limits.

Figure A6 reports yearly mean regulated-PFAS concentrations under alternative non-detect handling rules. The post-adoption decline in regulated PFAS is not driven by a single imputation choice.

A.10 Replication files and data documentation

The replication archive contains the cleaned PFAS monitoring panel, the state policy database, the public water-system crosswalks, the hydrologic classifications, and the code used to reproduce all tables and figures. The archive documents chemical-name harmonization, unit conversion, reporting-limit checks, non-detect imputation, county aggregation, and construction of the state-year policy variables.

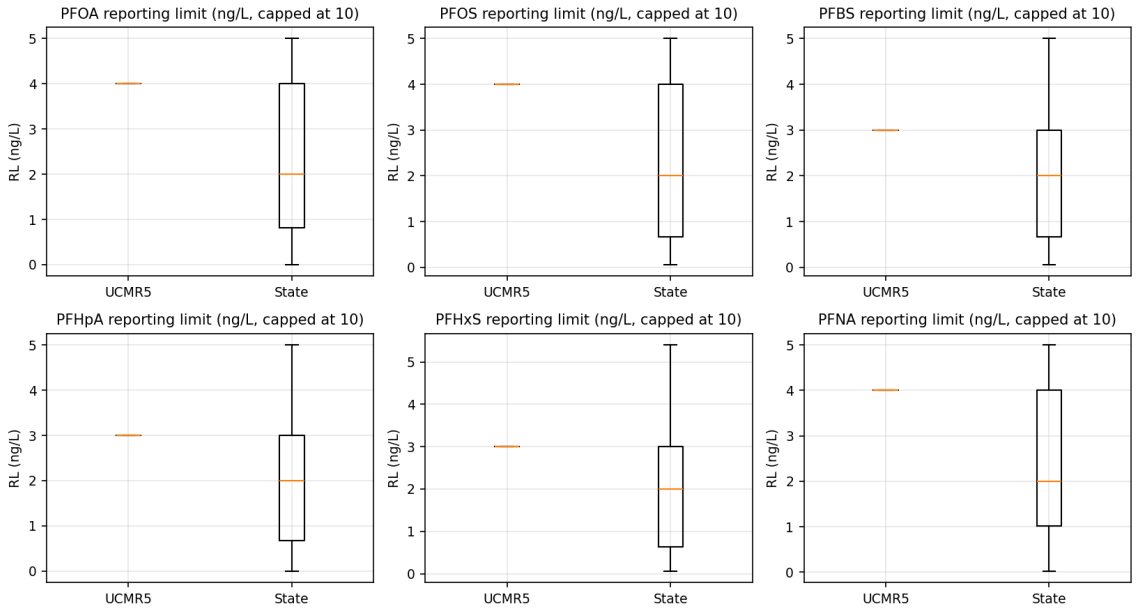


Figure A5: Reporting-limit heterogeneity. The figure reports reporting-limit distributions by PFAS chemical and source. UCMR5 has standardized reporting limits by chemical, while state monitoring data have heterogeneous reporting limits across laboratories, methods, and years.

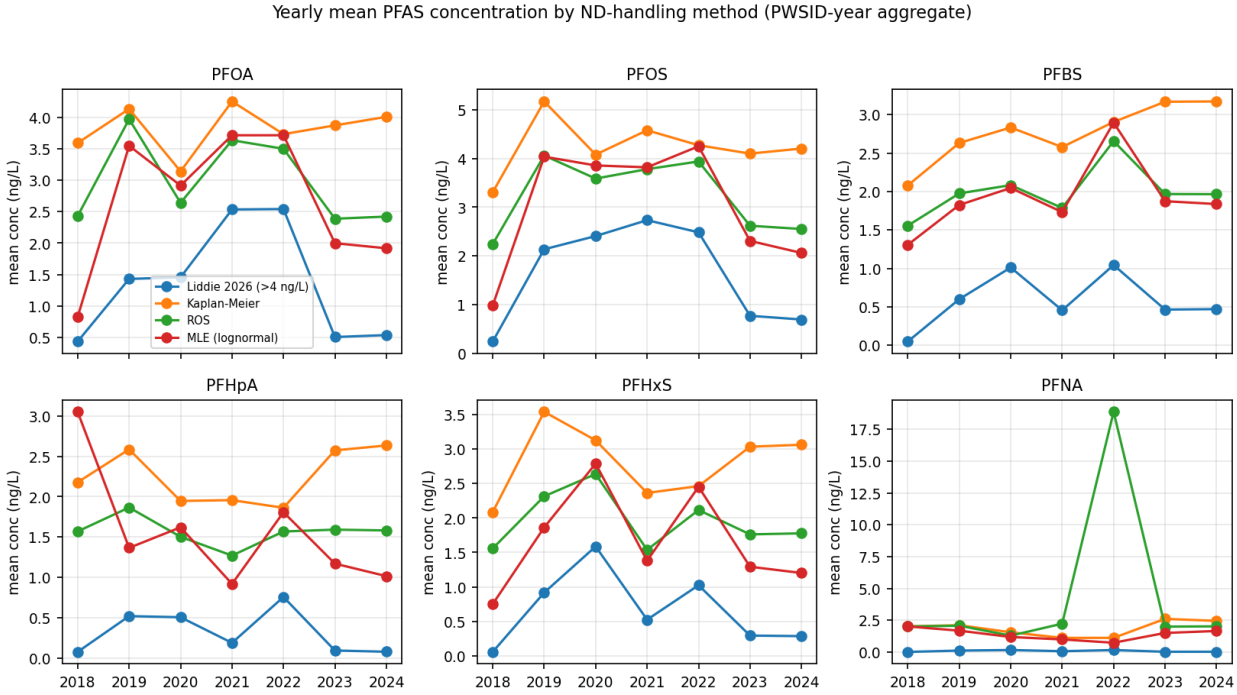


Figure A6: Regulated PFAS trends by non-detect handling method. The figure reports yearly mean concentrations for the regulated PFAS6 group under alternative treatments of left-censored observations.

Persistent Pollutants and Environmental Policy Misallocation: Theory and Calibration from PFAS

Shun Tong

April 2026

Dissertation Chapter 3 — Preliminary Draft

Abstract

Policy for persistent pollutants must allocate spending between cutting emissions and reducing exposure to existing contamination. I study this problem for PFAS. A stock-pollution model yields a sufficient statistic for the marginal return to prevention relative to remediation, separating instrument effectiveness from a horizon term governed by persistence, discounting, and remediation-capital life. Calibration to U.S. PFAS data implies that shifting about $200 million per year toward targeted source control raises annual net benefits by $5–8 billion under central assumptions. State policy evidence is consistent with the model: source-oriented rules reduce upstream releases, while drinking-water standards do not.

Keywords: persistent pollutants, prevention, remediation, sufficient statistic, PFAS, environmental policy, federalism. **JEL:** Q53, Q58, H23, H43, H77, K32.

1 Introduction

Persistent pollutants raise a distinct allocation problem in environmental policy. A regulator can spend resources upstream, preventing new releases, or downstream, reducing exposure to contamination already in the environment. For pollutants that dissipate quickly, current exposure is closely tied to current emissions, and the distinction between prevention and remediation is less important. For stock pollutants, emissions today affect the future stock path, while remediation typically protects particular populations and pathways for a finite period. The policy problem is therefore not only how much pollution to control, but where in the pollutant lifecycle control should occur.

PFAS provide a useful setting for studying this problem. Per- and polyfluoroalkyl substances are a large class of synthetic chemicals whose carbon–fluorine bonds make many compounds resistant to biological, chemical, and photolytic degradation [O'Hagan, 2008, Washington et al., 2010, Cousins et al., 2020]. Public and private commitments to address PFAS contamination are already large. Major settlements include 3M's $10.3 billion agreement and the DuPont/Chemours/Corteva $1.185 billion agreement. EPA estimates annual compliance costs of approximately $1.55 billion for the 2024 National Primary Drinking Water Regulation, and the Department of Defense projects more than $7 billion in cleanup obligations. Annualized identifiable public spending is approximately $3 billion, with most directed toward downstream treatment and cleanup rather than prevention of new releases.

A remediation-heavy allocation need not be inefficient. Inherited contamination can be large, current exposure can be severe, and drinking-water treatment can generate immediate benefits. The relevant question is marginal. Given the current policy mix, does an additional public dollar have a higher social return when spent on source control or on remediation? I answer that question with a stock-pollution model and a calibration to U.S. PFAS policy.

The model considers a planner who allocates resources between source control, denoted F, and remediation, denoted B. Source control reduces the emissions flow; remediation reduces exposure to the existing stock. A local comparison of marginal benefits yields the sufficient statistic

$$\mathcal{R} \equiv \frac{MB_F}{MB_B} = \underbrace{\frac{(1-m(B))g'(F)}{m'(B)S}}_{\text{relative effectiveness}} \cdot \underbrace{\frac{r}{(r+\delta)(1-e^{-rL})}}_{\text{relative horizon multiplier}} . \tag{1}$$

The first term compares the exposure reduction generated by the marginal dollar of source

control with the exposure reduction generated by the marginal dollar of remediation. The second term compares the payoff horizon of avoided emissions with the service life of remediation capital. Persistence enters through the dissipation rate δ. Holding relative effectiveness fixed, lower dissipation raises the return to prevention relative to remediation.

Equation (1) is a local condition, not a claim that prevention should replace remediation. As source-control spending rises, its marginal product may fall. As remediation spending falls, the marginal value of additional treatment may rise. An interior optimum equalizes marginal returns. The statistic identifies the direction of reallocation at a given allocation: if $\mathcal{R} > 1$, the marginal dollar has a higher social return upstream; if $\mathcal{R} < 1$, the marginal dollar has a higher return downstream.

Figure 1 illustrates the horizon component of the statistic. With $r = 0.03$ and $L = 30$ years, the relative horizon multiplier rises as δ falls. The figure is intended to show the comparative static; the pollutant labels should be read as visual anchors rather than parameter estimates.

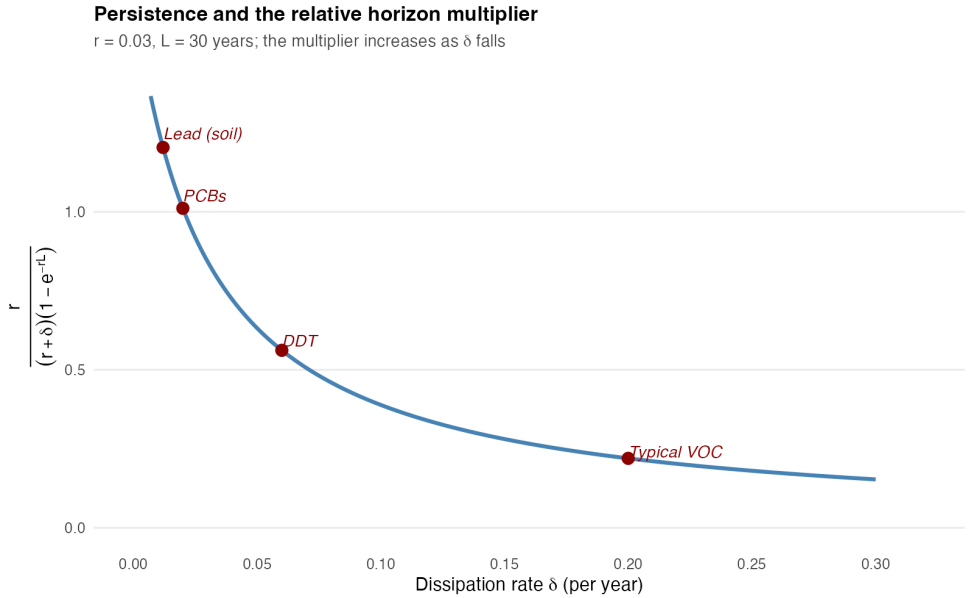


Figure 1: Persistence and the Relative Horizon Multiplier

Notes. The figure plots $r/[(r+\delta)(1-e^{-rL})]$ as a function of the dissipation rate δ, evaluated at $r = 0.03$ and $L = 30$ years. Reference pollutants are approximate visual anchors.

Private and decentralized decisions need not reproduce the planner's allocation. Source-control benefits are delayed, durable, and spatially diffuse. Firms may internalize only the portion of future damages that survives bankruptcy, restructuring, mergers, or in-

complete successor liability [Boomhower, 2019]. State regulators may internalize only damages borne within their own jurisdictions, even when emissions, contaminated products, or waterborne pollutants cross borders [Sigman, 2014, Gibson, 2019]. These wedges affect prevention more strongly than remediation because remediation benefits are often immediate, observable, and geographically concentrated.

The empirical exercise takes the sufficient statistic to U.S. PFAS data. I combine EPA administrative data on releases, drinking-water contamination, treatment costs, and facility characteristics with state regulatory timing, the Liddie-Schaider-Sunderland community-water-system panel [Liddie et al., 2023], and manually collected information on PFAS-related settlements. The exercise is a calibrated welfare calculation rather than a full structural estimation of all primitives. Its purpose is to discipline the relevant margins, quantify their plausible orders of magnitude, and show which assumptions determine the policy ranking.

Under the baseline assumptions, reallocating approximately $200 million per year from remediation spending beyond the drinking-water saturation margin toward targeted source control raises annual net social benefits by about $8 billion. A more conservative specification with a lower ongoing-flow share yields a gain of about $5 billion. The welfare gain remains positive in most sensitivity exercises, although its magnitude depends on the aggregate damage anchor, the share of future damages attributable to ongoing emissions, and the marginal return to source-control interventions.

Contributions. The paper makes three contributions. First, it provides a sufficient-statistic approach to the prevention–remediation allocation problem for stock pollutants. The statistic separates relative instrument effectiveness from the horizon effect generated by persistence. That separation is useful because policies that look similar for flow pollutants can have different welfare rankings for persistent pollutants.

Second, the paper links the allocation problem to liability and federalism. Finite liability horizons and jurisdictional spillovers reduce the private or decentralized return to source control relative to its social return. These wedges provide an economic explanation for why remediation-heavy allocations can arise even when the marginal social return to prevention is high.

Third, the paper provides a quantitative application to PFAS. The calibration combines emissions data, drinking-water monitoring, treatment-cost estimates, state policy timing, and settlement information. The results indicate that the current identifiable U.S. PFAS policy mix is strongly downstream-oriented and that modest reallocation toward targeted source control can generate large welfare gains under central assumptions. Reduced-form

evidence from state regulation is consistent with the distinction between source-oriented and downstream policies, but I interpret it as supportive rather than causal evidence.

Related literature. The paper relates to four strands of work. First, a large literature estimates the health and economic damages of pollution exposure, including local pollution shocks and changes in environmental quality [Currie et al., 2015, Currie and Walker, 2011, Deryugina et al., 2019]. I take damage estimates from this literature and ask how policy spending should be allocated across points in the pollutant lifecycle.

Second, work on abatement costs and the spatial incidence of environmental policy studies how regulation changes emissions, compliance costs, and welfare across firms and locations [Fowlie, 2010, Holland et al., 2016]. Much of that literature treats pollution reductions as contemporaneous flow changes. For stock pollutants, avoided emissions also change the future exposure path.

Third, research on environmental federalism and spillovers shows how decentralized regulation can underprovide pollution control when damages cross jurisdictions [Sigman, 2014, Gibson, 2019]. Persistence strengthens this concern because benefits from prevention accrue over long horizons and across space.

Finally, the paper connects to the theory of stock pollutants and discounting [Keeler et al., 1972, Plourde, 1972, Falk and Mendelsohn, 1993, Weitzman, 1998, 2001, Newell and Pizer, 2003, Arrow et al., 2013, Cropper et al., 2014]. The model is deliberately simple, but it isolates a policy-relevant statistic that can be disciplined with administrative, engineering, and settlement data. The empirical application also contributes to the emerging economics literature on PFAS [Marcus and Mueller, 2024, Jacqz et al., 2024, **?**, Obsekov et al., 2023, Cordner et al., 2021].

2 Institutional Background and Stylized Facts

2.1 The Chemistry of PFAS Persistence

PFAS comprise a family of synthetic organofluorine compounds first developed in the 1930s. The defining structural feature is the carbon–fluorine bond, with dissociation energy approximately 18 percent higher than the C–H bond and 40 percent higher than the C–C bond [O'Hagan, 2008]. This makes PFAS resistant to the ambient degradation pathways available to many organic compounds, including biodegradation, hydrolysis, photolysis, and thermal decomposition under environmental conditions [Cousins et al., 2020]. Washington et al. [2010] report PFOA half-life in groundwater above 1,000 years;

Ruyle et al. [2023] show that PFAS precursors at military fire-training sites can continue to generate terminal compounds for more than a century.

2.2 The Composition of Current PFAS Spending

A central stylized fact motivating the paper is the composition of current U.S. public PFAS spending. Table 1 compiles identifiable federal and state expenditures. Of approximately $3.0 billion in annualized public spending, most targets remediation and a small share targets prevention.

Table 1: Identifiable Annual U.S. Public PFAS Spending

Category	Annual spending ($M)	Share	Source
Remediation (back-end):			
NPDWR drinking-water compliance	1,548	51.6%	U.S. Environmental Protection Agency [2024b]
DoD installation cleanup	1,000	33.3%	GAO-25-107401
Superfund / EPA cleanup actions	320	10.7%	EPA CERCLA records
State water-system support	80	2.7%	State DEQ budgets (sample)
Prevention (front-end):			
EPA TRI compliance program	30	1.0%	EPA OECA budget
DoD/FAA AFFF transition program	8	0.3%	GAO-21-421; FAA F3 plan
Federal waste-segregation enforcement	12	0.4%	ECHO RCRA records
Total identifiable	**2,998**	**100%**	
Prevention share		**1.7%**	
Remediation share		**98.3%**	

Notes. Values in 2023 USD. The table reports directly observable public spending. Private capital expenditures on process substitution and emission reductions are not separately observable in administrative records and are excluded. Litigation settlements are excluded as transfers rather than allocative public spending.

A separate question is what candidate prevention interventions cost relative to the damages they avert. Table 2 summarizes three source-control interventions identified in the engineering and policy-analysis literature, together with drinking-water treatment as a remediation benchmark. These are modeled intervention costs and benefits, not observed expenditure flows.

Table 2: Modeled Prevention and Remediation Interventions

Intervention	Annual cost ($M)	Damage averted ($M/yr)	Implied ROI	Source
Prevention:				
TRI emission abatement at major facilities	30	770	25.7×	U.S. Environmental Protection Agency [2020]
AFFF replacement at DoD/FAA installations	2.8	6,029	2,143×	GAO-21-421
Industrial waste segregation	81	7,481	92.6×	Interstate Technology and Regulatory Council [2023]
Remediation:				
Drinking-water GAC under NPDWR	1,548	4,300	2.8×	U.S. Environmental Protection Agency [2024b]

Notes. Costs are annualized in 2023 USD. Damage-averted estimates draw on EPA, GAO, and ITRC sources. These estimates depend on epidemiological dose-response and valuation assumptions; Section 6 reports robustness checks that cap or exclude high-return source-control interventions.

The contrast between Table 1 and Table 2 motivates the empirical exercise. The observed allocation is remediation-oriented, while several candidate source-control interventions appear to have high modeled returns. The question is whether that gap survives transparent sensitivity analysis.

2.3 The Federal and State Regulatory Landscape

Federal PFAS regulation evolved from voluntary phaseouts and health advisories to the binding 2024 NPDWR, which set maximum contaminant levels of 4 ng/L for PFOA and PFOS [U.S. Environmental Protection Agency, 2024a]. State action has been heterogeneous. Some states adopted enforceable drinking-water MCLs before the federal rule; others adopted source-oriented restrictions, especially bans on PFAS-containing aqueous film-forming foam (AFFF) and intentionally added PFAS in selected consumer products. The model predicts different effects across these instruments. Source-oriented regulations should affect upstream emissions, while drinking-water standards should primarily affect downstream exposure.

3 A Model of Prevention and Remediation

3.1 Setup

Time is continuous. Let $S_t \geq 0$ denote the environmental stock of a pollutant. The stock evolves according to

$$\dot{S}_t = e_0 - g(F_t) - \delta S_t, \qquad S_0 \text{ given}, \tag{2}$$

where e_0 is the unregulated emission rate, $F_t \geq 0$ is source-control spending, and $g(\cdot)$ is the abatement function, with $g(0) = 0$, $g'(F) > 0$, and $g''(F) \leq 0$. The parameter $\delta \geq 0$ is the dissipation rate.

Exposure depends on the stock and remediation effort:

$$X_t = (1 - m(B_t))S_t, \tag{3}$$

where $B_t \geq 0$ is remediation spending and $m(\cdot) \in [0, \bar{m}]$ is the fraction of stock exposure removed by remediation, with $m(0) = 0$, $m'(B) > 0$, $m''(B) \leq 0$, and $\bar{m} < 1$. Damages are linear in exposure, $D_t = dX_t$.

The planner's problem is

$$\max_{\{F_t, B_t\}_{t \geq 0}} \int_0^\infty e^{-rt} \left[-d(1 - m(B_t))S_t - F_t - B_t\right] dt \tag{4}$$

subject to equation (2).

3.2 Marginal Returns and the Sufficient Statistic

A local comparison is sufficient for the policy question considered here. The planner need not solve for the entire transition path in order to ask whether the next dollar should be spent upstream or downstream. I therefore evaluate marginal benefits at a current allocation (F, B, S). Throughout the comparison, F and B should be interpreted as annualized dollars at the relevant margin. A marginal dollar of source control lowers the emissions flow; a marginal dollar of remediation purchases treatment or cleanup services that reduce exposure over a finite service life.

Consider first an additional dollar of source control. The spending increment lowers the emissions flow by $g'(F)$. Because the pollutant stock dissipates at rate δ, the stock reduction at horizon t is $g'(F)e^{-\delta t}$. Holding remediation fixed, only the unremediated

share of the stock contributes to exposure. The present value of avoided damages is

$$MB_F(F,B) = d(1-m(B))g'(F)\int_0^\infty e^{-(r+\delta)t}dt = \frac{d(1-m(B))g'(F)}{r+\delta}. \tag{5}$$

Source control therefore has a stock value. It does not merely reduce contemporaneous emissions; it prevents additions to the stock that would otherwise generate exposure over future periods. The relevant horizon is governed by both discounting and dissipation.

Now consider an additional dollar of remediation. Remediation does not affect the emissions flow. It reduces exposure to the existing stock. At stock S, the exposure reduction from the marginal dollar is $m'(B)S$. If the treatment service remains productive for L years, the present value of avoided damages is

$$MB_B(B,S) = dm'(B)S\int_0^L e^{-rt}dt = dm'(B)S\frac{1-e^{-rL}}{r}. \tag{6}$$

Finite service life matters. Drinking-water treatment, filtration media, and cleanup infrastructure require maintenance and replacement. A dollar spent today does not provide permanent protection without future expenditure. Coverage also matters: remediation reduces exposure only for the pathways and populations reached by the technology.

Dividing equation (5) by equation (6) gives

$$\boxed{\mathcal{R} \equiv \frac{MB_F}{MB_B} = \frac{(1-m(B))g'(F)}{m'(B)S}\cdot\frac{r}{(r+\delta)(1-e^{-rL})}.} \tag{7}$$

The first factor is an effectiveness ratio. It compares how much exposure-adjusted damage is avoided by the marginal dollar of source control relative to the marginal dollar of remediation. The second factor is a horizon ratio. It compares the discounted lifetime of avoided emissions with the discounted service life of remediation capital.

An interior optimum equalizes marginal benefits, so $\mathcal{R} = 1$. If $\mathcal{R} > 1$, the current allocation places too little weight on source control at the margin. If $\mathcal{R} < 1$, the current allocation places too little weight on remediation. The condition is local. It does not require corner solutions, and it does not imply that one instrument dominates everywhere. Concavity of $g(\cdot)$ and $m(\cdot)$ implies that reallocation changes the ratio as spending moves across instruments.

Persistence enters the allocation rule through the second factor. Holding F, B, S, r, and L fixed, lower dissipation raises the relative value of source control because avoided emissions remain relevant for longer. This channel is distinct from engineering effectiveness. Two pollutants may have the same abatement technology and the same treatment

technology, yet differ in the optimal policy mix because one pollutant persists longer in the environment.

Proposition 1 (Persistence raises the relative return to prevention)**.** *Holding F, B, S, r, and L fixed, the prevention–remediation ratio $\mathcal{R}$ is strictly decreasing in the dissipation rate δ.*

Proof. Differentiating equation (7) with respect to δ gives

$$\frac{\partial \mathcal{R}}{\partial \delta} = -\frac{\mathcal{R}}{r+\delta} < 0.$$

□

The proposition identifies the channel through which persistence affects policy allocation. A lower dissipation rate increases the present value of emissions that never enter the stock. Remediation may still be valuable, especially when the inherited stock is large. The point is marginal: persistence increases the social return to preventing new additions to the stock relative to treating the stock after exposure occurs.

3.3 Institutional Wedges

Social and private returns differ when future damages are not fully attached to the actors who can prevent emissions. Prevention is especially exposed to this problem because its benefits are delayed and often accrue outside the jurisdiction or firm boundary. Remediation is less exposed because many of its benefits are contemporaneous, observable, and tied to specific water systems or cleanup sites.

Liability horizon. Suppose a firm internalizes prevention benefits only through an effective liability horizon $T_f < \infty$. Its perceived marginal benefit is

$$MB_F^{\text{firm}}(F,B) = d(1-m(B))g'(F)\int_0^{T_f} e^{-(r+\delta)t}dt = \frac{d(1-m(B))g'(F)}{r+\delta}\left(1-e^{-(r+\delta)T_f}\right). \tag{8}$$

Relative to the planner, the firm discounts all benefits after T_f to zero. The social-to-private wedge is

$$W_T(T_f,\delta) = \frac{MB_F}{MB_F^{\text{firm}}} = \frac{1}{1-e^{-(r+\delta)T_f}}. \tag{9}$$

The wedge is larger when the effective liability horizon is shorter. For persistent pollutants, truncation is particularly important because a larger share of damages occurs after long delays.

A finite horizon is a reduced-form representation of liability survival rather than a literal managerial planning horizon. Let $q(t)$ be the probability that damages at horizon t remain attached to a solvent firm or successor. Then

$$MB_F^{\text{firm}}(F, B) = d(1 - m(B))g'(F) \int_0^\infty e^{-(r+\delta)t} q(t) dt. \tag{10}$$

If $q(t) = 1$, private and social valuations coincide. If $q(t) = e^{-\lambda t}$, the social-to-private wedge is

$$W_T(\lambda, \delta) = \frac{r + \delta + \lambda}{r + \delta}. \tag{11}$$

Bankruptcy, restructuring, mergers, spin-offs, and incomplete successor liability can all be interpreted as forces that lower $q(t)$.

Jurisdictional spillovers. A state regulator may internalize only a fraction of the damages from emissions in its jurisdiction. Let $\bar{\pi}_{ii}$ denote the share of damages from emissions in state i borne within the same state. The state-valued marginal benefit of source control is $\bar{\pi}_{ii} MB_F$, so the jurisdictional wedge is

$$W_J(\bar{\pi}_{ii}) = \frac{1}{\bar{\pi}_{ii}}. \tag{12}$$

PFAS exposure can arise through product supply chains, waste flows, contaminated media, and water systems that need not be located in the emitting state. A decentralized regulator therefore understates the value of prevention whenever $\bar{\pi}_{ii} < 1$.

Combined perceived return. Combining the liability and jurisdictional wedges gives

$$W_{\text{inst}}(T_f, \delta, \bar{\pi}_{ii}) = \frac{1}{(1 - e^{-(r+\delta)T_f})\bar{\pi}_{ii}}. \tag{13}$$

The perceived prevention–remediation ratio is

$$\mathcal{R}^{\text{perceived}} = \frac{\mathcal{R}^{\text{social}}}{W_{\text{inst}}}. \tag{14}$$

A remediation-heavy allocation can therefore arise even when the social marginal return to prevention is high. The wedge does not operate symmetrically across instruments. It falls most heavily on prevention because prevention avoids future and spatially dispersed damages, while remediation usually addresses observed exposure at identifiable sites.

3.4 Why High-Return Prevention May Remain Underfunded

The calibration below implies high marginal returns for some source-control interventions. That raises a natural question: why would these interventions remain underfunded? The model points to a simple answer, but the institutional setting adds several additional reasons.

First, the actor who can reduce emissions is often not the actor who bears the long-run cost of exposure. This is the direct implication of the firm-horizon and jurisdictional wedges. A manufacturer deciding whether to redesign a process, segregate waste, or substitute away from PFAS does not necessarily face the full present value of future damages. A state regulator deciding whether to restrict PFAS use may not internalize damages borne by downstream states or by consumers elsewhere. In both cases, the privately or locally valued return to source control can be below the national social return.

Second, source control often requires sunk investment before benefits are realized. Process redesign, chemical substitution, AFFF replacement, and waste-segregation systems require planning, capital expenditure, and operational changes. Once firms incur these costs, regulators may revise standards or impose additional requirements. Anticipation of such hold-up can delay investment even when the expected social return is high. Remediation is less exposed to this problem because it is usually purchased as a treatment service or cleanup activity after contamination is already observed.

Third, the measurement environment favors remediation. Drinking-water treatment costs are relatively well characterized. Engineering designs, operating costs, and removal rates are observable. Prevention benefits are more difficult to measure because the counterfactual stock path is not observed. A regulator can verify that a treatment system was installed and that measured concentrations fell at the tap. It is harder to verify the long-run damages avoided by a release that did not occur. This asymmetry in measurement can make downstream treatment politically and administratively easier to fund.

Fourth, remediation has a clear constituency. A contaminated water system, a military installation, or a Superfund site is geographically identifiable. The beneficiaries of cleanup can be named. Prevention benefits are dispersed across future populations and across exposure pathways. The political economy of appropriations therefore tends to favor visible cleanup over less visible source control.

These mechanisms are not alternatives to the sufficient statistic. They explain why an allocation with $\mathcal{R}^{\text{social}} > 1$ can persist. The empirical exercise below should therefore be read in two steps. The first step asks whether the social marginal return to prevention appears high at the observed allocation. The second asks whether institutional frictions are large enough to make the observed remediation-heavy allocation plausible.

4 Data and Parameter Construction

The calibration draws on EPA administrative datasets and supplementary sources covering emissions, drinking-water monitoring, treatment costs, enforcement, state policy timing, and liability. Table 3 summarizes the main data inputs.

Table 3: Data Sources

Database	Program	Records	Use
TRI Releases	EPA TRI	1,455	Emissions panel for δ
Discharge Monitoring	EPA NPDES	60,584	Emission validation
Spills (NRC)	EPA NRC	1,337	AFFF events
Drinking Water (UCMR)	EPA SDWIS	409,021	Contamination outcome
Drinking Water (State)	State DEQs	211,519	Robustness
Environmental Media	EPA WQX	310,273	Stock validation
Industry Sectors	EPA ECHO	206,596	Enforcement controls
Federal Sites	EPA FRS	756	Site location controls
Superfund Sites	EPA CERCLA	456	Remediation cost
GHG Emissions	EPA GHGRP	11,178	Plant-level controls
Transfers	EPA TRI	12,446	Waste flow
Production (CDR)	EPA OPPT	1,643	Volume context
Supplementary:			
Liddie et al. Dataverse	Harvard [Liddie et al., 2023]	44,111	CWS panel
Bloomberg PFAS Tracker	GitHub	465	State regulation panel
ITRC PFAS-1	ITRC	varies	MCL numeric values
SEC EDGAR 10-K filings	SEC	8 firms	Liability disclosures
PFAS settlements	Manual	25 events	Firm horizon T_f

Notes. The table reports approximate record counts after cleaning. The Liddie et al. replication package provides a community-water-system panel linked to PFAS sources and sociodemographic covariates.

The primary analysis sample combines EPA UCMR monitoring with the Liddie et al. replication sample. For each public water system, I construct PFAS detection and concentration measures, distance to TRI PFAS-reporting facilities, counts and emissions within fixed radii, analogous measures for airports and Superfund sites, population served, water source type, and state regulatory status.

4.1 Parameter Construction

Table 4 summarizes the structural objects used in the counterfactual exercise.

Table 4: Parameter Construction

Object	Baseline	Range / Source	Interpretation
δ^{eff} (/yr)	0.032	[0, 0.085]; lag profile	Reduced-form attenuation, not laboratory decay
D_0 (\$B/yr)	20	[5.5, 62.6]; Obsekov et al. [2023]	Aggregate annual U.S. PFAS damage
σ_W	0.55	[0.40, 0.70]	Share addressable by drinking-water remediation
σ_F	0.70	[0.25, 0.90]	Share of future damage avoidable by stopping new emissions
$\text{ROI}_F(F)$	piecewise	cap 10×/25× sensitivity	Source-control marginal returns
$\bar{B}$ (\$M/yr)	1,548	U.S. Environmental Protection Agency [2024b]	NPDWR saturation cost
$\bar{\eta}_B$	0.69	0.77 coverage × 0.90 removal	Maximum drinking-water treatment effectiveness
T_f (years)	31.2	[26.7, 34.2]	Settlement-timing proxy for liability horizon
$\bar{\pi}_{ii}$	0.196	state cost share / emission share	Self-internalization proxy
r	0.03	[0.05, 0.07]	Social discount rate
L (years)	30	GAC lifetime	Remediation-capital horizon

Notes. The parameters enter the welfare calculation in Section 6. Sensitivity ranges are reported in Table 8.

The effective attenuation parameter δ^{eff} is a reduced-form object. It combines chemical persistence, hydrologic dilution, transport away from monitoring locations, changes in source mix, and reporting-threshold dynamics. The source-control return schedule is calibrated from Table 2. The first tier has ROI 125 because the three targeted interventions cost about \$114 million annually and avert about \$14.28 billion in modeled damages. Subsequent tiers impose diminishing returns, and cumulative source-control benefits are capped at $\sigma_F D_0$. The back-end benefit function is

$$B_{\text{averted}}(B) = \bar{\eta}_B \sigma_W D_0 \min\left(\frac{B}{\bar{B}}, 1\right). \tag{15}$$

5 Calibration and the Sufficient Statistic

5.1 Persistence

I discipline plausible values of δ using the lag profile of emissions and contamination. For each emission year $\tau \in \{2014, \ldots, 2023\}$, I estimate

$$y_j = \alpha_\tau + \beta_\tau \ln(1 + e_{j,\tau}^{100\text{km}}) + X_j' \gamma_\tau + u_{j,\tau}, \tag{16}$$

where y_j is PFAS detection at water system j and $e_{j,\tau}^{100\text{km}}$ is TRI PFAS releases within 100 km in year τ. Under a stable transport kernel, the meta-regression $\log |\hat{\beta}_\tau| = a + b(T - \tau)$ estimates $b = -\delta$.

Table 5: Lag Profile of Emission–Contamination Correlations

Emission year τ	Lag $T - \tau$	$\hat{\beta}_\tau \times 10^4$	SE $\times 10^4$	Ratio to 2023
2014	9	5.97	1.95	0.65
2015	8	5.54	1.99	0.61
2016	7	10.01	2.20	1.10
2017	6	11.63	2.34	1.27
2018	5	11.92	2.39	1.31
2019	4	11.67	2.44	1.28
2020	3	9.09	1.80	1.00
2021	2	10.27	1.80	1.13
2022	1	7.70	1.69	0.84
2023	0	9.13	1.72	1.00

Notes. $N = 8,723$ water systems per row. The lag profile is consistent with slow attenuation but does not statistically distinguish $\delta = 0$ from moderate positive values. Meta-regression: $\hat{\delta} = 0.032$/yr (SE 0.026, $p = 0.26$); 95% upper bound 0.085.

5.2 Plant-Closure Evidence

The lag-profile exercise is correlational. Plant closures offer a more direct, though data-limited, way to study attenuation. Figure 2 compares PFOA detections near 3M's Cottage Grove facility before and after closure, using UCMR 3 and UCMR 5 monitoring harmonized to the UCMR 3 minimum reporting level. The treated sample is small, so I use this exercise as a descriptive cross-check rather than as a replacement for the baseline attenuation parameter.

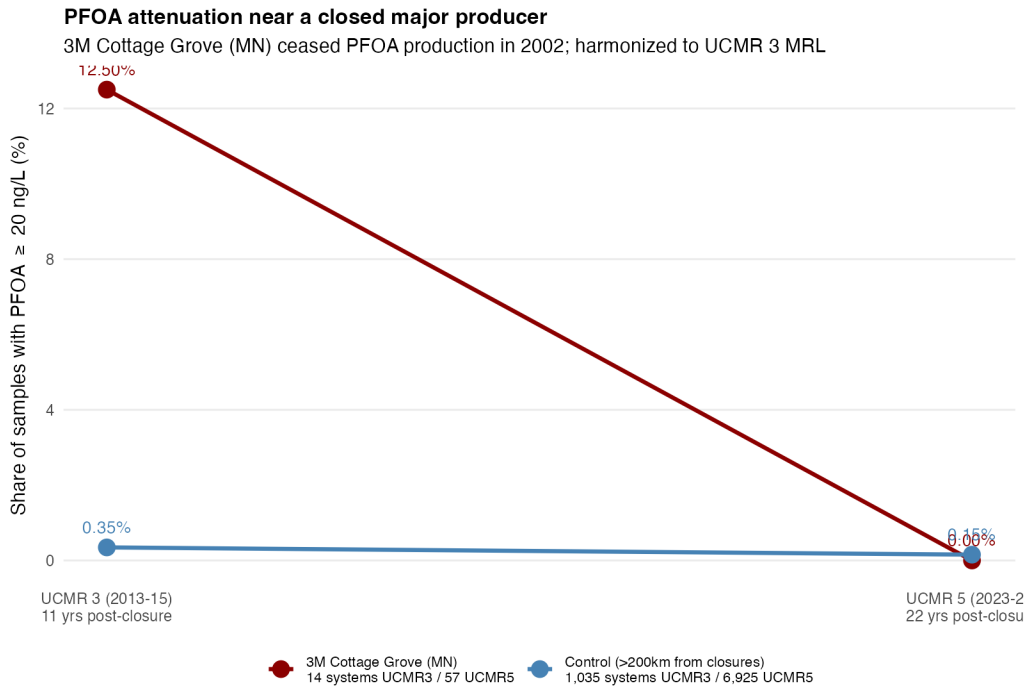


Figure 2: PFOA Attenuation Near a Closed Major Producer

Notes. Share of public-water-system PFOA samples at or above 20 ng/L near 3M Cottage Grove and in a control group more than 200 km from major closure sites. The figure is descriptive because the pre-period treated sample is small.

5.3 Firm Horizon and Jurisdictional Self-Internalization

I calibrate T_f from the timing of 25 major PFAS-related settlements, totaling $18.7 billion between 1999 and 2025. Treating settlement events as a reduced-form measure of liability survival, the amount-weighted mean lag from exposure-vintage midpoint to settlement date is 31.2 years, with bootstrap 95 percent confidence interval $[26.7, 34.2]$. At $r = 0.03$ and $\delta = 0.032$, this implies $W_T = 1.43$.

I proxy the state-level self-internalization share using the ratio of state treatment-cost share to state emission share. The emission-weighted mean is $\hat{\pi}_{ii} = 0.196$. The value should not be interpreted as a physical transport share. It is a reduced-form proxy for the gap between where emissions occur and where treatment costs are borne.

5.4 Implications for the Sufficient Statistic

At the current observed allocation, $F^{SQ} \approx \$30$ million and $B^{SQ} \approx \$2.97$ billion. The relative-horizon factor at $r = 0.03$, $\delta = 0.032$, and $L = 30$ is approximately 0.82. The relative-effectiveness factor is large because the next dollar of source control lies in the

high-return tier, while baseline remediation spending exceeds the drinking-water saturation point in the NPDWR calibration.

Table 6 collects the calibrated values. The baseline social prevention–remediation ratio is above one; under a 10× ROI cap, it remains above one but is much smaller. The institutional wedge is also sizable, which is consistent with the observation that private and decentralized actors may not supply the socially preferred amount of source control.

Table 6: Calibrated Parameters and Implied Sufficient Statistic

Parameter	Calibrated value	Source
δ^{eff} (/yr)	0.032 [0, 0.085]	Lag profile
Source-control ROI, first tier	125	Schedule in Section 4.1
Remediation ROI at saturation	2.8×	EPA NPDWR EA
T_f (years)	31.2 [26.7, 34.2]	Settlement timing
$\bar{\pi}_{ii}$	0.196	Cost-share / emission-share proxy
r	0.03	Baseline discount rate
W_T	1.43	Baseline T_f, δ, r
W_J	5.10	$1/\bar{\pi}_{ii}$
$\mathcal{R}^{\text{social}}$	≈ 5.3	Baseline counterfactual
$\mathcal{R}^{\text{social}}$ with 10× ROI cap	≈ 1.4	Sensitivity

6 Counterfactual Policy Allocation

A planner faces a fixed spending envelope $E = \$3$ billion per year, anchored to the identifiable annual spending in Table 1. The planner chooses F and $B = E - F$ to maximize total damage averted minus the fixed envelope cost. The baseline aggregate damage anchor is $D_0 = \$20$ billion per year, the midpoint of the range in Obsekov et al. [2023]. Two scalar shares govern how each instrument maps into damages: σ_W, the share of damage mediated by drinking-water exposure, and σ_F, the share of future damage attributable to ongoing emissions rather than legacy stock.

I set $\sigma_W = 0.55$ and use $\sigma_F = 0.70$ in the baseline. The latter is best interpreted as a high-flow specification; Section 6 reports sensitivity to lower values. The front-end benefit function follows the piecewise ROI schedule described above and is capped at $\sigma_F D_0$. The back-end benefit function follows equation (15). I also impose a modest overlap correction to avoid double-counting pathways affected by both instruments.

Table 7: Status Quo vs. Optimal Allocation

	Status quo	Optimum	Change
Front-end share of envelope	1.0%	7.6%	+6.6 pp
Front-end spending ($M/yr)	30	228	+198
Back-end spending ($M/yr)	2,970	2,772	−198
Front-end damage averted ($B/yr)	3.75	13.30	+9.55
Back-end damage averted ($B/yr)	7.60	7.60	—
Overlap adjustment ($B/yr)	−0.56	−1.90	−1.34
Total damage averted ($B/yr)	10.79	19.00	+8.21
Envelope spent ($B/yr)	3.00	3.00	—
Net benefit ($B/yr)	**7.78**	**16.00**	**+8.22**

Notes. In the baseline, the reallocated dollars come from remediation spending beyond the NPDWR drinking-water saturation margin $\bar{B} = \$1.55$ billion. Sensitivity to positive marginal returns on remediation above $\bar{B}$ is reported in Table 8.

Figure 3 summarizes the policy allocation curve and the posterior distribution of welfare gains under uncertainty in the damage anchor.

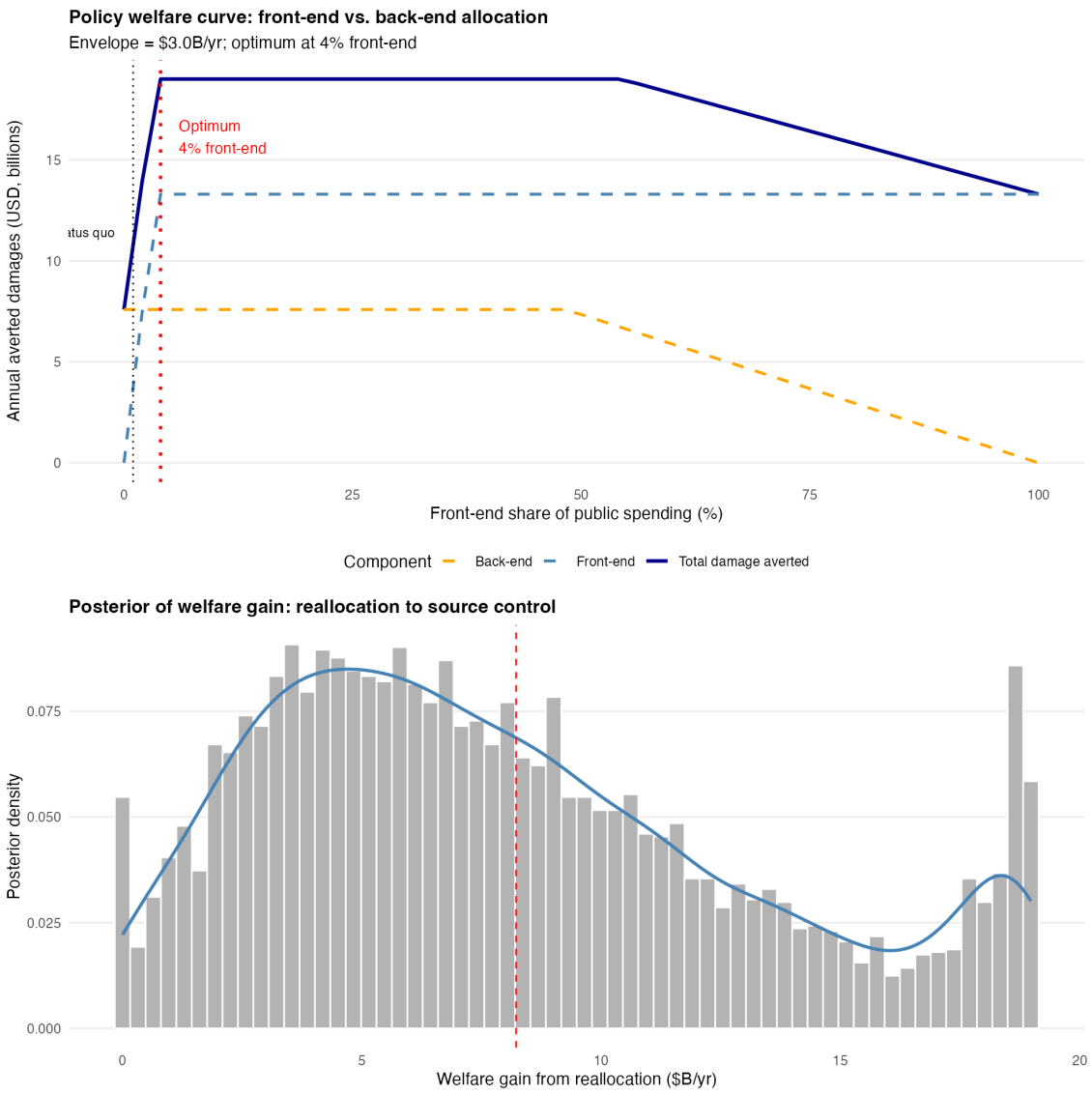


Figure 3: Policy Allocation Curve and Posterior Welfare Gain

Notes. The upper panel plots annual avoided damages against the front-end share of public spending. The lower panel plots the posterior distribution of welfare gains from reallocating spending toward source control.

6.1 Sensitivity

Table 8 reports sensitivity to persistence, discounting, the aggregate damage anchor, source-control returns, exposure-pathway shares, and residual returns to remediation above the NPDWR saturation point.

Table 8: Sensitivity of the Welfare Gain

Specification	Welfare gain ($B/yr)	ROI of optimum	Optimal F share
Baseline	**8.2**	5.3	7.6%
$\delta = 0$	8.4	5.4	8.0%
$\delta = 0.10$	7.5	5.1	6.5%
$r = 0.05$	7.7	5.2	7.2%
$r = 0.07$	7.1	5.1	6.8%
$D_0 = \$5.5$B	2.0	1.7	4.8%
$D_0 = \$62.6$B	28.1	11.2	12.5%
Exclude AFFF intervention	4.1	3.0	5.0%
Cap source-control ROI at 25×	3.7	2.7	4.5%
Cap source-control ROI at 10×	1.5	1.4	2.8%
Source-control benefits reduced 90%	0.4	0.6	1.5%
$\sigma_W = 0.40$	9.3	5.7	8.5%
$\sigma_W = 0.70$	6.9	4.7	6.4%
$\sigma_F = 0.25$	1.8	1.6	3.5%
$\sigma_F = 0.50$	5.4	4.0	6.2%
$\sigma_F = 0.90$	9.6	6.1	8.4%
Marginal $\text{ROI}_B = 1\times$ above $\bar{B}$	7.8	5.1	7.2%
Marginal $\text{ROI}_B = 2\times$ above $\bar{B}$	6.5	4.5	6.0%
ROI cap 10×, $D_0 = \$5.5$B, $\sigma_F = 0.25$	−0.3	0.9	1.0%

Notes. "Exclude AFFF" drops the AFFF replacement intervention. "Cap source-control ROI" replaces all prevention ROI values exceeding the cap with the cap.

The main conclusion is qualitative rather than exact. Modest reallocation toward source control has positive welfare gains in most specifications. The magnitude is most sensitive to the aggregate damage anchor, the source-control ROI schedule, and the share of future damages attributable to ongoing emissions.

7 Reduced-Form Evidence from State PFAS Regulations

Source-oriented regulation provides a useful empirical contrast for the calibration. The welfare calculation treats prevention and remediation as distinct policy margins. State PFAS policies allow a limited check of that distinction because they intervene at differ-

ent points in the lifecycle. Drinking-water maximum contaminant levels (MCLs) regulate delivered water quality and impose compliance obligations on water systems. AFFF restrictions and restrictions on intentionally added PFAS operate closer to the emissions or use margin. If the classification is meaningful, source-oriented policies should be more closely related to upstream releases than MCLs are.

I do not use state policy variation to estimate the welfare gain from reallocation. The TRI PFAS panel is too small, and mandatory PFAS reporting begins too late, to support that role. The purpose is narrower: to ask whether the available emissions data distinguish policies that act upstream from policies that act downstream. A finding that all PFAS policies have similar associations with TRI releases would weaken the prevention–remediation distinction. A finding that only source-oriented policies are associated with lower releases would support the allocation interpretation used in the calibration.

7.1 Specification

Let e_{ist} denote TRI-reported PFAS releases for facility i in state s and year t. I estimate

$$\log(1 + e_{ist}) = \alpha_i + \lambda_t + \beta^S \text{Source}_{st} + \beta^M \text{MCL}_{st} + X'_{ist}\gamma + u_{ist}, \tag{17}$$

where α_i are facility fixed effects and λ_t are year fixed effects. The source-policy indicator equals one after a state adopts a source-oriented PFAS restriction, such as an AFFF restriction or a restriction on intentionally added PFAS in selected products. The MCL indicator equals one after a state adopts an enforceable drinking-water standard. Facility fixed effects absorb time-invariant differences across reporting facilities, while year fixed effects absorb national changes in reporting, enforcement, and PFAS salience.

The coefficient β^S measures the within-facility association between source-oriented policy adoption and reported releases. The coefficient β^M measures the analogous association for drinking-water standards. The model gives a sign prediction for β^S: source-oriented policies should reduce releases if they bind on emissions or use. No analogous sign prediction applies to β^M, because MCLs regulate downstream water quality rather than releases from TRI-reporting facilities.

Staggered adoption raises the usual concerns about two-way fixed-effects estimates when treatment effects are heterogeneous. I therefore also estimate group-time average treatment effects following Callaway and Sant'Anna [2021] and compare them with imputation-based estimates in the spirit of Borusyak et al. [2024]. These specifications are not a substitute for broader treatment variation. They are used to check whether the sign pattern is sensitive to the estimator.

7.2 Results

The estimates separate the two policy margins. Source-oriented regulation is associated with lower TRI PFAS releases. Drinking-water MCL adoption is not associated with a statistically distinguishable reduction in reported releases. The contrast is consistent with the institutional design of the policies: restrictions on PFAS use or release operate upstream, while drinking-water standards require treatment after contamination is present.

Several features make the comparison informative despite the small sample. Facility fixed effects rule out level differences between facilities in states that adopt source-oriented policies and those that do not. Year effects absorb national changes in TRI reporting and PFAS attention. Including MCL adoption in the same regression separates source-directed policies from downstream regulatory activity. The absence of a similar release response after MCL adoption reduces the concern that the source-policy coefficient merely captures a general increase in state PFAS salience.

The evidence remains limited. The TRI panel contains 76 PFAS-reporting facilities, and only a small number are exposed to source-oriented policies during the sample window. Pre-trend tests have low power because mandatory PFAS TRI reporting begins late in the period. Standard errors account for within-state dependence, but sparse treatment limits precision. For these reasons, the estimates should not be interpreted as a structural estimate of $g'(F)$.

The useful inference is narrower. Upstream policies are the policies associated with changes in upstream releases; downstream drinking-water standards are not. That pattern supports the empirical relevance of the prevention–remediation distinction used in the welfare calculation. It also suggests that pooling all PFAS policies into a single treatment indicator would obscure economically meaningful differences across policy margins.

8 Limitations and Identification Priorities

The results should be interpreted as a calibrated welfare exercise rather than as a fully identified structural estimate. Three sources of uncertainty are especially important.

First, the persistence parameter is disciplined by a lag-profile exercise rather than by a clean plant-closure design. The estimate is consistent with slow attenuation, but it does not sharply distinguish $\delta = 0$ from moderate positive values. Second, the source-control ROI schedule is drawn from engineering and policy documents rather than from plant-level revealed-cost estimates. Third, the jurisdictional wedge is proxied by the ratio of

treatment-cost shares to emission shares, which captures more than physical interstate transport.

These limitations point to clear identification priorities. Plant-closure event studies could estimate effective attenuation directly. Facility-level variation in technology, production, and regulatory exposure could be used to estimate abatement-cost curves. Hydrological and product-flow matrices could replace the reduced-form self-internalization proxy with a physical spillover matrix.

9 Conclusion

Persistent pollutants make the timing and location of control central to policy design. For PFAS, current public spending is concentrated downstream, while the calibrated marginal return to targeted source control appears high under a wide range of assumptions. The quantitative magnitudes should be interpreted with caution. The exercise relies on external damage estimates, engineering cost information, and reduced-form proxies for liability and jurisdictional spillovers. Better plant-closure designs, facility-level abatement-cost estimates, and physical transport matrices would sharpen the estimates.

A more limited conclusion is therefore appropriate. Drinking-water standards and cleanup programs address important exposure margins, especially where legacy contamination is already present. Yet for long-lived pollutants, remediation alone leaves the future stock path largely unchanged. Marginal public resources may have high value when directed toward identifiable source-control margins, including AFFF replacement, abatement at major PFAS-reporting facilities, and waste-segregation requirements for PFAS-using manufacturers. The broader lesson is that persistence turns environmental policy into a lifecycle allocation problem.

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2 Research Timeline

This section summarizes the remaining research plan and expected submission timeline. The dissertation is organized around three papers. Each chapter is intended to be submitted as a standalone journal article.

Table 1: Dissertation Research and Submission Timeline

Chapter	Target Date	Planned Work
Chapter 1	June 2026	Finalize manuscript and submit to a journal. Polish introduction, tighten identification discussion, finalize robustness checks, and prepare submission materials.
Chapter 2	June 2026	Revise empirical design, clarify contribution, update tables and figures.
Chapter 2	July 2026	Complete manuscript revisions and submit to a journal.
Chapter 3	July 2026	Revise PFAS chapter: calibration, institutional background, and interpretation of the sufficient statistic.
Chapter 3	August 2026	Add supplementary empirical evidence on state PFAS policies and source-oriented regulation.
Chapter 3	End of August 2026	Submit the PFAS chapter to a journal.

The main priority for the coming months is to move each chapter from proposal draft to journal-ready manuscript. Chapter 1 will be prioritized first. By the end of August 2026, all three dissertation chapters should be submitted or ready for submission.